

1 Old and New Euclidean Geometry and Analysis

Contents

1.1	Preliminaries	2
1.2	Distance Geometry	2
1.2.1	A Basic Formula	2
1.2.2	The Length of a Path	3
1.2.3	The First Variation Formula and Application to Billiards	4
1.3	Plane Curves	9
1.3.1	Length	9
1.3.2	Curvature	12
1.4	Global Theory of Closed Plane Curves	18
1.4.1	“Obvious” Truths About Curves Which are Hard to Prove	18
1.4.2	The Four Vertex Theorem	20
1.4.3	Convexity with Respect to Arc Length	22
1.4.4	Umlaufsatz with Corners	23
1.4.5	Heat Shrinking of Plane Curves	24
1.4.6	Arnol’d’s Revolution in Plane Curve Theory .	24
1.5	The Isoperimetric Inequality for Curves	26
1.6	The Geometry of Surfaces Before and After Gauß	29
1.6.1	Inner Geometry: a First Attempt	30
1.6.2	Looking for Shortest Curves: Geodesics	33
1.6.3	The Second Fundamental Form and Principal Curvatures	45
1.6.4	The Meaning of the Sign of K	52
1.6.5	Global Surface Geometry	55
1.6.6	Minimal Surfaces	58
1.6.7	The Hartman-Nirenberg Theorem for Inner Flat Surfaces	62
1.6.8	The Isoperimetric Inequality in $\mathbb{E}^3$ à la Gromov	63
1.7	Generic Surfaces	66
1.8	Heat and Wave Analysis in $\mathbb{E}^2$	70
1.8.1	Planar Physics	70
1.8.2	Why the Eigenvalue Problem?	71
1.8.3	Minimax	75

	1.8.4	Shape of a Drum	78
	1.8.5	Heat	87
	1.8.6	Relations Between the Two Spectra	91
1.9		Heat and Waves in $\mathbb{E}^3, \mathbb{E}^d$ and on the Sphere ..	94
	1.9.1	Euclidean Spaces	94
	1.9.2	Spheres	95
	1.9.3	Billiards in Higher Dimensions	97
	1.9.4	The Wave Equation Versus the Heat Equation	98

1.1 Preliminaries

Some (not all) of the following generalizes to any dimension; this will be left to the reader. We will only give references for special topics. General references could be Berger and Gostiaux 1988 [175], Coxeter 1989 [409], do Carmo 1976 [451], Klingenberg 1995 [816], Spivak 1979 [1155], Sternberg 1983 [1157], and Stoker 1989 [1160]. For those who like computer programming, Gray 1998 [584] will be of interest. We will assume elementary calculus and also that functions are differentiable as often as needed.

1.2 Distance Geometry

1.2.1 A Basic Formula

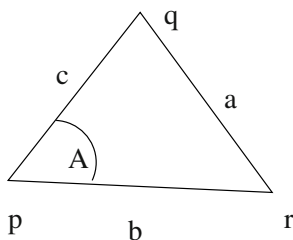


Fig. 1.1. A triangle in $\mathbb{E}^2$

In the Euclidean plane $\mathbb{E}^2$ we have distances between points and angles between smooth curves wherever the curves meet. The equation

$$a^2 = b^2 + c^2 - 2bc \cos A \quad (1.1)$$

should be kept in mind, where a, b, c are the pairwise distances between three points, as in figure 1.1 and A is the angle opposite the side of length a . The triangle inequality in $\mathbb{E}^2$:

$$a \leq b + c$$

is strict, in that equality forces the three points to be colinear, with p between q and r . Equation 1.1 is basic in telemetry, needed when your instant camera measures distances for you. To know the distance between any two of the points, you only need the angle (a local measurement) and two side lengths, not all three.¹ We will encounter other spaces like this later on. Think of mountains in your way along one side, as in figure 1.2. We use equation 1.1 in everyday life, in primitive geodesy, astronomy, automatic focusing, etc.

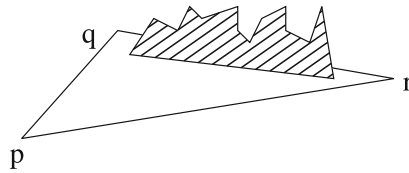


Fig. 1.2. One side is hard to measure

1.2.2 The Length of a Path

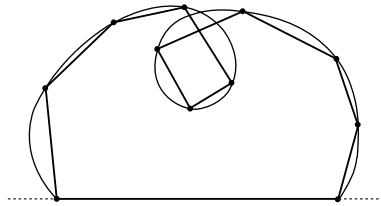


Fig. 1.3. Polygonal approximation

A *path* between two points is a continuous curve between those points. Its *length* is the supremum of the total lengths of polygons inscribed in the curve. The supremum need not be finite, e.g. a fractal curve like a snowflake. But if the curve, $c(t)$, is smooth enough² say with derivative $c'(t)$, then it has finite length $\int \|c'(t)\|$. The strict triangle inequality above ensures that the shortest path between two points is exactly the straight line interval between them.

For any metric space, one can define the length of a path as a supremum as above, and then consider the infimum of path lengths between two points.

¹ However, keep in mind the difficulty of making local measurements; angles between little pieces of straight lines are very sensitive to mismeasurement.

² More precisely, rectifiable; see Wheeden & Zygmund 1977 [1258].

If all points are connected by finite length paths, then this infimum is another metric on the same metric space. In general, it is a different metric, as in the case of the sphere in $\mathbb{E}^3$. The new metric is called the *inner metric*, and if the initial metric coincides with the inner metric, then the space is called a *length space*. Length spaces are becoming more important today. For example, see Burago, Gromov and Perelman 1992 [282] and Gromov 1999 [633].

1.2.3 The First Variation Formula and Application to Billiards

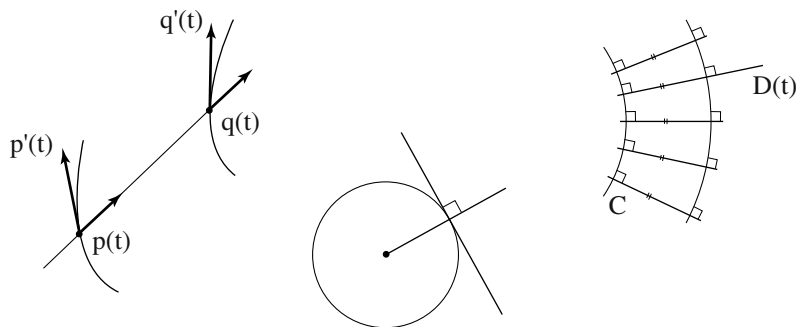


Fig. 1.4. Two points travelling through the plane

Imagine that we have two points $p(t)$ and $q(t)$ running along two curves in the Euclidean plane, as in figure 1.4 and we watch how the distance $d(p(t), q(t))$ varies with time t . The first variation formula is:

$$\frac{d}{dt}d(p(t), q(t)) = (q'(t) - p'(t)) \cdot \frac{q(t) - p(t)}{\|q(t) - p(t)\|} \quad (1.2)$$

Consequently, if we have a one parameter family of straight lines, then any two curves which are perpendicular to the lines must be at constant distance; for example, circles about the origin. So the tangent line to a circle is orthogonal to the radius.

A more elaborate application of this equation, which will be important later on, is the *billiard table*. Suppose that we have a convex domain in the plane, say D , and consider a small particle (think of a small ball, of infinitesimal radius) flying inside D with no force applied to it. The particle will travel on a straight line with constant speed through the interior of D , and when it strikes the boundary ∂D it will bounce and travel on a new straight line. The new straight line path and the old one will be reflections of one another about the normal line of the curve ∂D at the point of impact. We will refer to this as *mirror bouncing*, since we see the same trajectories if we have light rays travelling inside a curved mirror in the shape of ∂D . For the moment, we will

suppose that ∂D is differentiable and strictly convex. We will call our particle a *billiard ball*. The book of Tabachnikov 1995 [1175] is a complete reference on billiards; some other references are Sinai 1990 [1141], Katok et. al. 1986 [788], Arnoux 1988 [70], Veech 1989 & 1991 [1208, 1209], Katok 1987 [785], and Gutkin 1996 [672].

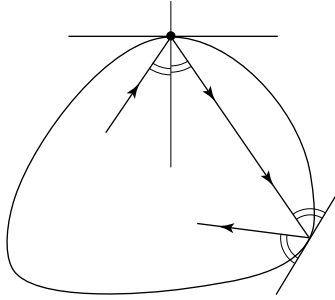


Fig. 1.5. A billiard ball

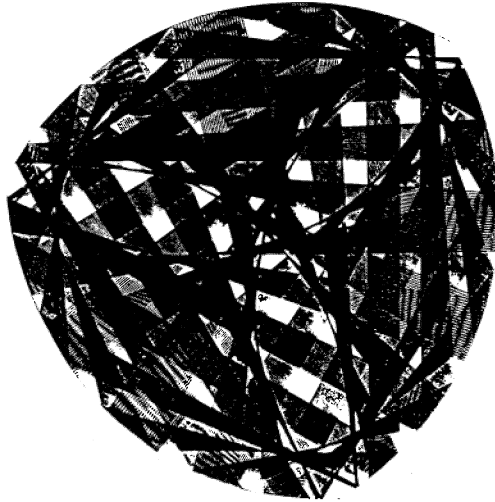


Fig. 1.6. The long term behaviour of the billiard ball

What happens to these trajectories over large periods of time? As we will see in chapter 10, this question is fundamental in rational mechanics and almost every field of physics. It may happen that a trajectory is periodic, returning eventually to its original position and velocity. The search for periodic trajectories is a corner stone in the study of long duration dynamics.

We devise the length counting function to try to count periodic trajectories, setting $CF(L)$ to be the number of periodic trajectories of length less than L . Let us now see how to exhibit some periodic trajectories.

In 1913, G. D. Birkhoff remarked that the first variation formula yields many periodic trajectories. Pick any number P of points of ∂D , in a fixed order (not necessarily consecutive along ∂D) as in figure 1.7, turning around the boundary Q times. Draw the polygon connecting those points in that order. Using the first variation formula, we find that mirror bouncing is exactly the condition that the polygon has critical length under perturbing its vertices along ∂D . But now for each choice of P and Q , we can take the polygon of greatest length, which exists by compactness of the space of polygons with given P and Q values. N.B. this construction does not yield all of the periodic trajectories. We will return to billiards in §§1.8.6.

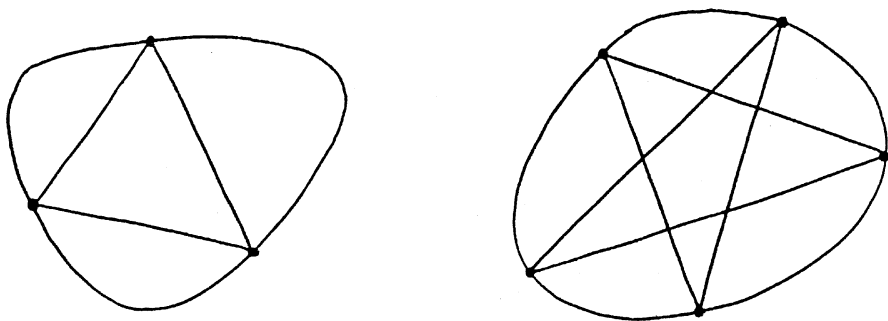


Fig. 1.7. Two of Birkhoff's configurations

Let us take our domain D to be a convex polygon, a case we expect to be simpler than a general convex curve. We shall be careful to eliminate trajectories which strike a vertex of the boundary. It is still an open question if there is even one periodic trajectory. This question is solved for polygons with angles rational multiples of π (see below), but is open for general polygons, even triangles. If the triangle is not obtuse, I leave it to you to find a (classical) periodic trajectory (hoping that you are not obtuse either). Everyone expects that infinitely many periodic trajectories exist, as for rational angled polygons and convex domains.

Rational angled polygons and convex domains are the only domains for which periodic billiard trajectories are well understood. To the rational angled polygons, one can associate a compact space, a generalized Riemannian manifold, locally Euclidean but for $2\pi k$ singularities, k an integer. See Arnoux 1988 [70] for more information. One knows that $CF(L)$ is asymptotically quadratic in L , as $L \rightarrow \infty$. See Vorobets, Gal'perin, and Stëpin 1992 [1224] and Gal'perin and Chernov 1991 [545].

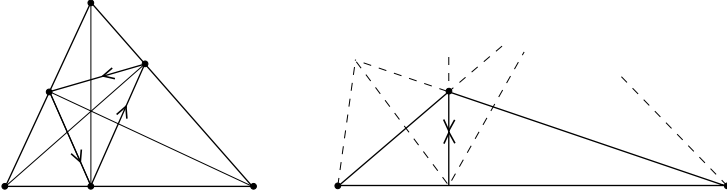


Fig. 1.8. Two triangular billiard tables

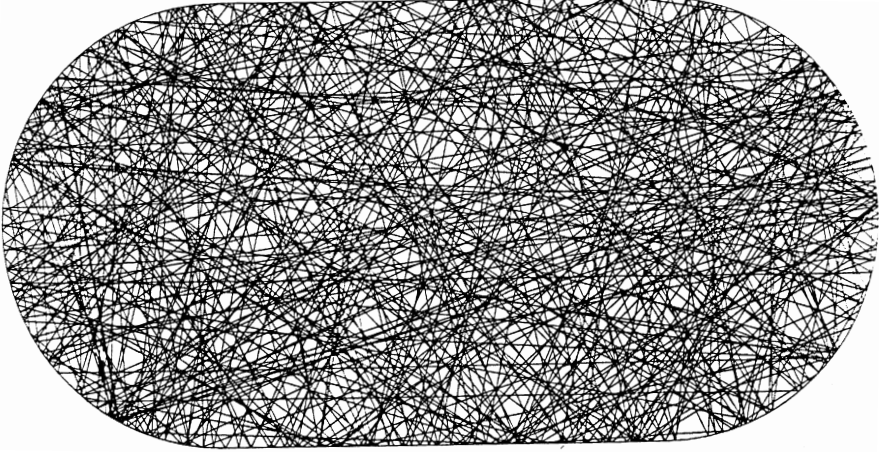


Fig. 1.9. The Bunimovich Stadium

In concave domains, beams of light or pencils of trajectories are dispersed when reflected from the boundary, as in figure 1.10 on the following page and this implies *ergodicity* (see §§§10.5.1.1 for the definition). See Katok et. al. 1986 [788] for the proof. Finding a simple proof is an open problem. However, a good deal is known about concave domains; for example let us just mention that $CF(L)$ is asymptotically exponential in L . Also see §§1.8.6 below for a heuristic relation between concave billiard tables, statistical mechanics, and manifolds with negative curvature.

For both rational angled polygons and convex domains, rigorous proofs of existence of periodic trajectories are quite difficult. There is a third case, a generalization of the *Bunimovich stadium* which is not completely mysterious; see section 5.3 of Tabachnikov 1995 [1175] for a modern proof that there are periodic trajectories, and also see the references therein. In this case we have ergodicity and $CF(L)$ is exponential in L . These examples are convex (but with flat parts) and came as a shock to the ergodic community in 1979 when most people were convinced that some concavity was required for ergodicity, or if you prefer, that convex domains had too much focusing to achieve ergodicity.

Defocusing is apparent in figure 1.10. In §§1.4.3 we will see an interesting curvature condition needed to construct these examples.

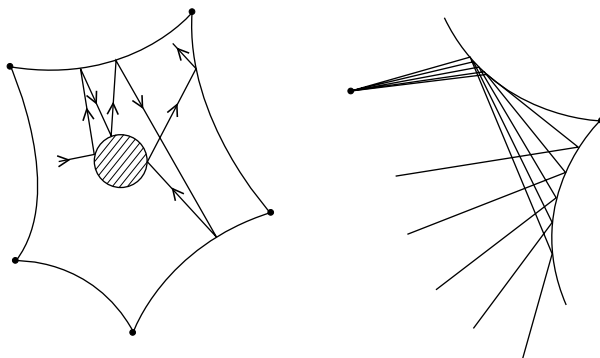


Fig. 1.10. Spreading rays in a nonconvex billiard table

A strictly convex billiard table can not be ergodic, because its flow has caustics (at least if its boundary is six times differentiable, or perhaps less—the optimal result is open), by a fundamental result of Lazutkin, discussed on page 92. But it is an open problem whether there are smooth ergodic billiard tables (the Bunimovitch and Wojtkowski examples are not smooth, having flat parts joined to curved parts with nonvanishing curvature).

One does not know how general these examples are. Let us mention two general results. The first, due to Katok, asserts that for polygonal billiards, $CF(L)$ grows more slowly than exponentially. See Katok's 1987 [785] and also the nice survey of Gutkin 1996 [672]. The proof is extremely interesting and consists of two parts. The first part studies the trajectories using successive reflexions on the sides, unwrapping the trajectory to make it a straight line on the plane. But this is not enough: the second part resorts to a strong theorem in ergodic theory connecting topological and metric entropy. We will define entropy in §§§10.5.1.2.

The second result is a density theorem in the set of all convex billiard tables, found in Arnoux 1988 [70]. For concave billiard tables, the length counting function $CF(L)$ is very well understood. One has the precise asymptotic expression

$$CF(L) \sim \frac{\exp hL}{hL}$$

as $L \rightarrow \infty$ with h the entropy, which is known to be positive. For information on higher dimensional billiards, see §§ 1.9.3 on page 97.

1.3 Geometry of Plane Curves and Two Dimensional Point Kinematics

1.3.1 Length

We will briefly recall some classical observations concerning plane curves, but also some less classical ones. Our style will probably appear different from standard expositions. This is intentional, in order to introduce the viewpoint of Riemannian geometry. So even if you think you know everything about plane curves and point kinematics, you might do well to glance at what follows.

We have a plane curve c with a point $c(t)$ running along it, a function of a parameter t , as smooth as needed (i.e. differentiable as many times as needed). The velocity of this curve c is $c'(t)$. As we have already noted, the length of c from $t = a$ to $t = b$ is

$$\int_a^b \|c'(t)\| dt .$$

This gives a distance on the curve, which is the time needed to traverse from $c(a)$ to $c(b)$ with unit speed. This inner geometry is the same as the geometry of an interval. By *inner* we mean that you care only about what happens on the curve itself, and not what happens outside. Consequently, all curves having the same length are *isometric*, namely they are the same as metric spaces. Equivalently, there is a one to one map which is an *isometry* between them (i.e. preserves distances). Moreover all of them are isometric to an interval of the same length on the Euclidean line $\mathbb{E}^1$. Surfaces will be completely different; see §§1.6.1.

Alternately, look at the triangle inequality. For the inner metric of a plane curve, at least locally,

$$d(p, r) = d(p, q) + d(q, r)$$

for any three points p, q, r with q between p and r . On a circle, between is a local notion, as you see in figure 1.11. Or think about the word “between”. If we call a curve *closed* when it is periodic, then such a curve has a total length, and two closed curves are isometric exactly when they have the same length.

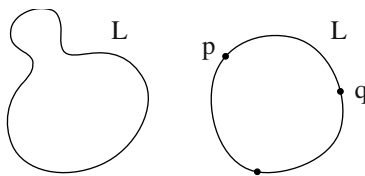


Fig. 1.11. These two curves are isometric

An important remark is in order: the attentive reader might have been puzzled, even outraged, that we never distinguished between a curve as a map $t \rightarrow c(t)$ from an interval of the real line, and its image as the set of points $c(t)$ when t ranges through the values of that interval. It is not easy to have a clear view of these two notions, and the relations between them; they are obscure even in many books. We refer the interested reader to chapter 8 of Berger and Gostiaux 1988 [175] for a very detailed exposition of this point. Now we will just say a few words about this distinction.

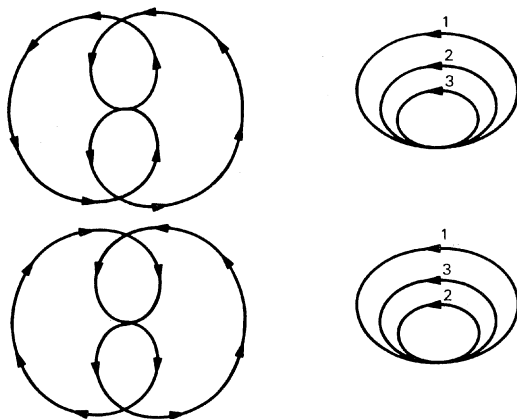


Fig. 1.12. Geometric images you can travel through on different routes

Maps $t \mapsto c(t)$ from an interval of $\mathbb{E}^1$ into $\mathbb{E}^2$ should be called *parameterized curves*. Most often only *regular curves* are allowed, i.e. the speed $c'(t)$ is never zero. Note that here, and also in the sequel, we use two different notions of how quickly something moves: the *speed* $\|c'(t)\|$ which is the norm of the *velocity* $c'(t)$ (see Feynman, Leighton and Sands 1963 [516], page 9-2). We skip intermediate developments (again, see Berger and Gostiaux 1988 [175]) and jump directly to the concept of a *geometric curve*: a line in the plane (not necessarily a straight line). By this we mean a subset of the plane which, near any point, can be reshaped into a straight line, i.e. taken by diffeomorphism of an ambient region of the plane into a straight line. In modern mathematical jargon, such an object is called a *one dimensional submanifold of the plane*. The link between the two notions is that the image set made by all of the $c(t)$ is always, but in general *only locally*, a geometric curve, and conversely a geometric curve admits regular parameterizations, in particular by arc length (i.e. with constant unit scalar speed). In the sequel, we will most often leave it to the reader to understand if parameterized or geometric curves are considered.

A word about orientations of curves: parameterized curves are automatically oriented, while geometric curves can be given two orientations. When you change parameterizations of a curve, two invariants appear: the first is

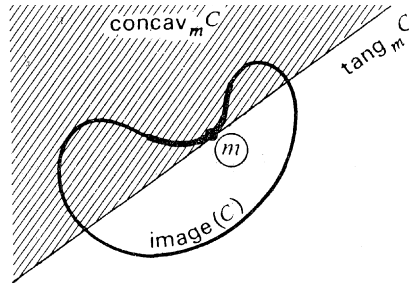


Fig. 1.13. Concavity and tangent line

the *tangent line* to which every $c'(t)$ belongs. The second is defined only for *biregular* curves, i.e. curves for which the acceleration $c''(t)$ is not proportional to the velocity $c'(t)$. The acceleration will then remain in the same open half plane under any reparameterization, and this half plane is called the *concavity* of the curve. It does not depend on orientation. If you parameterize the curve by arc length, the two orientations yield two opposite unit velocity vectors, but one acceleration vector, which is then attached canonically to the curve.

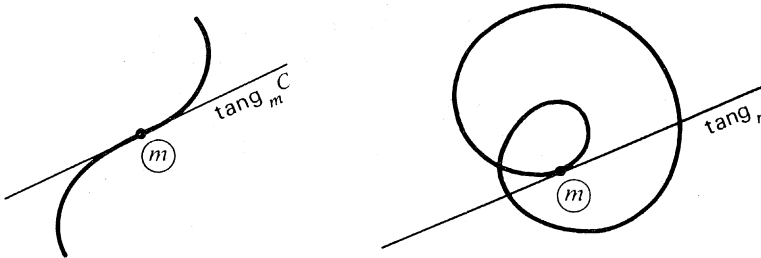


Fig. 1.14. Tangent lines

Everything said here is valid with obvious modification for curves in any Euclidean space $\mathbb{E}^d$.

A closed (periodic) plane curve is said to be *simple* if it is a one-to-one map up to the period (this for a parameterized curve). For a geometric curve, *simple* means a curve which has the global topology of a circle (in the jargon, it is a differentiable embedding in the plane of the circle seen as an abstract one dimensional manifold). We also will assume the speed never vanishes, indeed that it has unit speed.

Do not think that everything is known today about plane curves. A famous problem is to prove that in any plane closed curve, one can inscribe a square. For this problem, and other open ones, see the book Croft, Falconer & Guy 1994 [410]. The oldest unsolved problem in plane geometry is the *equichordal problem*, from 1916: prove that no plane curve can have two equichordal points,

i.e. points so that all chords through them have constant length. The problem is studied extensively in note 6.3, chapter 6 of Gardner 1995 [548]. Do not miss the recent revolution in the theory of plane curves: see the end of §1.4.

1.3.2 Curvature

We now look at the outer geometry of plane curves. The inner geometry of curves does not differ from that of straight lines, but the geometry is radically different if we look at the way a curve sits in the plane. We are going to introduce a concept of *curvature* which measures how much a curve differs from a straight line. For a curve seen as a kinematic motion, curvature is directly linked with the *acceleration* vector $c''(t)$. For a geometric curve, we can introduce it by looking at the variation of the length of the curves drawn at a constant distance, called the *parallel curves* (recall the first variation equation 1.2 on page 4). If the absolute value of the infinitesimal change of length of these equidistant curves, close to a point $m = c(t)$, is the same as for a circle of radius r we say that the curve c has *radius of curvature* r at $m = c(t)$, and that its curvature is $K = 1/r$. The formula to compute curvature is

$$K(t) = \frac{1}{r(t)} = \frac{|\det(c'(t), c''(t))|}{\|c'(t)\|^3}$$

where t is any parameter, and boils down to

$$K(t) = \frac{1}{r(t)} = \|c''(t)\|$$

when t is a unit speed parameterization (i.e. arc length parameterization). Of course a circle of radius R has constant curvature equal to $1/R$. Another way to look at it is the following: the circle $C(t)$ which is defined as tangent to the curve at the point $c(t)$ and has radius $r(t)$ is the circle which has the most intimate contact with the curve (technically, the contact is third order, meaning that the curve and the circle have the same first three derivatives at that point). This circle is called the *osculating circle* to the curve at $m = c(t)$.

To rest a little, we mention an interesting fact rarely found in textbooks. When at a point m of the curve c , the curvature varies, that is to say $dK/dt(m) \neq 0$, then the osculating circle at m crosses the curve (guess where it is inside, and where outside). A baffling consequence is that the osculating circles of a curve whose curvature is never critical never intersect one another. See the picture in figure 1.15. This picture gives two counterexamples. First, consider the unit tangent vectors to this family of circles. We get a continuous vector field, alias a differential equation in the plane. Now at every point of the curve there are at least *two* different integral curves of that vector field: namely the curve and the circle. The moral is that uniqueness of solutions of ordinary differential equations does not follow only from continuity. Second, consider the envelope of this family of circles. In many books it

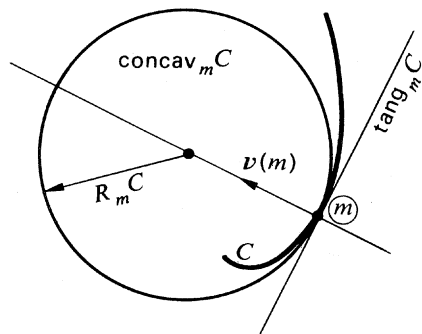


Fig. 1.15. Finding tangent line, concavity and curvature

is “proven” that the envelope is the limit of the so-called characteristic point, the point of intersection of close-by curves. This is wrong (unless we invoke “complex circles”).

There is another geometric definition of the curvature. Parameterize the curve by arc length, so that $\|c'(t)\| = 1$ identically. Then $c'(t)$ is a unit vector, running through the unit circle of the plane. The curvature is the speed of its rotation. More precisely, one needs first to orient the plane, and choose an oriented direction in the plane. Then one can define an angle $\alpha(t)$ for $c'(t)$ which is a real number, defined only up to integer multiples of 2π . The curvature is $K(t) = d\alpha/dt$.

To get rid of the 2π ambiguity, there is an important fact which we are going to use below. For regular curves, when t runs through the interval of definition, one can follow $\alpha(t)$ by continuity and then define a map $t \mapsto \alpha(t)$ into $\mathbb{R}$. Note that if the curve is closed with period T then $\alpha(t + T)$ will in general differ from $\alpha(t)$ by an integral multiple of 2π .

We saw above that straight lines are the shortest paths between two points. Let us look at this again, now in a more sophisticated way. We start with a curve c with ends $p = c(a)$ and $q = c(b)$. Suppose it is as short as any curve can be with these extremities. Consider any one parameter family of curves c_μ with the same ends, with $c = c_0$, and compute their lengths. The fact that the length should be a minimum implies in particular that the first derivative of the length as a function of the parameter μ has to vanish. Computation yields

$$\left. \frac{d \text{length}(c_\mu)}{d\mu} \right|_{\mu=0} = - \int_a^b \langle f(s), c''(s) \rangle ds \quad (1.3)$$

where the vector valued function $f(s)$ (which we can choose to be orthogonal to the curve everywhere) estimates the derivative of the normal displacement of the family of curves. This is also a first variation formula. Note that this formula is nothing but the integrated version of the definition we previously gave for curvature. A classical calculus trick (which is not obvious) proves that if we want all of these derivatives to vanish, namely the above integral

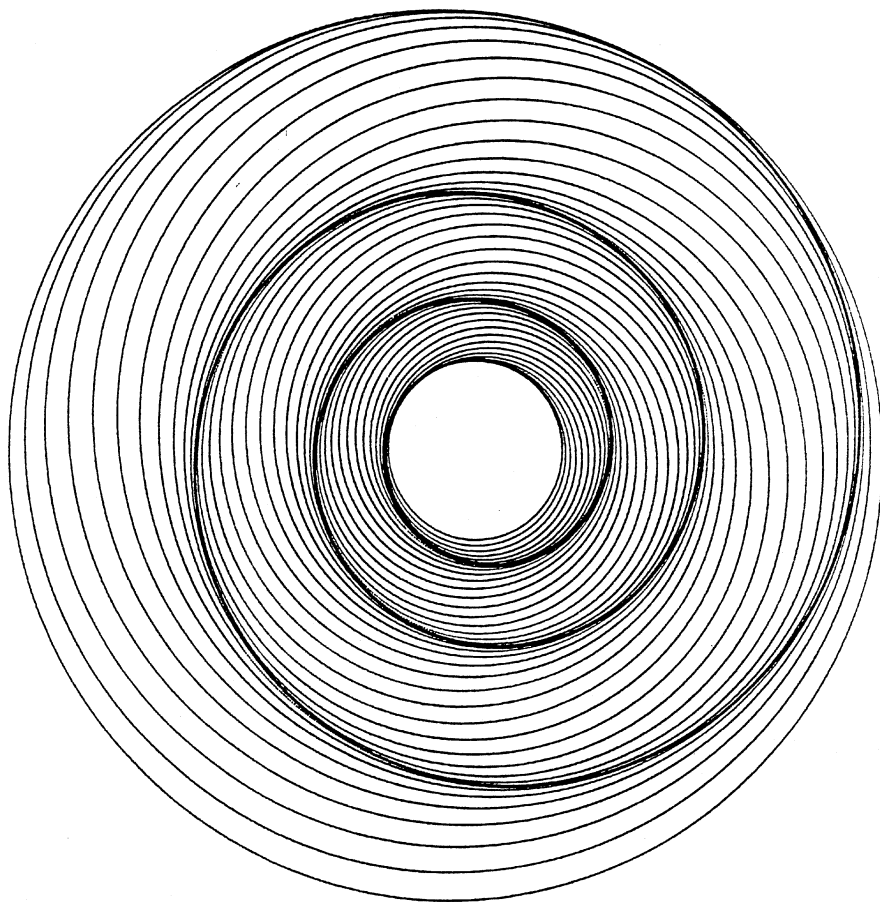


Fig. 1.16. Picture drawn for us by Etienne Ghys

to be 0 whatever the function f vanishing at $s = a$ and $s = b$, then $c''(t) = 0$ identically, and in particular our curve should have everywhere vanishing curvature. Curves with everywhere vanishing curvature are, of course, straight lines. Beware: what we have proven up to now by this method is that shortest paths can only be found among straight lines. Not every zero of the derivative of a function is a minimum. For a local minimum, a sufficient condition is to have positive second derivative. But we still have to search for an absolute minimum. The above proof is interesting because it is the simplest model of a field called *the calculus of variations*, and gives a general scheme to start searching for shortest paths. We will use it soon in §1.6 but principally in chapter 6. To conclude this story for the Euclidean plane, we know that if shortest paths exist, they should be straight lines. Then we check directly as above.

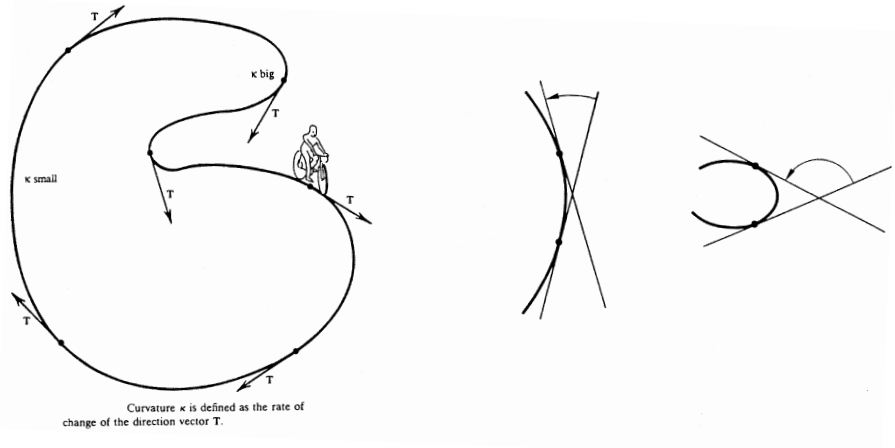


Fig. 1.17. Different definitions of curvature

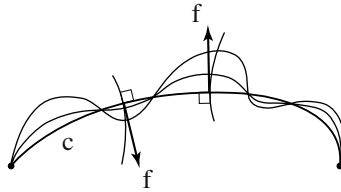


Fig. 1.18. Planar calculus of variations

There is a notion of *algebraic curvature* k as opposed to the scalar curvature K , linked by $K = |k|$. This is defined as soon as the curve is regular and oriented and the plane is also oriented, or (equivalently) if one side of the geometric curve is chosen, or (equivalently) if one of the two possible unit normal vectors is chosen. This side need not be the concavity side. Denote by $n(t)$ the unit normal vector you have chosen. Remember now that $c''(t)$ is an invariant. The algebraic curvature of c at t is the real number $k(t)$ so that

$$c''(t) = k(t)n(t)$$

A nice theorem about algebraic curvature:

Theorem 1 (More tightly curved curves) *If two curves c_1, c_2 are arc length parameterized, and have respective curvatures satisfying $k_1(t) \geq k_2(t)$ and start at the same initial point in the same direction, then the curve c_1 lies entirely inside the curve c_2 .*

(because it is “more curved” as intuition tells you). For a proof see Chern 1989 [366] and exercise 7, section 5-7 of do Carmo 1976 [451]. Beware that you need simple curves if you want a global result. If not, you just get a local one. The

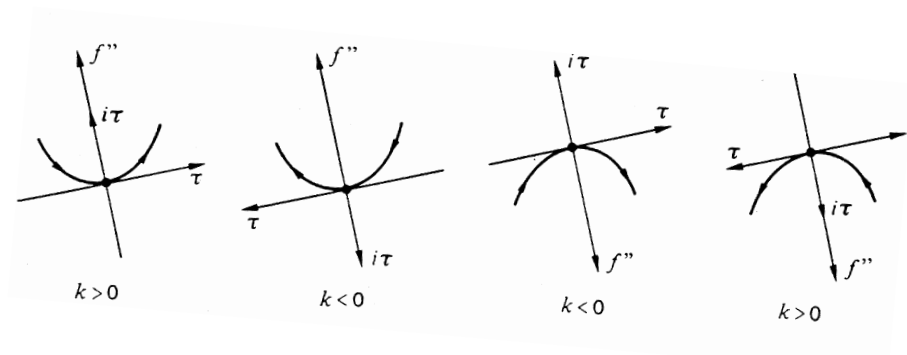


Fig. 1.19. Picturing algebraic curvatures

above theorem is one of the simplest ones in *control theory*. You will see many, many others later on. Note the following amusing consequences. Consider a simple closed plane curve, and the point where the curvature is maximum. Then the osculating circle at that point can keep rolling inside the curve all the way around. Similarly, the curve itself can roll inside its osculating circle of minimum curvature.

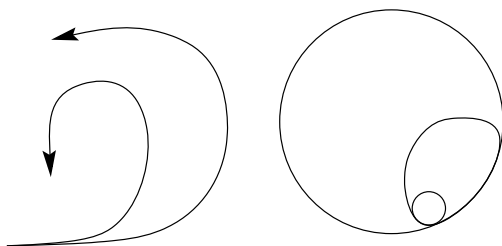


Fig. 1.20. (a) One is more curved than the other (b) The little circle can roll inside the closed curve, and the closed curve can roll inside the big circle

Let us make more precise our claim above that curvature is the only invariant of plane curves. More precisely the algebraic curvature is a characteristic invariant: if in the above inequality for curvatures we have equality ($k_1(t) = k_2(t)$ for every t) then the curves c_1 and c_2 are the same. Moral: if you know the value of the curvature as a function of the arc length then you know the curve completely. If the curves do not have the same initial point and direction, this just means that they are deduced one from the other by a Euclidean motion. The simplest example: a curve with non-zero constant curvature is part (or the whole) of a circle.

A spectacular application of this property of arcs of circle is the manufacture of a perfect straightedge. Take two pieces (of anything suitable for the job) and rub one against the other. The above theorem tells you that such a rubbing will eventually carve the two bounding curves into perfect circles. Now take three pieces, and rub them two by two. The boundaries will be circles, but the pieces will have to exhibit opposite convexity, i.e. equal algebraic curvature with opposite sign. For three, this is not possible unless the common curvature is zero. Naively, one might imagine that a laser can draw straight lines, but this is impractical. Today, lasers are used only to check the accuracy of the straightedge, after it is manufactured following the above recipe; once the laser has reported its evaluation of the straightedge, corrections are carried out by skilled hands using emery powder. Before the invention of the laser, accuracy was tested with interferences.

The same technique is still the only one available to make “perfect” planes: rub three surfaces against one another, and the result has to be a plane. To prove it, use the fact which we will quote on page 47 that pieces of a surface, all of whose points are umbilics, are pieces of spheres. To check accuracy, one doesn’t use a laser, but drags a straightedge around on the surface, or a comparison plane, or applies an interferometer.³ Rubbing only two pieces will yield pieces of a sphere, and is still the only method to sculpt spherical lenses.

In the same spirit, to fashion “perfect” balls, (which have thousands of practical applications, for example as ball bearings), one method was to put approximate balls inside a box, and stir them. Proof that the limit object is a sphere is not as elementary as proofs of the efficacy of the previous production methods, and was only recently established in Andrews 1999 [52]. Since 1907, balls for ball bearings have been produced by rolling rough balls between two plates with toroidal concentric grooves and random indentations. See Berger 2003 [173] or technical books for more details.

For the story of curves in $\mathbb{E}^3$ see Berger and Gostiaux 1988 [175], do Carmo 1976 [451], or Stoker 1989 [1160]. We will just mention briefly what happens there. This time there are two characteristic invariants, curvature and torsion, but the curves need to be biregular, i.e. the acceleration should never vanish. Things also extend similarly to any $\mathbb{E}^d$, this time with $d - 1$ invariants: see Spivak 1979 [1155]. But the first invariant is always the curvature and is defined simply as the norm $\|c''(t)\|$ for any arc-length parameterization. Only straight lines have everywhere zero curvature. For the kinematician this is the old fact that points with no force applied to them move along straight lines and at constant speed.

Take helices for example. They can be characterized as the curves with constant curvature and torsion. A practical application, in the spirit of the one just above: graduate a straightedge to make ruler; this is equivalent to

³ An *interferometer* is a device which uses the interference of two waves (radio, acoustic, or light waves will do) to make very precise distance measurements.

making a protractor, dividing a circle in equal parts, etc. It is clear that you can do it if you can produce a screw (a helix). This is even simpler than a straightedge. Just start with an approximately good pair of a bolt and a nut and rub one against the other. At the end we will get a perfect bolt, and a perfect nut, because the common boundary will be made up of a union of curves with necessarily constant metric invariants, hence easily seen to be helices.

1.4 Global Theory of Closed Plane Curves

1.4.1 “Obvious” Truths About Curves Which are Hard to Prove

Systematic references for what follows are Stoker 1989 [1160], do Carmo 1976 [451], Berger and Gostiaux 1988 [175], among others. See also the new events in Arnol’d 1994 [65] and other references in §§1.4.6.

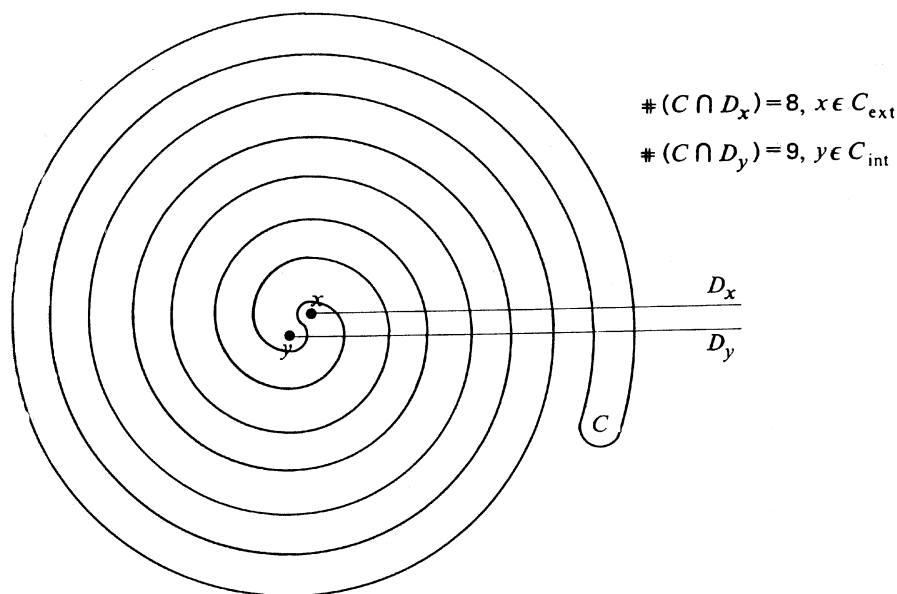


Fig. 1.21. A trick to find when a point belongs to the interior or to the exterior of a curve

Theorem 2 (Jordan curve theorem) *A simple closed plane curve has an interior and an exterior.*

This implies the more common statement that a simple plane closed curve separates the plane in two regions. More precisely the plane with the curve

deleted from it has two connected components and moreover one is bounded whilst the other is not. The bounded one is of course the one called the interior. An immediate corollary is that a simple closed plane curve has a given side, the *interior* one, and by the above this implies that its algebraic curvature is defined, independently of an orientation of the curve and an orientation of the plane. More generally, closed plane curves have a well defined algebraic curvature. Note again that the interior side need not be concave. From the viewpoint of algebraic topology the interior is homeomorphic to a disk and so it is simply connected.

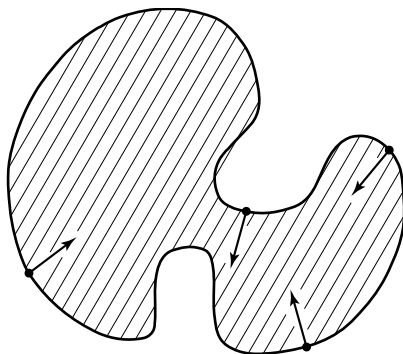


Fig. 1.22. Inner normals

This first result is valid even for curves which are not smooth but only continuous. The next result we will cite needs two derivatives and claims that any simple plane closed curve satisfies

$$\int k(s) ds = 2\pi . \quad (1.4)$$

The proof cannot avoid using the so-called Umlaufsatz (the turning tangent theorem) which asserts intuitively that the total rotation of the velocity (or of the tangent line to the curve) is equal to 2π . If you visualize this velocity vector as a point of the unit circle of $\mathbb{E}^2$ it means that when the parameter t runs through a complete period of definition of the curve the continuous determination $\alpha(t) \in \mathbb{R}$ which was considered above runs exactly from 0 to 2π . This does not mean that $\alpha(t)$ is monotonically increasing: it means that the unit circle is at the end algebraically covered once. Beware that the same point of the unit circle can be obtained many times but that, for example, if you come back a second time at a value previously passed you will have to come again a third time to “erase” it. To make all this mathematically precise one needs the notions of universal covering, simple connectivity, etc. The Umlaufsatz was essentially known to Riemann but a rigorous proof is pretty hard (try one if not convinced). Simplicity of the curve is necessary

here. For closed (periodic) curves the *turning number*, defined as the number of rotations of the tangent line, hence

$$\text{turning number} = \frac{1}{2\pi} \int k(s) ds$$

can be any integer including 0. See the pictures below as well as Berger & Gostiaux 1988 [175] for an elementary proof and interesting theorems about that number. Also see §§1.4.4.

A basic remark is in order: the turning tangent theorem is a very particular case of the Gauß–Bonnet theorem given in §§3.1.2 for general surfaces, when moreover the curve can admit corners. See §§1.4.4.

The Umlaufsatz has many other corollaries. The first is a surprising explicit tube formula. Consider a simple plane closed curve c and take the tube around it of radius ε . Beware that the boundary of this tube might not be smooth if ε is too large. But for ε small enough,

$$\text{area of this tube} = 4\pi\varepsilon \cdot \text{length}(c)$$

Please check this on some examples. The length of the inner boundary curve is

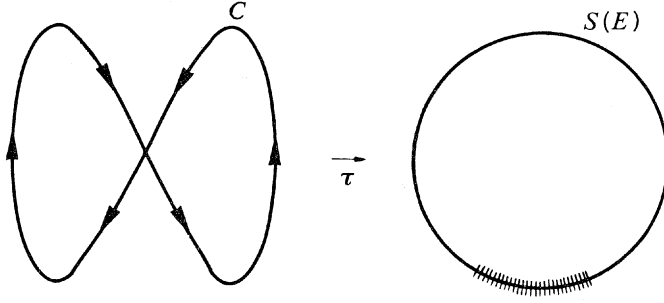
$$\text{length}(c) - 2\pi\varepsilon$$

An amusing consequence: consider the belt highway of a big town and a car running through it completely. Is it more economical to drive on the inside lane rather than the outside? Not much, because the above shows that the difference of elapsed distance is always $2\pi\varepsilon$ where ε is the distance between the two lanes (say something like less than 100 meters at the end), this being true whatever the shape and the total length of the belt.

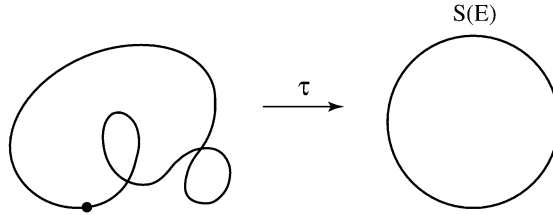
The second is a convexity result: suppose a simple closed plane curve satisfies $k(t) < 0$ for every t . Then its interior is a convex set. Try to prove it your own way before dismissing this result.

1.4.2 The Four Vertex Theorem

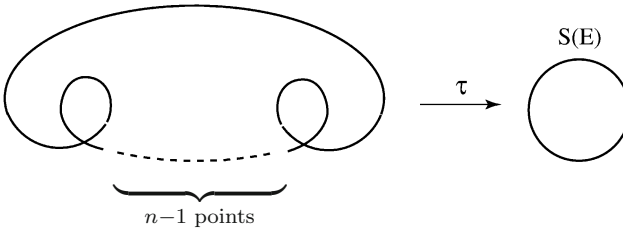
On a simple closed plane curve the curvature $k(t)$ is a continuous function (a periodic one). So it admits at least one maximum and one minimum. At such a point the derivative $k'(t)$ vanishes. Points where $k'(t) = 0$ are called *vertices* (think of ellipses if you are interested in the etymology). So any closed plane curve has at least two vertices. Now try to draw a curve with only two vertices: you will have a hard time (this is no reason not to try it seriously) because the four-vertex theorem asserts precisely that every closed plane curve has at least four vertices. In fact the proof of Osserman (see Osserman 1985 [983]) shows that generically it has at least six vertices. Ellipses have exactly four vertices. The converse of the four-vertex theorem was proven only recently in Gluck 1971 [569]: a function on a circle can be realized as the curvature



Turning number equal to 0



Turning number equal to 1



Turning number equal to n

Fig. 1.23. Turning numbers equal to 1, 0 and n

of a closed curve exactly if it admits at least two maxima separated by two minima.

Recently the four vertex theorem was applied to physics as follows. In a planar world with no gravitational forces, a plane convex body D stands in equilibrium between two liquids of different capillary constants and with a straight line common boundary. From the physics, at both of the two points of separation the angle θ between the tangent to the curve and the line of separation of the two liquids satisfies

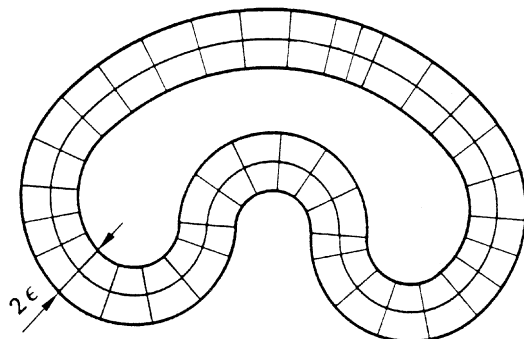


Fig. 1.24. The belt highway theorem

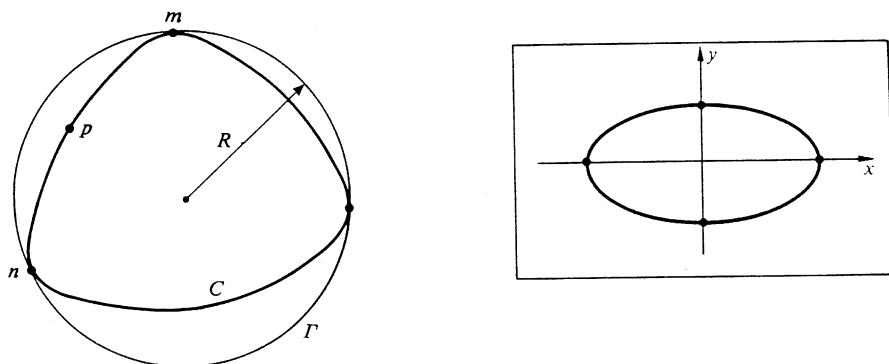


Fig. 1.25. A curve with six vertices

$$\cos \theta = \frac{\gamma_{13} - \gamma_{23}}{\gamma_{12}}$$

where γ_{ij} is the energy per unit area between phases i and j , the three phases here being 1 for the first liquid, 2 for the second liquid and 3 for the body. Using the four-vertex theorem and being quite tricky one can prove that there are at least four equilibrium positions, two of them moreover being stable ones (and two unstable): see Raphaël, di Meglio, Berger & Calabi 1992 [1048] or the latest on the story in Berger 1996 [168].

1.4.3 Convexity with Respect to Arc Length

Very recently an interesting local condition on a curve appeared in the literature, namely that the radius of curvature should be a concave function of the arc length:

$$\frac{d^2 r}{ds^2} \leq 0.$$

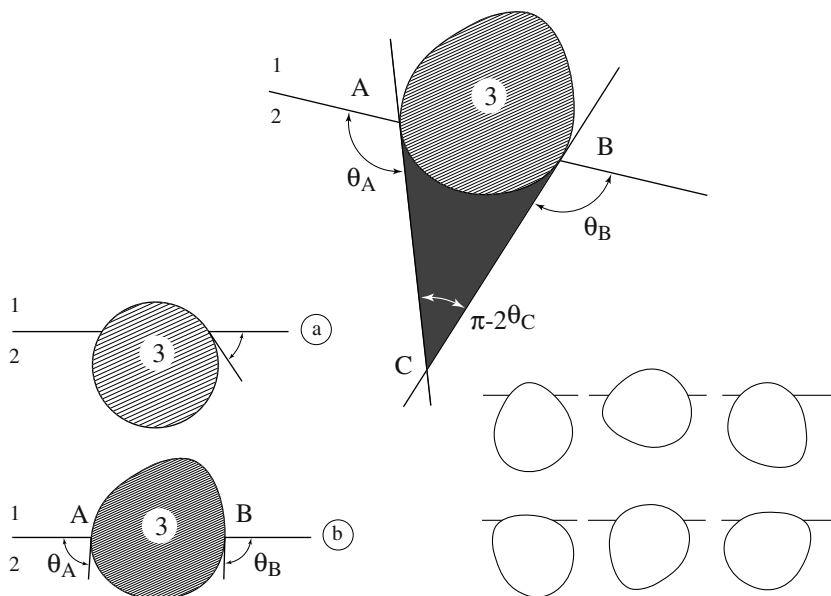


Fig. 1.26. (a) Angles of contact. (b) A spherical particle and convex particle at a fluid–fluid interface. (c) A convex particle with six equilibrium positions

As an exercise compute which parts of an ellipse obey it. This inequality allows one to build convex billiard tables (see §§1.2.3 above, and Wojtkowski 1986 [1274]) for which almost every nonperiodic trajectory is everywhere dense in space as well as in phase (they are called *ergodic*). The inequality controls geometric optics: it insures divergence after two reflections (see figure 1.27).

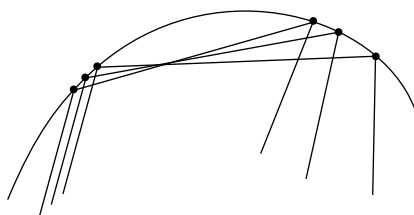


Fig. 1.27. Convexity with respect to arc length

1.4.4 Umlaufsatz with Corners

An important remark is in order, which might have already occurred to the reader. Consider a triangle in the plane and think of it “à la *Umlaufsatz*”, that is to say: drag the unit tangent vector along as you traverse each side. As a

vector it keeps being constant but when you arrive at a vertex of angle A you have to turn by exactly $\pi - A$. Coming back to the origin you have finally turned from

$$(\pi - A) + (\pi - B) + (\pi - C) = 3\pi - (A + B + C) = 3\pi - \pi = 2\pi$$

As you might guess there is a formula covering both cases, namely for curves with reasonable singularities. We leave to you to write it, because we will give a much more general one in lemma 18 on page 112.

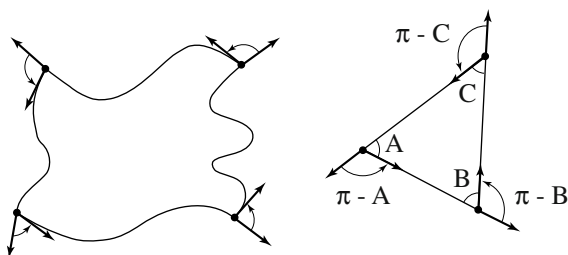


Fig. 1.28. Umlaufsatz with corners

1.4.5 Heat Shrinking of Plane Curves

Recently the global theory of plane curves became richer with results on the following evolution problem (sometimes also called the *heat shrinking curve equation*). This is one instance of the modern invasion of geometry by dynamical ideas, to be compared also with the invasion by iteration and combinatorics. Consider a simply closed plane curve and try to make it more like a circle by a systematic dynamical flow. The idea is to deform the curve along its normal direction at a speed proportional to the algebraic curvature. This is natural; at points where the curve has large curvature, one should reshape it more. This leads to a parabolic partial differential equation, which one can call a heat equation, because it turns out to be very similar to the standard heat equation as in §1.8. But it is very hard to prove that evolution is possible, first for a short time and then forever. Moreover one can prove that the curve, suitably normalized, converges to a circle as expected. A good reference is the survey of Linnér 1990 [871] and Gage and Hamilton 1986 [537]. Related problems were studied by Andrews 1996 [51] and 1999 [52]. A practical application: when one drops some liquid on a hot plate of metal, the drop evolves according to this heat equation; see Wang 2002 [1242].

1.4.6 Arnol'd's Revolution in Plane Curve Theory

Is the global theory of curves just beginning? We alluded above to the turning number of a plane closed curve. The Whitney–Grauertstein theorem asserts

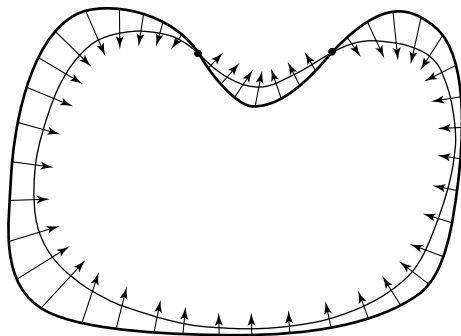


Fig. 1.29. A curve undergoing heat shrinking

that two immersed (not embedded in general) plane curves with the same turning number can be deformed one into the other through proper immersions. So one can say that the turning number is a characteristic invariant.⁴ Until lately, that theorem remained isolated, along with the four vertex one, as well as a mysterious formula of Fabricius–Bjerre 1962 [500] relating the number of double tangents to the numbers of inflexion points and crossings (see section 9.8 of Berger & Gostiaux 1988 [175]). Very recently, Vladimir I. Arnol'd started a revolution when studying plane curves, hammering out a general frame to encompass these results. This is a very active field of research today. There is a price to pay of course. It is quite reasonably expensive. The curve has to be considered together with its tangent lines, so that the object to study is the set of all oriented tangent directions to the Euclidean plane, an object of dimension three (not two) and inside it the curve consisting of the tangents of a given plane curve. The three dimensional space has the topology of the inside of a torus; an immersed circle in the plane will lift to a knotted closed curve inside that torus and then one will appeal both to the theory of knots and the field of contact geometry, intimately linked to symplectic geometry, both very active fields of research. Deformations of curves will then be interpreted as wave fronts in a geometrical optics language, following flows given inside that torus by a more or less canonical vector field. Some references: the initial bomb was Arnol'd 1994 [65], then Aicardi 1995 [13], Ferrand 1997 [513], and Chmutov & Duzhin 1997 [374]. One can find an informal presentation of these theories, along with a discussion of the open problems in the theory of real algebraic plane curves, in chapter V of the work in progress Berger 2003 [173].

⁴ For the historian, it was discovered by Pinkall (see Karcher & Pinkall 1997 [782]) that the Whitney–Grauert theorem appears in Boy 1903 [249] as a footnote. We will meet Boy's article again on page 136.

1.5 The Isoperimetric Inequality for Curves

The isoperimetric inequality for closed plane curves tells us that among all such curves bounding regions of fixed area, the circle, and only the circle, is the shortest. Or, among closed curves with fixed length, the circle encloses the greatest area. Explicitly, if A is the enclosed area, and L the perimeter,

$$L^2 \geq 4\pi A$$

with equality only for circles. This was known to the Greeks, but the history of the proof is fantastic. Let us just say that the first solid proof was written by Schwarz around 1875; references for this history are Osserman 1978 [982] and section 1.3 of the article by Talenti in Gruber and Wills 1993 [661]. One might also consult chapter 12 of Berger 1987 [164]. For the plane, and arbitrary dimensions as well, there is the excellent reference Burago & Zalgaller 1988 [283]. To our knowledge, it is the only source of not just one but all of the classical proofs of the isoperimetric inequality, handling the cases of equality with minimal regularity assumptions on the boundary.

Do not be surprised that we are going to spend quite a lot of time on isoperimetric inequalities. Generalizations of them to Riemannian manifolds are very important, and surprisingly recent. We believe that it is useful to explain the inequality in the plane because the general case is quite complicated.

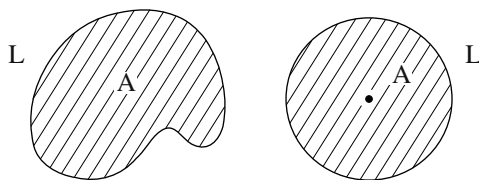


Fig. 1.30. Two planar domains

One of the most natural proofs should be one using Stokes' theorem. The desired inequality is a relation between the area, which is an integral over the interior of the region, and the perimeter: an integral over the boundary. Stokes' theorem is a basic equality between integrals over domains and other integrals over their boundaries. Astonishingly, it was not until 1957 that Knothe found a proof of the isoperimetric inequality using Stokes' theorem.⁵ This proof remained unnoticed for a long time, and was brought to light by Misha Gromov, who communicated it to the author in the 1980's. One can find it in Berger 1987 [164] and also in chapter VI of Sakai 1996 [1085].

⁵ Peter Petersen points out that Schmidt's proof in Schmidt 1939 [1105] uses Stokes' theorem, even if obliquely.

Since closed curves are periodic objects, an analyst or a physicist will naturally use harmonic analysis, i.e. Fourier series. Hurwitz established just such a proof in 1901 (see section 7.4 of Groemer’s contribution to Gruber & Wills 1993 [661]), and it is quite simple. Contrary to the Gromov–Knothe proof, it has the major drawback that, at least up to today, nobody has been able to extend it to higher dimensions using the natural extension of Fourier series, *spherical harmonics*. See §1.9 and Gruber & Wills 1993 [661] for more on spherical harmonics.

The other most natural proof should be as follows: pick up a curve which realizes the minimum of the ratio L^2/A and just prove that it is a circle. But in exchange it is the most expensive proof, because one needs very hard results from analysis to ensure first the existence of such a curve and second that such a minimizing curve is differentiable. But this is known now and more, in any dimension and also in general Riemannian manifolds (with the proviso that the minimizing object is differentiable only almost everywhere, but that is enough as we will see in §§§7.1.2.2). As for plane curves, let us consider our minimizing curve c . Follow the same scheme as above when we proved that straight lines minimize length among curves with fixed ends. Again we employ the (sophisticated but fruitful) notion of curvature. Here we have to express that the length is critical, among a family of curves deforming the given one, as long as all of the curves enclose the same area. The derivative is easily computed (using the first variation formula 1.3 on page 13) to show that the algebraic curvature satisfies:

$$\int_a^b k(t) f(t) dt = 0$$

for every function $f(t)$ such that

$$\int_a^b f(t) = 0 .$$

The classical Lagrange multiplier technique implies immediately that $k(t)$ has to be constant (a necessary, but perhaps not sufficient, condition). But we know that only circles have constant curvature (prove it directly). This comes from the above statement to the effect that curvature is a characterizing invariant. We will see in section §1.6 that this technique, with one trick more, extends to the isoperimetric inequality for surfaces in $\mathbb{E}^3$, and in §§§7.1.2.2 we will watch it extend to Riemannian manifolds with Ricci curvature bounded from below.

The reader will ask now what were the older, classical proofs. Proving that only circles can achieve equality turns out to be difficult in every demonstration of the inequality, which is otherwise simple (this for general dimension, see §§1.6.8). But for the plane case, there is a very tricky and completely elementary geometric “quadrilateral” argument due to Steiner which can be seen for example in the pictures below (see also Berger 1987 [164]).

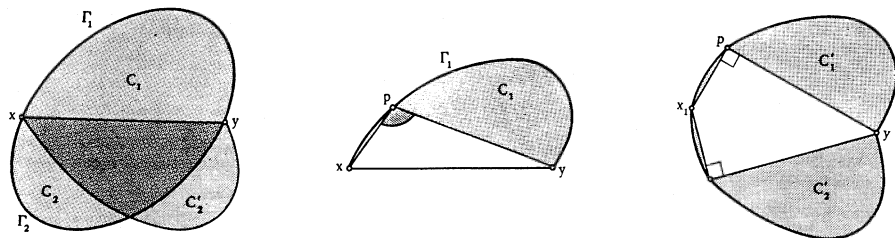


Fig. 1.31. Steiner's quadrilateral argument

Two proofs are classical and not too difficult: the one based on Steiner symmetrization and the one based on the Brunn–Minkowski theorem. We refer the reader to Berger 1987 [164] for both of them; see also the entry of Talenti in Gruber & Wills 1993 [661] and the very comprehensive reference Burago & Zalgaller 1988 [283]. Steiner symmetrization is too important, powerful and geometric an idea to be concealed. Briefly it goes like this. To every plane domain D and every straight line d we associate the symmetrization $d(D)$ of D with respect to d as depicted in figure 1.32. Slice D along every perpendicular line to d and then slide every slice in order that its middle sits on d . This yields by construction a domain which is symmetric with respect to d . One shows quite easily that the operation $D \mapsto d(D)$ keeps areas constant and can only lessen the perimeter. With some general topology one then concludes the inequality using the fact that disks are the only domains which are symmetric with respect to every direction.

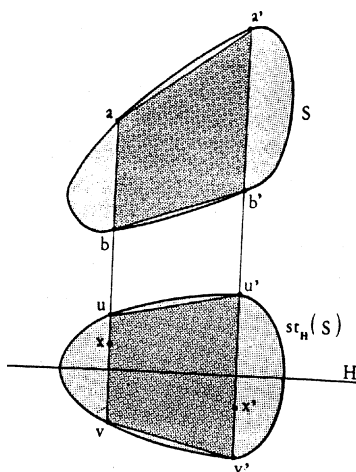


Fig. 1.32. Steiner symmetrization

Performing line symmetrizations with respect to suitably chosen directions is interesting to visualize: the convergence to a circle is very fast. We leave the interested and capable reader to see this on a computer.

Quite recently a very expensive proof has arisen, based on the evolution equation mentioned in §§1.4.5. Under this deformation the initial curve will look more and more like a circle; one discovers that, under the evolution, the isoperimetric ratio does not increase. See Linnér 1990 [871] for this. This proof is not really interesting when applied to a planar curve—its true purpose lies in the fact that it also works for Riemannian surfaces to yield the existence of simple periodic geodesics, a hard and expensive topic; see a lot about this in §10.4.

1.6 The Geometry of Surfaces Before and After Gauß

Unlike for curves, we will only define geometric surfaces in the space $\mathbb{E}^3$ (see why for yourself). Geometric curves are everywhere locally differentiable nice one-dimensional subsets in $\mathbb{E}^2$, while surfaces are the nice subsets of $\mathbb{E}^3$, nice in the sense that there is a local map from $\mathbb{E}^3$ to itself sending the local piece of the surface onto an open subset of a plane. That map must be differentiable, one to one and with a differentiable inverse. The jargon is: surfaces are two dimensional submanifolds of $\mathbb{E}^3$. Equivalently near any point, M can be written as $f^{-1}(0)$ where f is a smooth locally defined function $f : U \text{ open } \subset \mathbb{E}^3 \rightarrow \mathbb{R}$ whose derivative never vanishes. The fact that we do not introduce parameterized surfaces does not prevent our using local parameterizations of geometric surfaces, also called local charts. Moreover, unless explicitly stated, we require our surfaces to be connected. Alexandrov gave a very brief and accessible account of the theory of surfaces in chapter 7 of Aleksandrov, Kolmogorov & Lavrent'ev 1956 [15], translated into English in Aleksandrov, Kolmogorov & Lavrent'ev 1999 [16]; also see Morgan 1998 [936]. Monge gave a terse explanation of surface theory, translated to English in Struik 1969 [1165]. For an exposition of the submanifolds of $\mathbb{E}^d$ see Berger & Gostiaux 1988 [175].

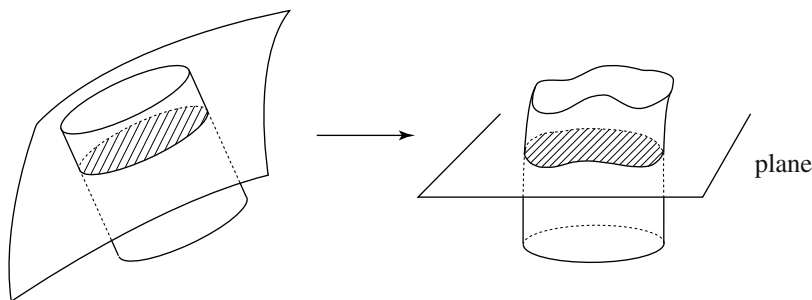


Fig. 1.33. Straightening a piece of a surface

1.6.1 Inner Geometry: a First Attempt

Following the scheme we used for plane curves we first study the *inner geometry* of our surface. The way that M sits in $\mathbb{E}^3$ will be studied next. Imagine people living on the surface M and ignoring things which happen outside of it. Those people walk, and they are interested in the distance needed to travel from one point p to another point q of M . Mathematically, we consider a curve drawn on M —it is forbidden to leave M and to go, even for one second, into the outside world. Because such a curve c lies in $\mathbb{E}^3$ it has a length, and of course the surface's inhabitants want to find the shortest path from p to q . We define the distance from p to q in M as the infimum of the lengths of all curves going from p to q in M , and denote it by $d_M(p, q)$ (or $d(p, q)$ if there is no need to be more precise):

$$d(p, q) = d_M(p, q) = \inf \{ \text{length}(c) \mid c \text{ is a curve in } M \text{ from } p \text{ to } q \} . \quad (1.5)$$

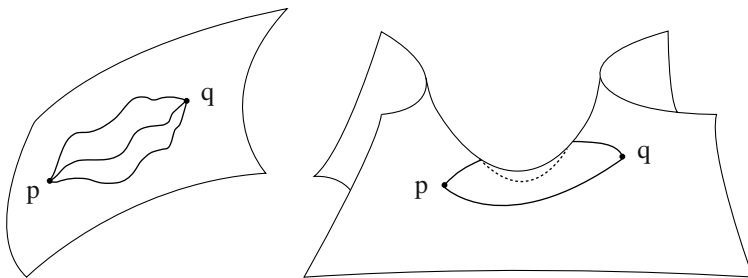


Fig. 1.34. Curves drawn on a surface

It is easy to see that $d = d_M$ is a metric on the set M (since M is connected). This metric is sometimes called the *inner* or *intrinsic* metric (see again §1.2.2). Beware that d_M is not what is called the *induced metric* on M from that of $\mathbb{E}^3$. The only surfaces M for which the induced metric and the inner metric coincide are portions of a plane. This inner metric is determined by the Euclidean structures on the collection of the tangent planes $T_m M$ when m runs through M , as we will see.

As a first example, consider the unit sphere $S^2 \subset \mathbb{E}^3$. How do we compute $d(p, q)$? We should find the shortest paths. Assume we know, or have made the right guess: the shortest paths on spheres are arcs of great circles (i.e. the intersections of the sphere by planes through its center). From this the distance is easily computed:

$$d(p, q) = \arccos \langle p, q \rangle \in [0, \pi].$$

To prove that the shortest paths are indeed great circular arcs, we take such an arc, less than half of a great circle, and arrange by rotation that it be

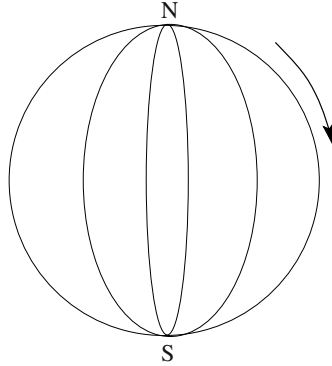


Fig. 1.35. Latitude grows at unit rate along longitude lines

a longitude line, and that it touches neither the north nor the south pole; see figure 1.35. The latitude (expressed in radians) increases at unit rate as we traverse the longitude line at unit speed, i.e. change in latitude is arc length. One now needs to see only that the rate of increase of latitude along any unit speed curve is slower than unit rate (i.e. slower than arc length) except where the curve is tangent to a longitude line (due south). We conclude that total arc length must be greater than difference in latitude, except along longitude lines. If we have a longitude line, and any other path with the same end points, then the other path has the same total change in latitude between its end points, and so must have greater length. This is an important idea because it is the basic scheme of the proof that geodesics locally minimize length on arbitrary surfaces; see equation 3.8 on page 116.

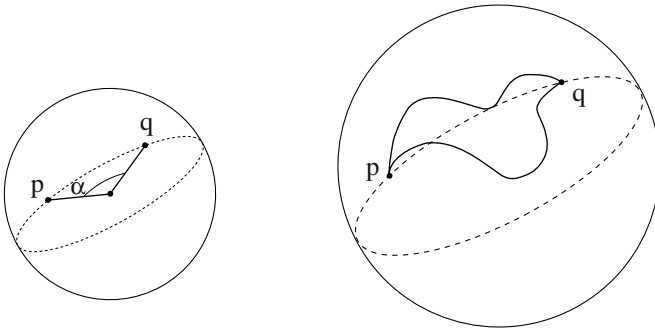


Fig. 1.36. Curves on spheres

The *principal formula of spherical trigonometry* provides the sphere an analogue of the Euclidean formula 1.1 on page 2:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A \quad (1.6)$$

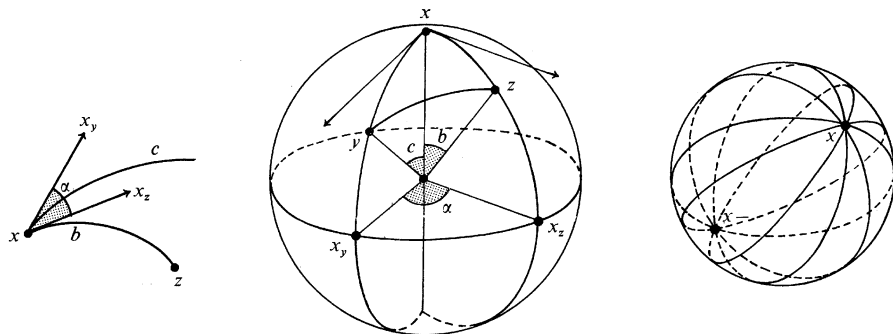


Fig. 1.37. Spherical trigonometry

for every spherical triangle $\{p, q, r\}$ with side lengths $a = d(q, r)$, $b = d(r, p)$, $c = d(p, q)$ and where A is the angle at the vertex p . It is a good exercise to extend this formula to a sphere of radius R , let R go to infinity and prove that the formula converges to the Euclidean one. Another exercise is to check that the sphere has the strict triangle inequality, which easily comes out of equation 1.6. Conclusion: shortest paths on the sphere are the arcs of great circles provided we keep them of length smaller than or equal to π . In particular, shortest paths between two given points are unique with the sole exception of *antipodal* points. The word “principal” is used because, as with equation 1.1 on page 2, all the other formulas for spherical triangles can be deduced from this one by pure trigonometrical computations.

Another consequence of the formula is that the sphere is never, even locally, isometric to a part of the Euclidean plane $\mathbb{E}^2$. From the inner geometry distance measurements, we can compute angles. But then the angle sums of a spherical triangle will be too large to be those of a planar triangle. Most surfaces are not even locally isometric to $\mathbb{E}^2$ (see §§3.1.1 for proof). The case of curves was radically different, remember §§1.3.1. This explains why cartography is a whole world in itself; if interested see chapter 18 of Berger 1987 [164].

Before leaving the sphere, recall an old formula whose importance for the sequel is not to be underestimated. The formula says that for a spherical triangle T with angles A, B, C its area is given by

$$\text{Area}(T) = A + B + C - \pi. \quad (1.7)$$

The real history of this formula seems to have come to light only very recently. It was discovered by Thomas Harriot (1560-1621) in 1603 and published (perhaps rediscovered) in 1629 by Albert Girard (1595-1632). See references on page 55 of Ratcliffe 1994 [1049], a fascinating and extremely informative book.

Note 1.6.1.1 (Space forms) Returning to formulas 1.6 on the previous page and 1.1 on page 2 we see that spheres share with Euclidean spaces the

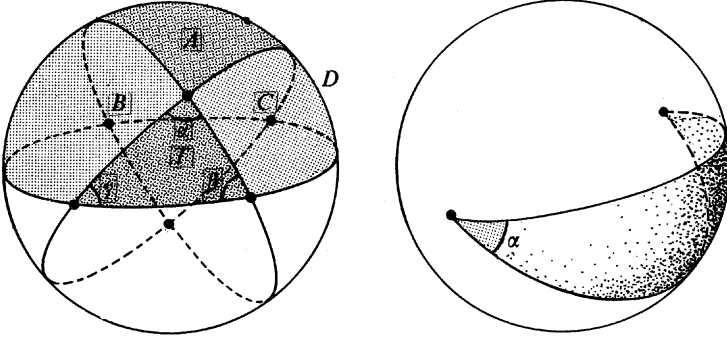


Fig. 1.38. Spherical areas and angle sums

basic philosophy of §§1.2.1. The tempting question to find all such spaces was in the minds of many mathematicians starting in the second half of the 19th century and thereafter. See §§4.3.2, note 4.3.2.3 on page 182 and §§6.3.2 for the continuation of this story. ♦

1.6.2 Looking for Shortest Curves: Geodesics

Let us talk to somebody who does not know how to find the shortest paths on the sphere. We want to apply the first variation technique of §§1.3.2 and find the formula which is the analogue of equation 1.3 on page 13. Because we can displace curves only under the proviso that the curves are traced entirely on our surface M , only the tangential part of a variation can be chosen freely. At a point $c(t)$ of the considered curve pick up a unit vector $n(t)$ tangent to the surface and at the same time normal to the curve. Note that (as for plane curves) there are two possibilities up to ± 1 . We parameterize c by its arc length s .

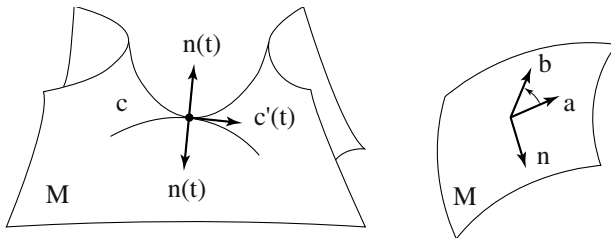


Fig. 1.39. M is oriented and $\{a, b, n\}$ is positively oriented

The first variation of the lengths of a one parameter family of curves c_α around $c = c_0$ is

$$\left. \frac{d \text{length}(c_\alpha)}{d\alpha} \right|_{\alpha=0} = - \int_a^b f(s) \langle c''(s), n(s) \rangle ds \quad (1.8)$$

where the variation of the curves is

$$\left. \frac{\partial c_\alpha}{\partial \alpha} \right|_{\alpha=0} = f(s)n(s)$$

We give a name to the important quantity $\langle c''(s), n(s) \rangle$, calling it the *geodesic curvature* of c . It will be denoted by $k_g = k_g c$. Geodesic curvature changes sign when a different choice is made for the unit normal vector. The *scalar geodesic curvature* is just its absolute value $|k_g|$. From equation 1.8 we see that

Lemma 3 *The geodesic curvature of a curve on a surface depends only on the inner metric of the surface, not on how the surface is embedded in $\mathbb{E}^3$.*

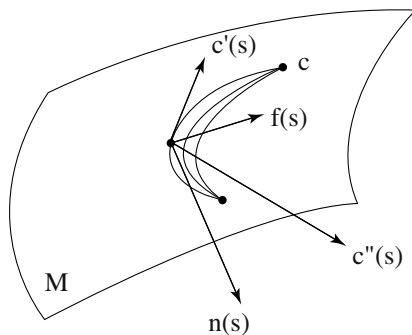


Fig. 1.40. Variation of a curve on a surface

As before we deduce from this formula that the curve c can be shortest only if its geodesic curvature vanishes identically: $k_g(t) = 0$ for every t . This amounts exactly to ask the curve c to have acceleration normal to the surface. This condition was known to Jacob Bernoulli I, who taught it to his student Euler in the beginning of the 18th century. It appeared for the first time in written form in Euler's work. Such curves are called *geodesics* of the surface. This justifies the name of geodesic curvature because k_g is a measure of how the curve under consideration differs from a geodesic. Unless otherwise stated, geodesics will always be parameterized by arc length.

In an intuitive sense, to follow a geodesic is just walking straight ahead in front of you. A more picturesque (approximate) way to get geodesics is to roll along the surface a small buggy made of two wheels of equal radii joined by a short axle, as in figure 1.41 on the facing page. We leave to the reader the proof of this fact. The kinematic interpretation of geodesics is

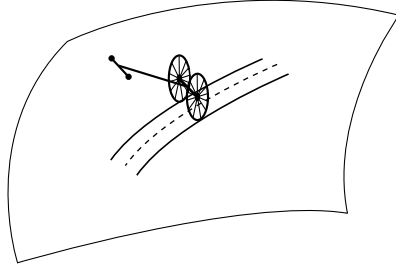


Fig. 1.41. A cart traverses geodesics

that they are the trajectories followed by a point moving on the surface with no force applied to it. In some sense you should feel no inner acceleration. Shortest paths are necessarily geodesics but geodesics are not in general the shortest paths between points, as we will see later. We will call a path a *segment* if it is the shortest path between its end points (some books call these *minimal geodesics*). Geodesics are segments only between points which are close enough to one another (think of the sphere). This comment is valid in any Riemannian manifold; see §§6.1.1. The distance between end points of a segment is exactly the length of the segment. To give the reader a flavor for the general Riemannian manifold, we study the theory of geodesics and of the shortest paths in a somewhat detailed way. This because there is not much difference in difficulty between surfaces and general Riemannian manifolds.

There are extremely few surfaces on which geodesics (and a fortiori shortest routes) can be more or less explicitly determined. But one can get a reasonably general statement first as follows. We want to find out if geodesics exist. For a general surface the simplest possible chart is the one given when one considers the surface (locally) as the graph of a function, and this is always possible (perhaps after relabeling the Cartesian coordinates). So we write M locally as the image of the map

$$(x, y) \mapsto (x, y, F(x, y)) \quad (1.9)$$

where (x, y) runs through some open set in $\mathbb{E}^2$.

Our curve will be c given by

$$t \mapsto (x(t), y(t), F(x(t), y(t)))$$

and we just write that its acceleration $c''(t)$ is normal to M at every point. Computation yields the equations

$$x''(t) + P \frac{(x')^2 R + 2x'y'S + (y')^2 T}{1 + P^2 + Q^2} = 0 \quad (1.10)$$

$$y''(t) + Q \frac{(x')^2 R + 2x'y'S + (y')^2 T}{1 + P^2 + Q^2} = 0 \quad (1.11)$$

where for simplicity we set

$$P = \frac{\partial F}{\partial x} \quad Q = \frac{\partial F}{\partial y}$$

$$R = \frac{\partial^2 F}{\partial x^2} \quad S = \frac{\partial^2 F}{\partial x \partial y} \quad T = \frac{\partial^2 F}{\partial y^2}$$

and general theorems about differential equations tell us immediately that geodesics exist with any given initial conditions, at least for small values of t , and are unique under these conditions. In particular there is one and only one geodesic parameterized by arc length, starting from a given point $m \in M$ and with given unit length velocity vector in the tangent plane to M at m .

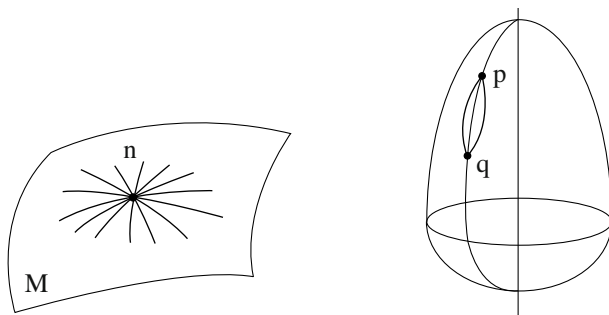


Fig. 1.42. (a) Geodesics through a point. (b) Meridians are geodesics by symmetry

From the above, general topology implies quite easily (using Gauß's lemma on page 116) that geodesics are shortest paths locally and that for any two sufficiently close points on M there is one and only one shortest path. But to find it explicitly is almost always impossible. Moreover globally uniqueness is false as already seen on the sphere. Let us now mention a few cases where geodesics are exceptionally easy to find.

The first case is when one has a plane of symmetry for the surface, because then the local uniqueness implies that the intersection with that symmetry plane is always a geodesic. Example: the great circles of the sphere are exactly its geodesics. For a surface of revolution every meridian will be a geodesic.

Surfaces of revolution are those obtained by rotating a curve around an axis. Geodesics on surfaces of revolution are a step easier to find because they satisfy a conservation law. The conservation law was discovered by Clairaut around 1730. We use two facts: the first is that the normal straight line to a surface of revolution always meets the axis of revolution. The second is that under projection to a plane orthogonal to the axis of revolution, the acceleration of the projection is the projection of the acceleration. This implies, in kinematic language, that the projected curve is a motion under a central force because the straight line containing the acceleration vector always goes through the origin of the plane. Then it is classical that such a motion obeys

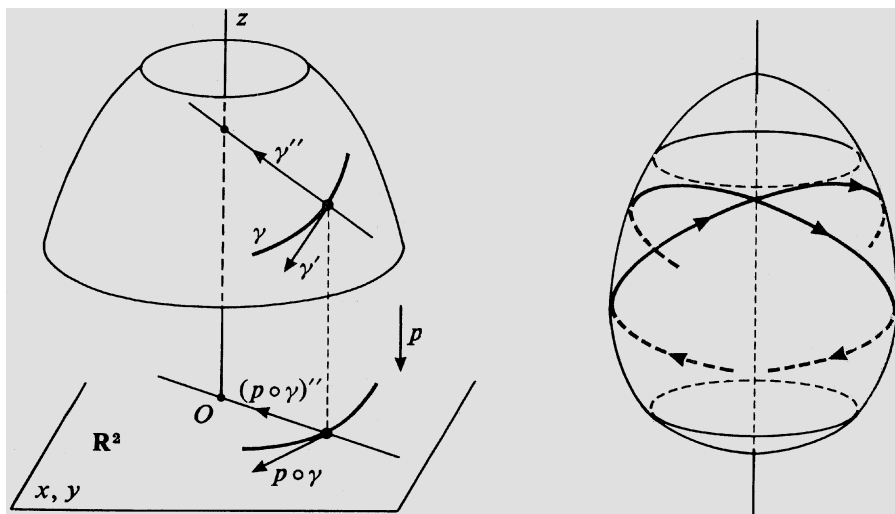


Fig. 1.43. Geodesics on surfaces of revolution

the law of area and from there the geodesic can be uncovered just by finding a primitive of a given function. For details and examples, see Berger & Gostiaux 1988 [175], Klingenberg 1995 [816] or do Carmo 1976 [451] and also §§10.2.1.

For example, the global behavior of geodesics is very simple when the meridian is convex: geodesics move between two parallels of the surface, parallels which moreover are at the same distance from the axis. In particular the behavior of a geodesic when the time goes to infinity is completely known. One can also find the periodic ones. For a torus of revolution we suggest that you study the different kinds of geodesics, or see Bliss 1902 [208].

The second type of surface whose geodesics are easily determined is a quadric. Quadrics are the surfaces of $\mathbb{E}^3$ given as the points where a quadratic form vanishes. The simplest ones are the compact ones, i.e. the ellipsoids:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

When $a = b = c$ we have a sphere; when only two of the three are equal we are on an ellipsoid of revolution and the determination of geodesics is easy to carry through completely (but the geodesics are given by elliptic functions).⁶ But for a, b, c all distinct, it is much harder. Jacobi succeeded in 1839 in integrating the equations for geodesics of ellipsoids by a quite difficult trick; explicit expression requires hyperelliptic functions. Details in textbook form are in Klingenberg 1995 [816] along with a lot of information about ellipsoids, but note that some statements are imprecise.

⁶ Do not be afraid of elliptic functions. They are just the “classical functions” which come next after polynomials, rational fractions, exponentials and logarithms, trigonometric functions and their combinations.

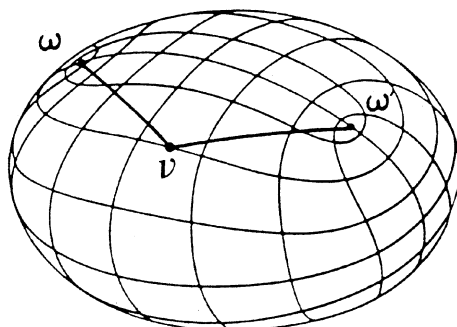
Jacobi's motivation to find ellipsoidal geodesics has some historical interest. He was asked to do this job by Weierstraß for the following reason. At that time, the earth was considered to be an ellipsoid of revolution, whose eccentricity was quite well measured. However, geographers found discrepancies and the conjecture was that the real shape is that of an ellipsoid with three distinct axis lengths. So there was enormous interest in finding the geodesics. The picture is now back to an ellipsoid of revolution, but with a little flattening around the poles.

The geometric version of Jacobi's result is too nice to be concealed. It is explained in Arnol'd 1996 [66] pp. 469–479. First we introduce the one parameter family of quadrics called *homofocal* to the above ellipsoid (or *confocal*). They are defined by the equation

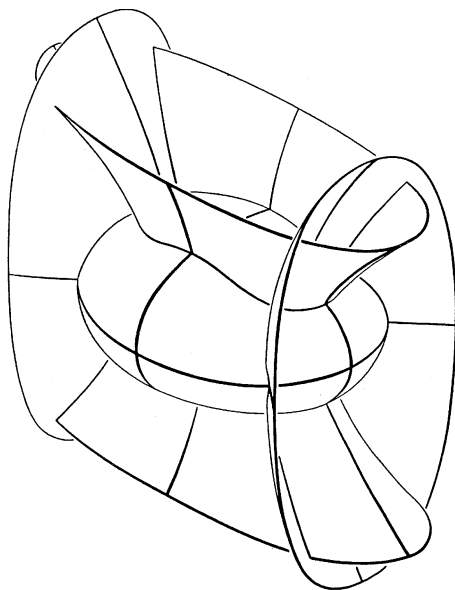
$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1$$

and are pictured in figure 1.44 on the facing page. A line will usually be tangent to exactly two quadrics of this family, at two different points of the line. Given any quadric of this family, we take the lines which are tangent to it at one point and to the original ellipsoid somewhere else. These lines form a geodesic on the ellipsoid, and all geodesics arise this way. As a consequence, a geodesic on the ellipsoid will either be periodic or everywhere dense in a region given by the corresponding quadric. Thus the global behavior of geodesics is completely understood. In particular, for example, there are infinitely many periodic geodesics, coming in one parameter families (bands). Note, for the geometer, that the intersection curves of the ellipsoid with the confocal quadrics are exactly the curvature lines of the ellipsoid: those lines are the integral curves of the two line fields given by the principal directions and play an important role in surface theory (see §§1.6.3). The family of confocal quadrics gives the ellipsoid nice coordinates which are basic for various studies, see Morse & Feshbach 1953 [945] or Courant & Hilbert 1989 [407] for example.

But the story of the shortest paths on ellipsoids is not finished, even when they are of revolution. Here is why. We continue to suppose that our axes have three distinct lengths. On our ellipsoid there are four points called umbilic points; they come in two pairs p, p' and q, q' . These four points are the intersection of the ellipsoid with the quadric of the confocal family which has degenerated into a hyperbola located in the $x = 0$ plane. Every geodesic from one umbilic goes after a time T to the other point of the pair. The time T does not depend on the initial direction because of the first variation formula for surfaces (see §§3.1.4). Caution: if you take a second lap for time T you will come back to your original point, but in general with a different direction, unlike on the sphere. This is of course completely proven. The complete picture of such a geodesic, infinite in both directions, is quite subtle: it will be used for an interesting application in §10.9.



(a)



(b)

Fig. 1.44. Homofocal quadrics

If p is not one of the four umbilics then associated to it is its *cut locus* (see §3.3 and §6.5 for the general definition of the cut locus and details). The cut locus of a point p is the closure of the set of points which can be joined to p by more than one segment. For nonumbilic points p of the ellipsoid, it was claimed by Braunmühl 1878 that the cut locus of a point p , as a subset of the ellipsoid, is homeomorphic to a compact interval (see Braunmühl 1882 [257]). The two extremities are joined to p by a unique shortest path, while points in the interior of this interval are joined to p by exactly two shortest paths. There is still no complete proof of this assertion, despite von Mangoldt

1881 [893]. Very recently, Sinclair & Tanaka 2002 [1142][1143] wrote software to study the cut loci of surfaces; moreover they proved the conjecture for the particular case of an ellipsoid of revolution for points sitting on the equator.

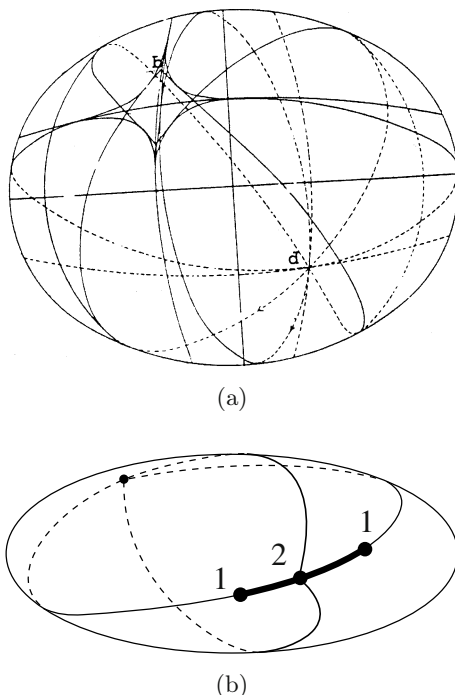


Fig. 1.45. Cut loci on the ellipsoid

Some clever readers will have been wishing to ask the author for some time why he did not address two natural questions. The first: can we extend geodesics indefinitely? The second: is there always at least one shortest path between any pair of points of a surface? Both answers are no in general; the simplest counterexample killing both questions is a plane with one point deleted.

The answers to these questions on general surfaces are neither easy to find nor to state. What is happening in the above example of a punctured plane is that a line running through this deleted point will arrive at it as to a boundary point. So we hope that things will be alright for surfaces with no boundary. Indeed they will, but for boundary in the sense of the inner metric of the surface. For example, if you draw in the plane a curve of infinite length, and then produce in three dimensional Euclidean space the “cylinder” on that curve (see figure 1.49 on page 42), i.e. the lines through the curve which are perpendicular to the plane, then on that surface every geodesic is indefinitely extendable and any two points can be joined by a unique geodesic

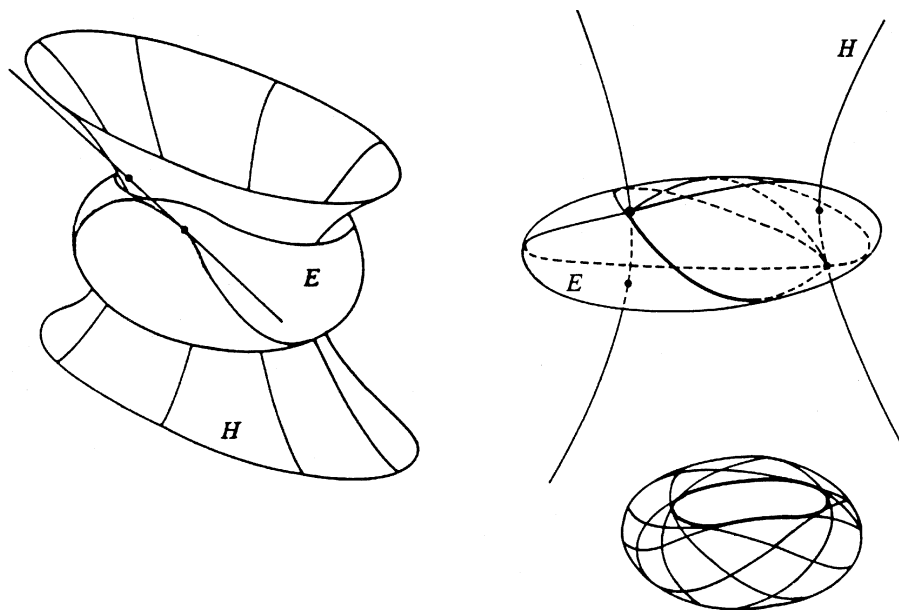


Fig. 1.46. Geodesics on the ellipsoid

which is moreover the shortest path between those points. However, in $\mathbb{E}^3$ this cylinder can have a boundary. The jargon answer is: the surface should be a complete metric space when endowed with the inner metric forced on it by its embedding in $\mathbb{E}^3$. The simplest case is when the surface is compact: then it is also complete in the above sense. We leave the subject here because we will study it intensively in the general case of Riemannian manifolds in §6.1.

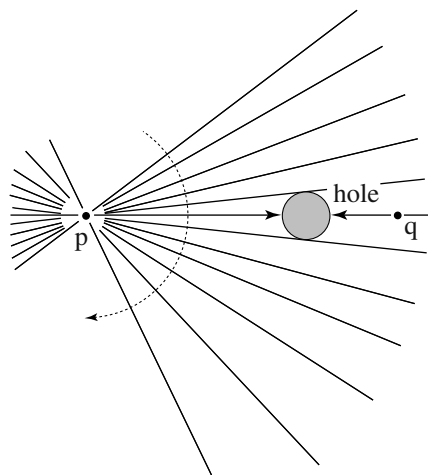
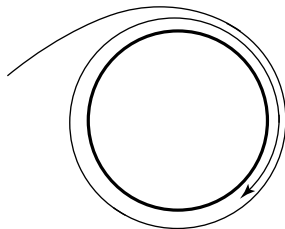
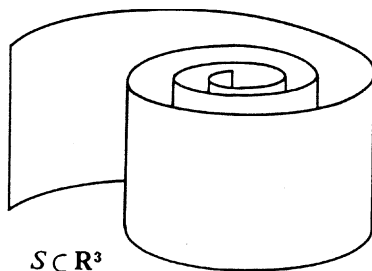


Fig. 1.47. There is no shortest path from p to q

**Fig. 1.48.** Curve asymptotic to a circle $S \subset \mathbb{R}^3$ **Fig. 1.49.** A complete surface, not closed

Another important question: do our geodesics provide the shortest passage? Assume the surface M is complete and pick up a geodesic c (parameterized by arc length) starting from a point $p = c(0) \in M$. We know that we can extend it indefinitely and that at first (for small t) it will be the shortest route between $c(0)$ and $c(t)$. But there is no reason that this will remain true for larger t ; think for example of any compact surface M . People like to control things, analysts like to have a priori inequalities. Here we would like to specify a positive real number I (attached to a given surface M of course) such that any geodesic of length smaller than I is a segment. As we will see later (in §6.5) for the general compact Riemannian manifold, the best possible I always exists (is positive) and has a name because of its basic importance: it is called the *injectivity radius* of M , written $\text{Inj}(M)$. For the unit sphere its value is π . Even for ellipsoids it is not easy to compute. Note that it can fail to exist (more precisely, it can be zero) for complete noncompact surfaces as shown in figure 1.50 on the facing page. Even for surfaces of revolution, cut loci and injectivity radius are impossible to find explicitly. A few exceptions are tori of revolution and Zoll's surfaces; see Besse 1978 [182] for Zoll's ones. For tori of revolution, the result is not found in textbooks. From what will be said in chapter 6 it can be left to the reader as a hard exercise, say already only for the case where the point is sitting in the outer meridian.

In §§1.2.3 we were interested in periodic trajectories of billiard balls, for example as a kinematic problem. Running along geodesics on a given surface is also a type of kinematics. Moreover it is a paragon for kinematics because many mechanical situations, as soon as they involve only two parameters,

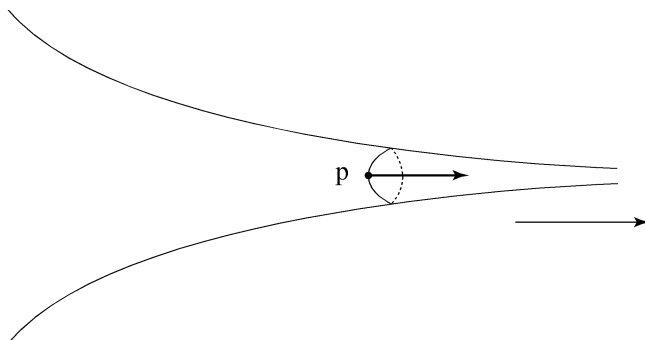


Fig. 1.50. $\text{Inj}(M) = 0$

have motions given exactly as the geodesics of a surface. Note, to be more careful, that the surface under consideration should most often be an abstract surface (not one embedded in $\mathbb{E}^3$) and that will be one of our important motivations to extend the notion of surface and of geometry to abstract Riemannian manifolds. In any event we now play billiards on a surface M in $\mathbb{E}^3$ with no boundary and so we look for periodic geodesics and ask ourselves the standard questions: are there many of them, or is there at least one? How many of them have length smaller than a given length L (this is called the *length counting function* and denoted by $CF(L)$).

We look first at the sphere and discover that all geodesics are periodic with the same period equal to 2π . Given a general surface, assuming moreover that it is strictly convex, even Poincaré could not prove there is at least one periodic geodesic on it. Birkhoff established it in 1917. Proof that there is an infinity for every surface had to wait until 1993; prior to that date it was not known even for strictly convex surfaces. We will come back to this more seriously in chapter 10.

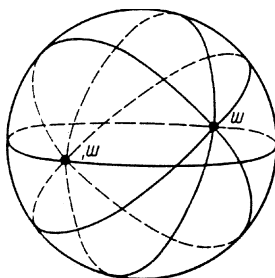


Fig. 1.51. Geodesics on the sphere

Be careful about infinity: we want an infinite number of *geometrically distinct* geodesics. As a kinematic motion, running twice along a periodic geodesic

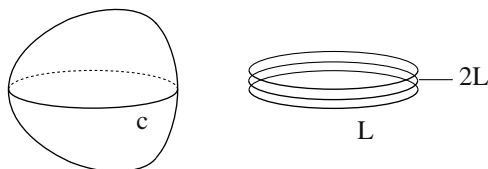


Fig. 1.52. Counting periodic geodesics

is different from running only once, but it is not geometrically distinct. For example when working on counting functions one should be careful to distinguish between the counting function for geometric periodic geodesics and that for parameterized ones. The question is difficult because the standard ways to prove existence of periodic geodesics consider them as motions. We will devote an entire chapter to this (and more) in the case of a general compact manifold, namely chapter 10.

We come back to the sphere. A typical mathematically minded question: are spheres the only surfaces all of whose geodesics are periodic? In kinematics this is the case for one of the simplest systems, the harmonic oscillator: one considers in Euclidean space a point which is attracted to the origin by a force positively proportional to the distance. Then every trajectory is an ellipse and thus periodic. So our question (at least as regards surfaces) asks for the harmonic oscillators of Riemannian geometry.

The question was taken up by various people around the end of the last century. They worked only with surfaces of revolution because their geodesics are easily computed as we saw above. The question was settled by Zoll 1903 [1310] when he proved that there are many surfaces of revolution all of whose geodesics are periodic and which are not spheres.⁷ These surfaces are not all convex and most remarkably some of them are even real analytic.⁸

This does not end the question. In fact mathematicians like to classify, if possible completely, objects with a given property. So we want to find all surfaces which are harmonic oscillators. Only partial results are known today. Moreover to have a clear view of the problem it is better to work with abstract Riemannian surfaces. We then refer the reader to §10.10 where we will see the current state of the art.

Let us come back again to the inner geometry of the sphere. Not only are all geodesics periodic, all of those starting from a given point pass after time π through the same point, its antipode. Are spheres the only surfaces having that property? This time the answer conforms to intuition: only spheres can do

⁷ Some people say *closed* instead of *periodic*, which can be ambiguous since it could also be used to mean only that a geodesic comes back to the same point, but not with the same direction; such a geodesic will not usually be periodic, and will be called a *geodesic loop*.

⁸ *Real analytic* functions are those equal to their convergent Taylor series, in an open set about each point. The definition of real analytic surfaces is analogous.

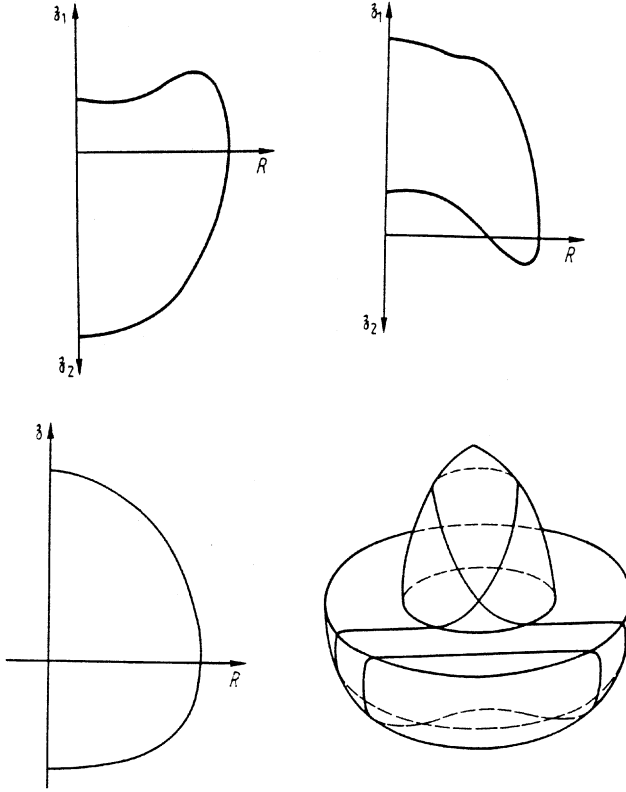


Fig. 1.53. Zoll's surfaces with concave parts

this. The question was raised by Blaschke in the 1930's but solved only in 1961 by Leon Green, see Green 1963 [586]. For all the above questions the standard reference is Besse 1978 [182], but it is now outdated, so see §10.10. We will just prepare the reader for the first variation formula for general Riemannian manifolds to be uncovered in §§6.1.1: this formula shows immediately that in an antipodal manifold the running time from one point to its antipode has to be a constant (for example, π for unit spheres).

We stop here because we need more powerful tools to study the inner metric, namely those invented by Gauß. They will provide the content of chapter 3.

1.6.3 The Second Fundamental Form and Principal Curvatures

We now are interested in how a surface M sits in $\mathbb{E}^3$. We will generalize the notion of curvature for plane curves seen in §§1.3.2, and erect an object called the *second fundamental form*, which will tell us how much the surface differs from its tangent plane.

We will look at the curvature of various curves sitting in M , parameterized by arc length. The first result on these curvatures is Meusnier's theorem which goes back to the end of the 18th century. To a curve c , to be studied at the point $m = c(t)$, we attach the curve d which is the cross section of M by the plane containing both the tangent $c'(t)$ to c at m and the normal straight line to M at m . Orient M near m (i.e. pick a unit normal vector to M at m), and orient d to agree with the orientation of c at m where c and d are tangent. In general, the acceleration vector $c''(t)$ is not normal to M at m , but makes an angle $\alpha \in [0, \pi/2]$ with it. Meusnier's theorem relates the two geodesic curvatures $k_c(t)$ and $k_d(t)$:

$$k_c(t) = k_d(t) \arccos \alpha.$$

Note those are essentially the lengths of the accelerations, not the radii of curvature, explaining why things are not too intuitive. But for osculating circles this relation is the one we expect: the osculating circle of c is the projection of the osculating circle of d on the plane determined by $c'(t)$ and $c''(t)$. It helps to draw curves on a sphere (see figure 1.54).

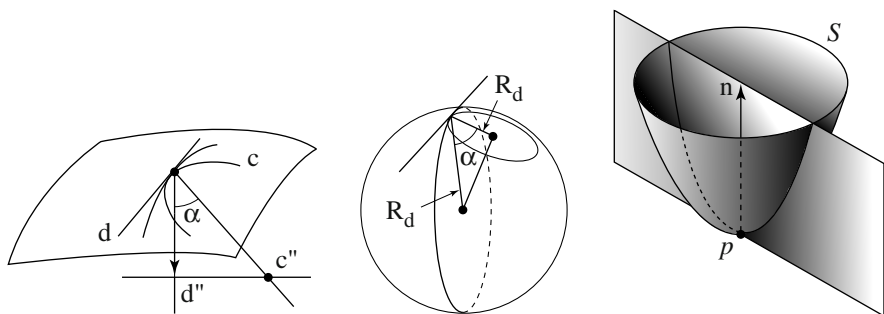


Fig. 1.54. Meusnier's theorem

Conclusion: to know the curvature of any curve, we need only know the curvatures of the normal cross sections of M .

The next theorem organizes these curvatures algebraically. The curvature of a normal section d at $m \in M$ depends obviously only on the unit tangent vector $u = d'(t)$. When u runs through the unit circle in the tangent plane $T_m M$ to M at m , the curvature k_u turns out to be a quadratic form in u , denoted by II_m and called the *second fundamental form* of M at m . With respect to the Euclidean norm on $T_m M$ this form has principal directions (which do not depend on the unit normal vector choice) and principal values for these principal directions. We denote them by k_1 and k_2 and call them the *principal curvatures* of M at m . Keep in mind our choice of an orientation for M : if we reorient M , then the sign of II_m flips, and so do the signs of k_1 and k_2 , although the principal directions are unchanged. We now have on $T_m M$ two

quadratic forms: the first one, the Euclidean norm induced by the embedding of $T_m M$ in $\mathbb{E}^3$ (which then depends only, so to speak, on the inner geometry) and the second, denoted by II which depends on the outer geometry (i.e. on the way that M sits in $\mathbb{E}^3$). The second fundamental form measures the defect of M to be a plane.

It turns out that for most standard examples, the elements of the second fundamental form are quite easy to compute (contrary to the inner geometry, where we had trouble computing distances and geodesics). For a unit sphere, at every point $k_1 = k_2 = 1$ (if the unit normal vector is the inward pointing one) and so the principal directions are not defined. For any surface the points where $k_1 = k_2$ are called *umbilics*. This is not a contradiction with the same word used for ellipsoids in §§1.6.2 because in fact these points of ellipsoids are also umbilics in the present sense. It is not hard to see that a surface all of whose points are umbilics is a part (or the totality) of a sphere or plane. But there is a conjecture of Carathéodory (made in the early 1920's) still open:

Conjecture 4 (Carathéodory) *Every compact surface in $\mathbb{E}^3$ which is homeomorphic to a sphere has at least two umbilics.*

This was proven between 1940 and 1959 by different authors (Hamburger, Bol, Klotz 1959 [818]) for surfaces which are strictly convex and real analytic. The proof is very hard and it seems that rigor was varying. For the state of affairs today see Scherbel 1993 [1102] and Gutierrez & Sotomayor 1998 [671]. It is natural to apply Hopf's index formula (see equations 15.9 on page 709 and 15.12 on page 713)

$$\chi(M) = \sum_x \text{index}_x(\xi) \quad (1.12)$$

where ξ is a vector field on M with a finite number of singular points. If M is topologically the sphere S^2 then one has $\chi(M) = 2$. Now at nonumbilic points of M one has the two principal directions. By continuity and orientation one hopes to use them to define a vector field, with singularities only at umbilics. In fact one can first extend Hopf's result to fields of directions; then indices are no longer necessarily integers but in general only half integers (rational numbers with denominator 2): for example the four umbilics of an ellipsoid have index equal to $1/2$. This does not prove that there are at least two umbilics, because one umbilic might have an index equal precisely to 2. The essence of the result above consists in proving, when the surface is moreover real analytic, that the index of an umbilic cannot be larger than one. The proof is extremely involved.

For very recent contributions to the subject, see Rozoy 1990 [1073] and Smith & Xavier 1999 [1150].

More generally for surfaces of revolution, using a symmetry argument, one sees immediately that the principal directions at a point $m \in M$ are the tangents to the meridian and the parallel through m . Then one of the principal

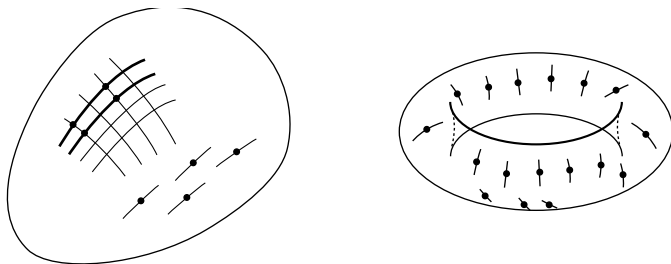


Fig. 1.55. (a) Impossible everywhere on a topological sphere. (b) Possible on a topological torus

curvatures is the curvature of the meridian, while the other is the inverse of the distance between m and the point of the axis of revolution where the normal to M at m meets this axis.

For a cone, a cylinder and more generally for the envelope of a one parameter family of planes things are as follows.⁹ Such surfaces contain through every point m a *generatrix* which is a portion of a straight line and also a geodesic. Along a generatrix the tangent plane does not turn, i.e. the unit normal vector is constant. The generatrix is always a principal direction and the associated principal curvature vanishes.

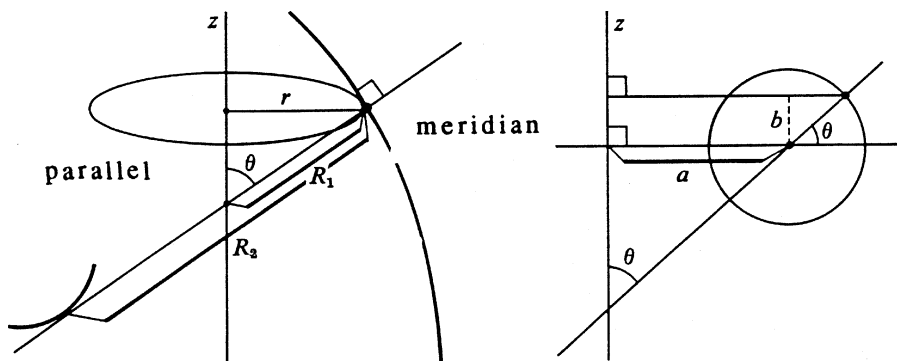


Fig. 1.56. The principal curvatures of a surface of revolution and the particular case of a torus of revolution

We leave the reader to compute the elements of the second fundamental form for ellipsoids. More generally principal curvatures are easy to compute either when M is given as the level set of a function or by some parameterization: see one case on page 52.

To every quadratic form on Euclidean space are attached two basic invariants: its trace and its determinant. It is thus most natural to attach to a

⁹ These surfaces are called *developable* for a reason to be seen below in §§1.6.7.

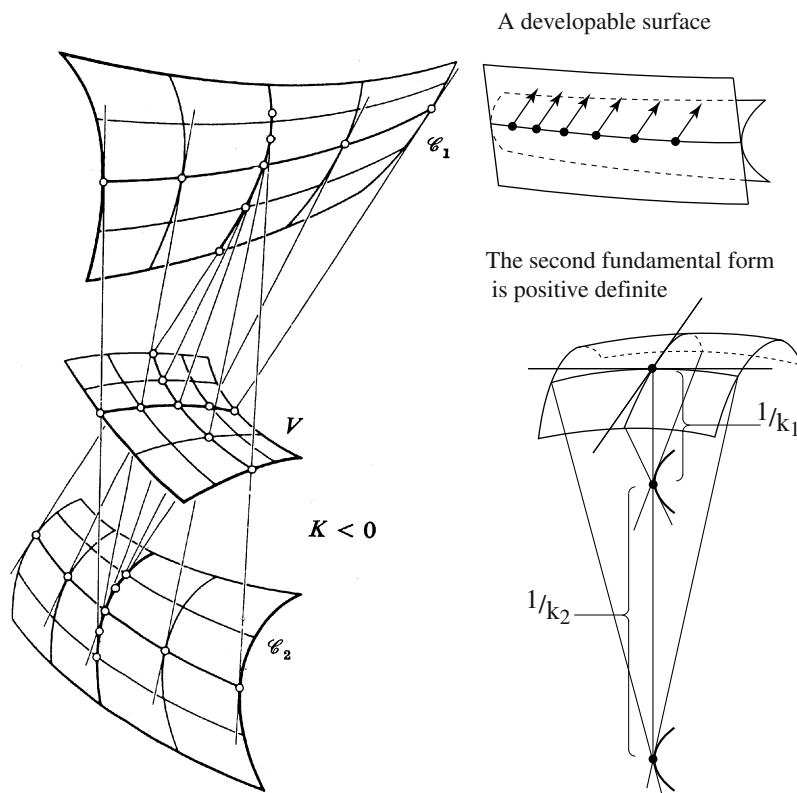


Fig. 1.57. The second fundamental form has opposite signs

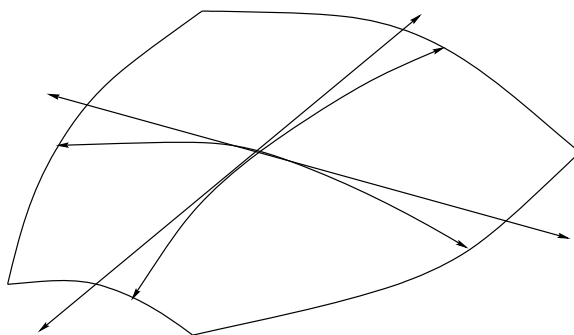


Fig. 1.58. The second fundamental form is positive definite

surface M the two following invariants:

$$K = k_1 k_2 \quad \text{the total or the Gauß curvature}$$

$$H = \frac{k_1 + k_2}{2} \quad \text{the mean curvature}$$

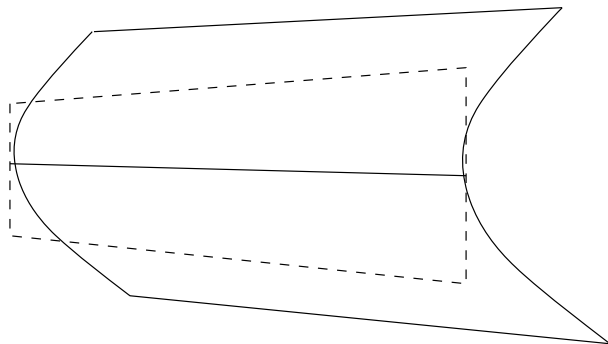


Fig. 1.59. A developable surface

Note that K does not depend on the choice of unit normal, while H changes sign if we change the unit normal. The reason for the name of Gauß will appear fully in the sequel. As a first example one sees from the above that K vanishes identically on developable surfaces. Obviously K and H are constant on spheres and planes. But there are many surfaces, different from spheres, with either H or K constant (not both). We will come back to this later on in §3.4. For the moment, we just mention the simplest examples of these. Surfaces of revolution with constant H or K can be explicitly determined, for the reason that (from what was said above about their principal curvatures) one can reduce the condition to an ordinary differential equation for their meridian curve. The shapes of some such surfaces are given in figure 1.60.

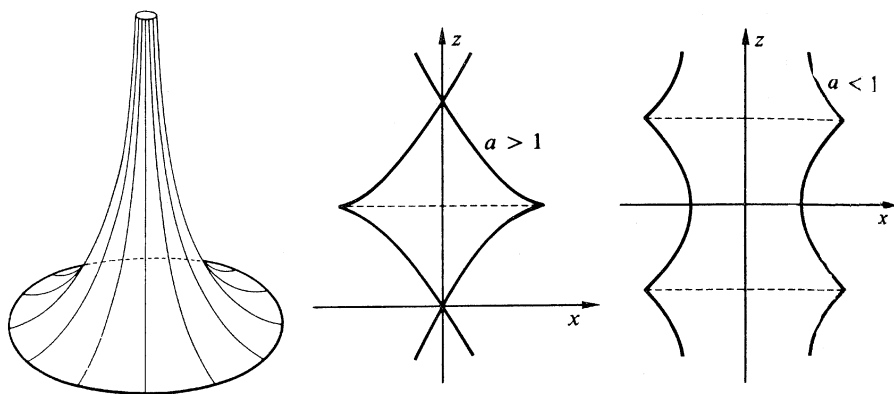


Fig. 1.60. (a) Beltrami's trumpet. (b), (c) Meridians of surfaces of revolution of constant negative Gauß curvature

Note that there are singularities (except for the standard sphere, and the standard cylinder). This is compulsory because there are theorems which say

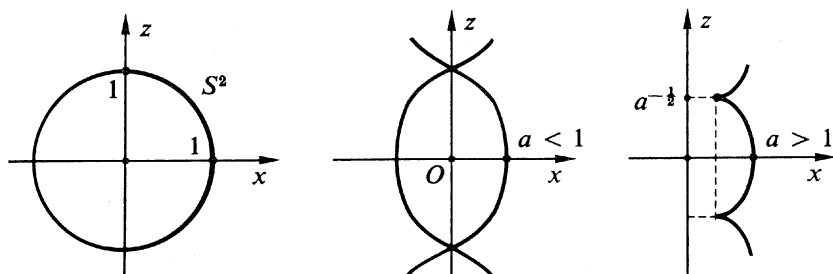


Fig. 1.61. Constant positive Gauß curvature surfaces of revolution

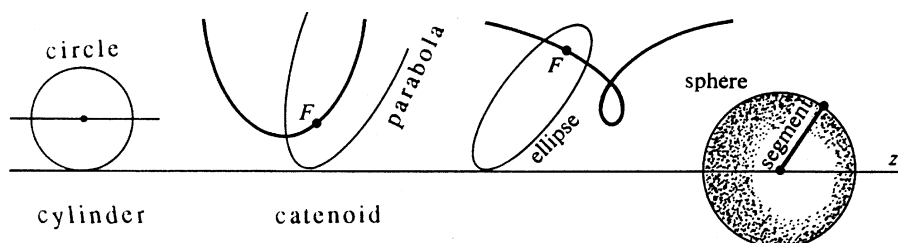


Fig. 1.62. Surfaces of revolution with constant mean curvature

that closed surfaces of either constant mean or constant Gauß curvature must be spheres; see §§3.4.1 and §§3.4.2. Note that these theorems are hard to prove. They were obtained only at the turn of the century. But the philosophy is that local pieces of surfaces of revolution are then easy to find. Now take a small cap of sphere or take some piece out of a tennis ball. Intuitively you might think it is easy to deform it in many ways keeping either K or H constant, because we have lot of freedom and are asking only for one condition. Rotational symmetry is exceptional. Indeed this flexibility with fixed mean or Gauß curvature turns out to be true for “general” surfaces, e.g. when K is not zero, but one needs extremely hard analysis to prove it. Moreover it is false for some surfaces which have exceptionally flat points. Indeed, some surfaces with such a strange flat point on them can not be bent without stretching (i.e. deformed through isometries other than rigid motions), even after cutting away all but an arbitrarily small neighborhood of that point.

For K a positive constant, it is natural to ask for closed surfaces, because positive K makes the surface shaped a little like an ellipsoid at each point. But how about $K \equiv -1$? It should be shaped more like a saddle. We know from §§1.6.4 that if it is complete for the inner metric, then such a surface cannot be compact. But is it possible to find an inner-complete surface in $\mathbb{E}^3$ with $K \equiv -1$? The answer is no, and this was proved by Hilbert in 1901. The proof is hard. Hilbert proved even more: the answer is still no even if you permit the surface to cross itself as in figure 1.63 on the following page. Such surfaces are called *immersions* as opposed to *embeddings*, see Berger &

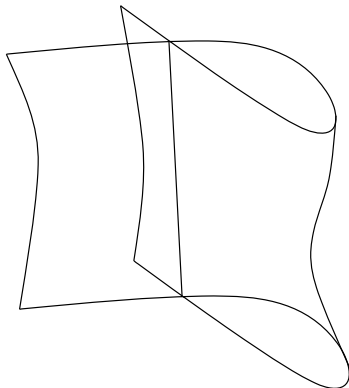


Fig. 1.63. An immersion

Gostiaux 1988 [175], do Carmo 1976 [451], Gallot, Hulin, & Lafontaine 1990 [542] for the jargon. Hilbert's theorem is of fundamental historical importance. It explains why hyperbolic geometry has to be defined abstractly, and can never be obtained as the inner geometry of a surface in $\mathbb{E}^3$. The interested reader can find its proof in section 5.11 of do Carmo 1976 [451] or in chapter 5 of volume III of Spivak 1979 [1155]. A recent survey of the state of affairs for the various generalizations and conjectures connected with Hilbert's result is presented in chapter II of Burago & Zalgaller 1992 [284].

Note the following kinematic interpretation of the meridian of surfaces of revolution with H constant (see figure 1.62 on the previous page), called *Delaunay surfaces*. They are obtained by rolling a conic in a plane along one of its tangents and the desired curve is that described during that rolling by one of its foci. This is not only a graceful depiction; there are deeper things behind it in connection with harmonic maps (see §14.3 and V.3 of Eells & Rato 1993 [480]).

We mention that computing K and H is easy even in the more general form

$$(u, v) \mapsto (x(u, v), y(u, v), z(u, v)).$$

When M is locally the graph $(x, y) \mapsto (x, y, F(x, y))$ of a function F , then K is

$$K = \frac{RT - S^2}{(1 + P^2 + Q^2)^2} \quad (1.13)$$

(with notation as in equation 1.10 on page 35).

1.6.4 The Meaning of the Sign of K

A first interpretation of K concerns the outer geometry. To say that $K(m) > 0$ is to say that the principal curvatures are not zero and of the same sign.

You can deduce easily that the surface M is locally strictly convex at m , and therefore that locally all points of M stay on one side of the tangent plane $T_m M$ with only the point m being in $T_m M$. If $K(m) < 0$ the principal curvatures are non-zero and of different signs. Locally the surface is close to a hyperbolic paraboloid. This, for example follows from the Taylor expansion for M around m :

$$(x, y) \mapsto \left(x, y, \frac{1}{2}k_1(m)x^2 + \frac{1}{2}k_2(m)y^2 + o(x^2 + y^2) \right) \quad (1.14)$$

in a chart where $m = (0, 0, 0)$, where $T_m M$ is the plane $z = 0$ and the x and y axes are the principal directions. A similar expansion clarifies the positive case.

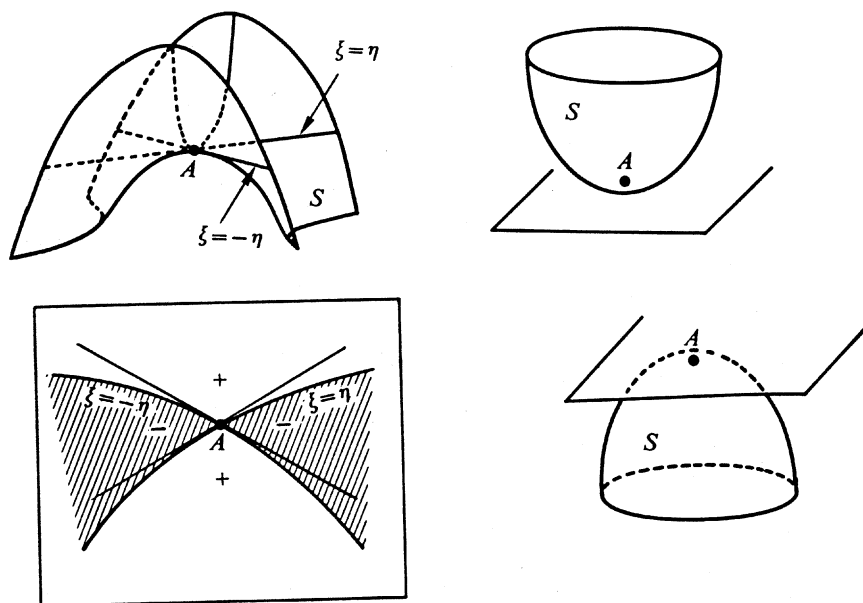


Fig. 1.64. (a) $K < 0$ (b) $K > 0$

It is important to remark that if $K(m) = 0$ nothing can be said about the shape of M locally at m without further assumptions. In some sense, every shape is possible. Consider some of the possibilities presented in figure 1.65 on the next page.

The second meaning will be treated in the most conceptually appropriate manner later on, but we can now give just the first zest of it. Consider a surface having the property that $K \geq 0$ everywhere. In particular, it is locally convex. Thinking, for example, of the spherical case, we might guess that if we have a triangle with sides a, b, c and an angle A , the basic formula 1.1 on

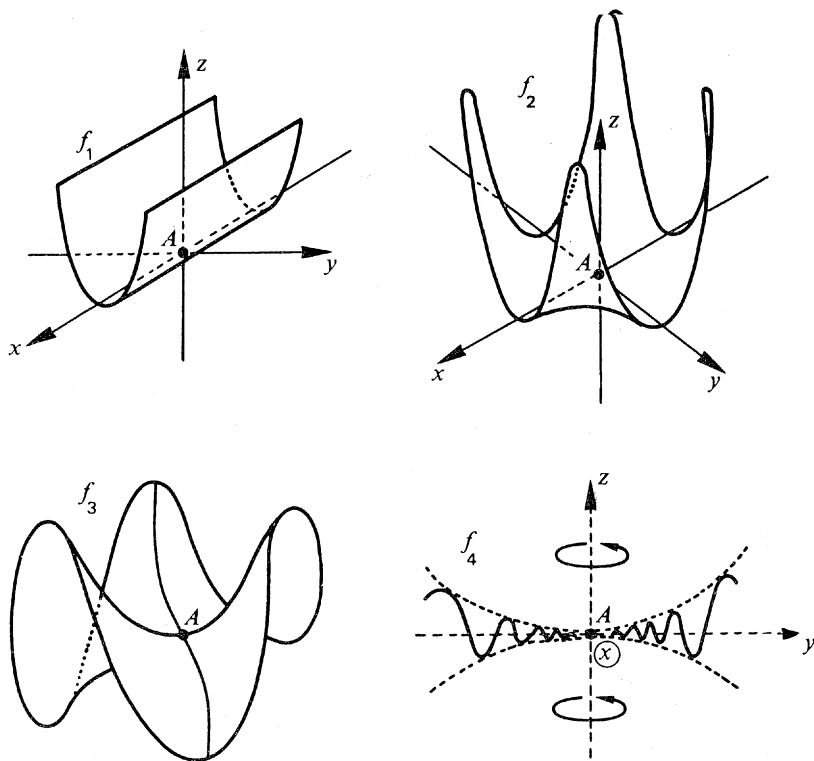


Fig. 1.65. Flat points

page 2 is bent into an inequality:

$$a^2 \leq b^2 + c^2 - 2bc \cos A \quad (1.15)$$

We will see in §3.2 that such an inequality is always valid under the sole assumption that $K \geq 0$ everywhere on M (provided, of course, that M is inner-complete).

How about $K \leq 0$? This time the surface is in some sense the opposite of convex¹⁰ in the sense that every point (at least when $K < 0$) is a saddle point. The reader can guess that at saddle points geodesics diverge, and possibly confirm her impression on some explicit examples. We will see in §3.2 that this intuition is correct, namely that for triangles on surfaces enjoying $K \leq 0$ everywhere one always has

$$a^2 \geq b^2 + c^2 - 2bc \cos A. \quad (1.16)$$

But beware that in the negative case, unlike the positive one, one has to restrict to small triangles, more precisely to triangles which can be filled up

¹⁰ Also the opposite of concave, since concave versus convex is just a change of orientation, which doesn't affect K .

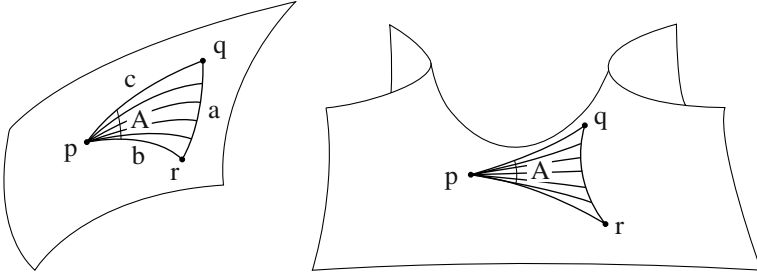


Fig. 1.66. Triangles on surfaces with positive and negative Gauß curvature

by a nice piece of the surface. Figure 1.67 shows a counterexample, the angle at the vertex being equal to π and the distance between q and r being as small as we like, for large b and c . This counterexample is linked with the notion of injectivity radius introduced on page 42. For the proof of these two theorems see §3.2.

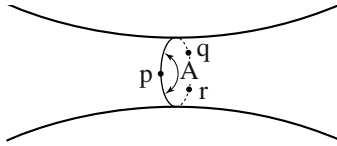
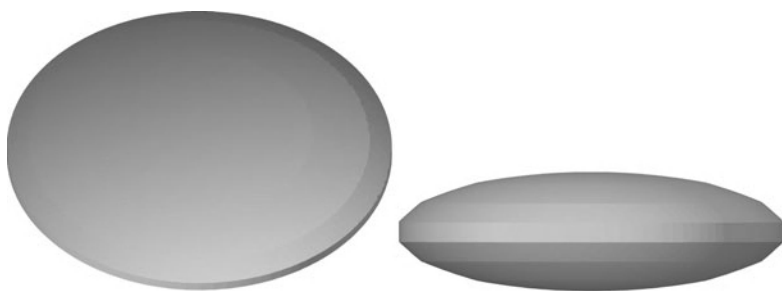


Fig. 1.67. Angle $A = \pi$

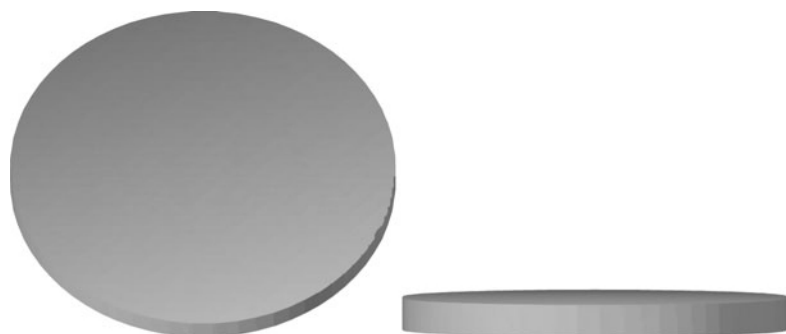
1.6.5 Global Surface Geometry

Let us mention some global results on surfaces, to give the reader an inkling of the subject. But only a few, much less comparatively than we did for plane curves in §1.4. There are two reasons: first, surfaces are more complicated objects than curves. Secondly, surfaces $M \subset \mathbb{E}^3$ are not our aim, which is Riemannian geometry. Surfaces are just used here as an introduction.

One open problem is the two umbilics conjecture mentioned above. Another open problem is *Alexandrov's conjecture*. Consider all compact surfaces with $K \geq 0$ and look for a kind of isoperimetric inequality, better an *isodiametric inequality*, specifying the maximal area of a surface with given inner diameter. The diameter of a metric space is the supremum of the distances between any two of its points. The inner diameter of a surface is its diameter for its inner metric. Check that the sphere is certainly not the best (just bend it). The conjecture is that the best bound is for the (flat) double disk for which, if the diameter is equal to D , the area is equal to $\pi D^2/2$. Note: this a bound which is not realizable by a smooth surface, but probably only by an object which is a limit of smooth surfaces with better and better ratios for Area/D^2 , as in figure 1.68 on the next page. See partial results in Croke 1988 [417].



a Very thin disk



b Double flat disk

Fig. 1.68. The limit of a flattening disk

A simple theorem:

Theorem 5 *On any compact surface there is at least one point m with $K(m) > 0$.*

You will enjoy finding the proof for yourself. Another result (which to our knowledge appears in no modern textbook and was told us by Gromov) extends the rolling theorem 1 on page 15.

Theorem 6 (Rolling surface theorem) *Take a compact surface M and let $\sup k$ be the largest of the absolute values of its principal curvatures. Then a sphere of radius $\sup k$ can roll everywhere inside M . And the analogous result for the infimum value of the principal curvatures. Consequence: every closed surface can be squeezed between two spheres, one of radius $1/\sqrt{\inf k}$ and the other of radius $1/\sqrt{\sup k}$.*

We now explain Hadamard's theorem, which is the analogue for surfaces of the plane curve global convexity theorem we described in §1.4.

Theorem 7 (Hadamard, 1898) *Let M be a compact surface with positive Gauß curvature everywhere. Then M is the boundary of a convex body and consequently as a topological surface is a sphere.*

Proof. Remember the proof in the case of the plane. What is needed is a result ensuring here that there is only one point of M where the unit inner normal vector has any particular value (the existence of one being trivial). In the plane case this intermediate result was a consequence of the Umlaufsatz. Here things are simpler. The idea is to introduce the normal map, most often called the Gauß map. It is a map going from the surface M into the unit sphere $S^2 \subset \mathbb{E}^3$ obtained, as you have already guessed, by attaching to the point m the unit inner normal vector to M at m . The positivity of K (its convexity if you prefer) ensures that this map is a covering map. Now algebraic topology says that a sphere cannot have a covering map unless the map has a continuous inverse, because the sphere is simply connected (see §§4.1.3 for example).

Note 1.6.5.1 Various people discovered more or less recently that this map appeared explicitly before Gauß in the work of Rodrigues in 1815, so we propose to call it in this book the *Rodrigues–Gauß map*. Rodrigues used it to prove, among other things, that the integral of the curvature of an ellipsoid is equal to 4π (i.e. without any explicit computation). See references in §3.4.5.

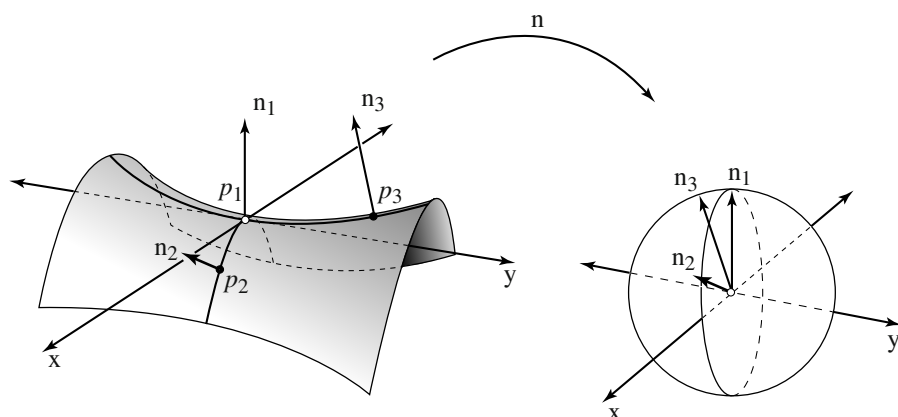


Fig. 1.69. The Rodrigues-Gauß map

Note 1.6.5.2 For closed (but not necessarily simple) curves in the plane, it is possible to have examples where the curve is everywhere locally strictly convex but is not the boundary of a convex domain of the plane. The turning number can be any integer as we saw on page 20. In $\mathbb{E}^3$ things are simpler: consider an immersion in $\mathbb{E}^3$ of an abstract surface M (for the jargon, if needed, any differential geometry book will do¹¹). It is clear how to define the Gauß curvature K of an immersed surface. Assume that $K > 0$ everywhere. Then this immersion has to be one-to-one, i.e. an embedding, and of course the image is the boundary of a convex body; in particular, topologically M is a sphere. This follows with no change from the argument above because the Gauß map is still defined and is still a covering of S^2 by S^2 . See more details in §§3.2.1. ♦

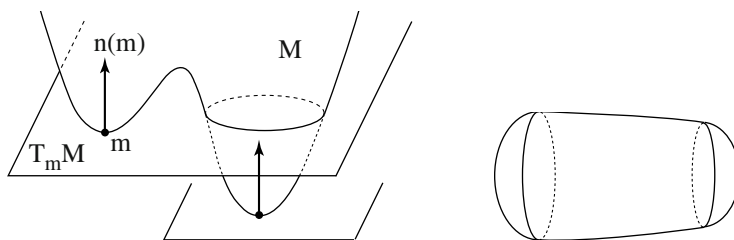


Fig. 1.70. (a) If M were not entirely on one side of $T_m M$. (b) Flat parts are permitted

Note 1.6.5.3 It seems intuitive that for a compact surface in $\mathbb{E}^3$ the condition $K \geq 0$ should be enough to insure that it is a topological sphere which is the boundary of a convex body: why not admit “flat” parts, or flat points, as in figure 1.70? You expect a sphere, but proof had to wait for Chern and Lashof in 1958. For references see Berger & Gostiaux 1988 [175] section 11. 3. 2, do Carmo 1976 [451] 5. 6-B and Burago & Zalgaller 1992 [284], chapter I, for generalizations. ♦

1.6.6 Minimal Surfaces

In this section and the next one we are going to address the natural question

Question 8 *What are the surfaces for which one of the two fundamental invariants K or H vanishes identically?*

We begin with the mean curvature $H = (k_1 + k_2)/2$. The mean curvature happens to have a very simple geometrical meaning: exactly the extension to

¹¹ Except, perhaps, this one.

surfaces of the curvature of plane curves when this curvature is seen as yielding the first variation for the length of curves (see equation 1.3 on page 13).

More precisely consider a surface M , a piece Ω of it and a one parameter family of neighboring surfaces, parameterized by a parameter t , and given by a normal variation function f . Then

Lemma 9 *The first derivative at M of the area of the piece of surface that Ω stretches into is*

$$\frac{d}{dt} \text{Area}(\Omega_t) = 2 \int_{\Omega} H(m) f(m) dm \quad (1.17)$$

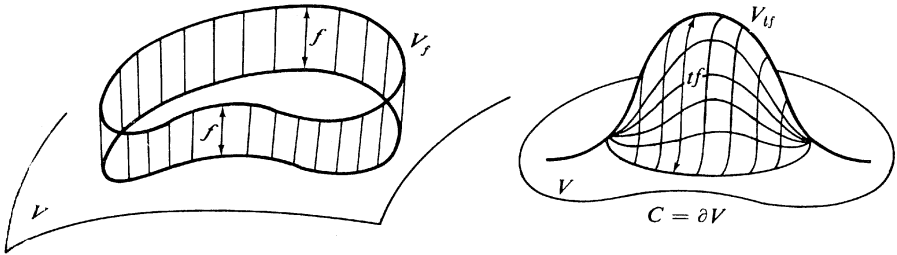


Fig. 1.71. Variation of a piece of a surface

We will use this first variation formula in §§1.6.8 and in §§3.4.2. It implies that the surfaces with $H \equiv 0$ everywhere are in some sense candidates for minimizing area. For this reason, surfaces with $H \equiv 0$ are called *minimal surfaces*. Consider the so-called *Plateau problem*: given a curve C in $\mathbb{E}^3$ find a surface S whose boundary is C and which has smallest area. This you can solve experimentally by dipping your curve, as realized by a piece of wire, in soapy water (these are not soap bubbles which correspond to surfaces for which H is not zero but only constant, see §§1.6.8). The study of minimal surfaces is a mathematical topic in itself which has been extremely active since the 1960's. Before then, there were only a few results, due to the lack of analysis tools. We refer to the two complementary bibles (and of course the references there): Dierkes, Hildebrandt, Küster, & Wohlrab 1992 [444, 445] and Nitsche 1989 [968], and to the excellent survey Meeks 1981 [911], and the recent Osserman 1997 [984]. Some minimal surfaces are drawn in figure 1.72 on the next page. Minimal surfaces can sometimes be found by an evolution flow; see Chopp 1993 [376] and Wang 2002 [1242].

Examples of minimal surfaces with no boundary condition are easy to build up; $H \equiv 0$ is a weak condition in that case. To get examples systematically there is a trick linking minimal surfaces with complex functions of a one variable: see section 10.2.3.6 of Berger & Gostiaux 1988 [175].

Minimal surfaces of revolution are straightforward to produce and analyze, because we saw in §§1.6.3 how to find principal curvatures of surfaces of

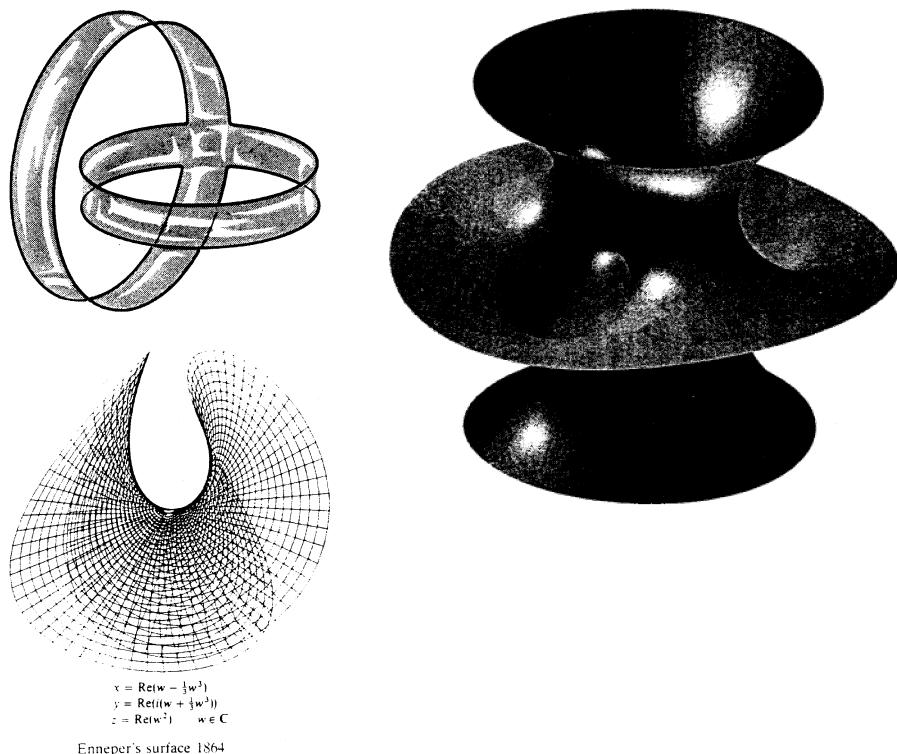


Fig. 1.72. Some minimal surfaces

revolution. From this, an exercise will prove that surfaces which are at the same time minimal and of revolution are necessarily *catenoids*, namely those whose meridian is the graph of the cosh (hyperbolic cosine) function. It is also easy to prove that the only ruled minimal surfaces are *helicoids* and planes. A surface is *ruled* when it is the set made up by a one parameter family of straight lines called the *generatrices* of the surface; beware that for a general ruled surface the tangent plane $T_m M$ turns when m moves along a generatrix. A helicoid is the image of a straight line moving under a spiral staircase motion. Computing the principal curvatures is especially easy for ruled surfaces. For catenoids and helicoids, if needed, see section 4. 2 of do Carmo 1976 [451]. An important technique in thinking about mean curvature is the *mean curvature flow*; see §§1.4.5, Ecker 2001 [477], Chopp 1993 [376] and Chen & Li 2001 [361].

We can slice a catenoid to wrap it into a helicoid, as in figure 1.73 on the facing page. This is a famous example of a one parameter family of surfaces (a deformation) having two properties. The first is that every surface in the family is minimal. The second is that all the inner metrics of these surfaces are isometric under a suitable point to point correspondence (this property to be

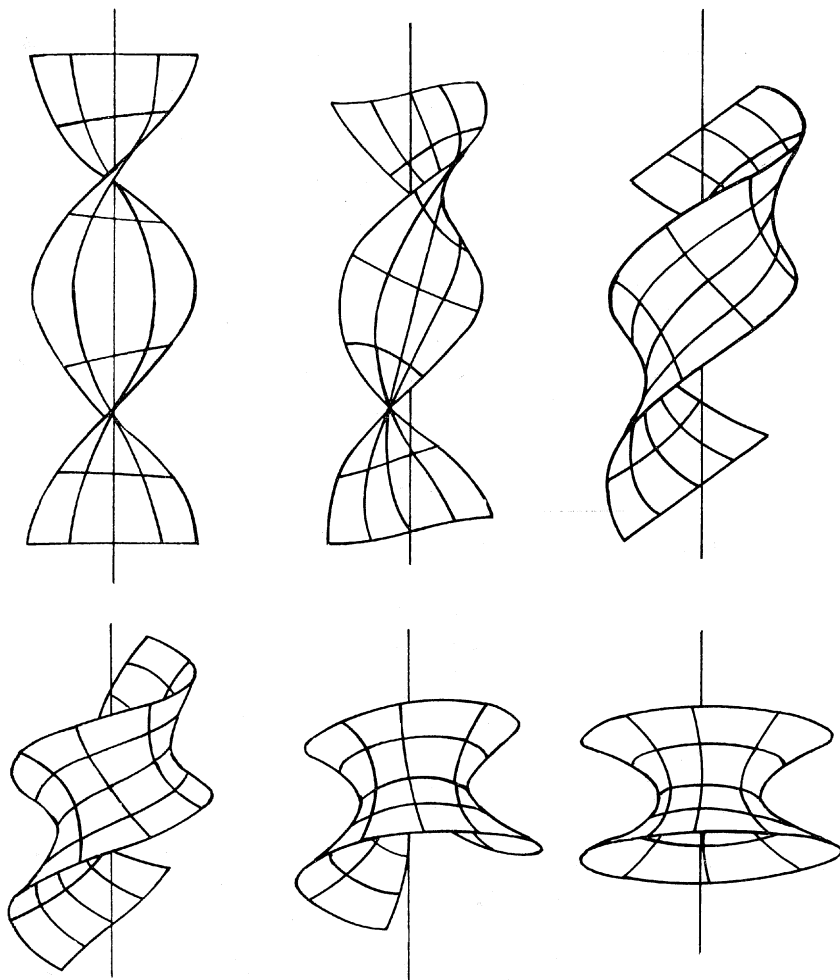


Fig. 1.73. Cutting a catenoid to make a helicoid

an isometric deformation is sometimes called bending [without stretching]). The fundamental theorem of Gauß which is going to be the keystone of chapter 3 implies then that the invariant K is preserved by isometries. But $H = 0$ throughout the bending, hence the two principal curvatures are preserved since their two values are $\pm\sqrt{-K}$.

Note 1.6.6.1 Recalling that the curvature of a plane curve, given as a function of the arc length, characterizes the curve up to Euclidean motion, could have suggested to you that the knowledge of the two principal curvatures will do the same for surfaces. This is false. However, if one knows the second fundamental form, that is, not only the principal curvatures but also the

principal directions, then the surface is determined up to rigid motion. In the family above these principal directions are not preserved by the isometric correspondence, they turn. For demonstration of the above see for example the theory of these families in 3.5 of Dierkes, Hildebrandt, Küster, & Wohlrab 1992 [444, 445]. ♦

1.6.7 The Hartman-Nirenberg Theorem for Inner Flat Surfaces

We want to study surfaces M with $K \equiv 0$. We saw in §1.6.3 that this is always the case for developable surfaces, i.e. for the envelope of a one parameter family of planes. Let us not forget the special case of a piece of plane for which the second fundamental form vanishes identically. One can build very wild examples of surfaces by attaching to pieces of planes pieces of cones and pieces of cylinders. This can be done as smoothly as we like using functions like $\exp(-1/x^2) \sin(1/x)$, etc. Or just smoothly bend the unbound upper corner of this page; now notice that you still have freedom to bend the unbound lower corner with your other hand. Such a surface will have both planar points (where $k_1 = k_2 = 0$) and non-planar points where one of the principal curvatures is non zero.

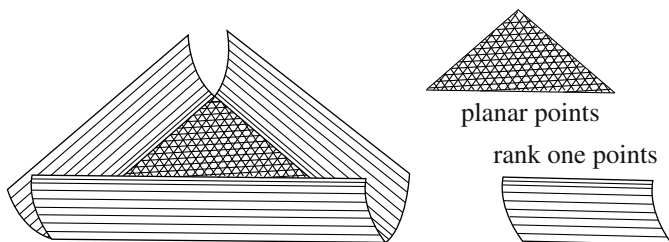


Fig. 1.74. Surfaces with $K = 0$

Locally at a non-planar point it is easy to prove (see section 5-8 of do Carmo 1976 [451] for details) that a surface with vanishing Gauß curvature is a piece of a developable one and second, that its inner metric is locally isometric to that of the Euclidean plane (this will also follow from §3.1.4). To get the generatrices, just follow the principal direction associated with zero principal curvature. A similar assertion being of course true at planar points, finally surfaces with $K \equiv 0$ have everywhere the same inner geometry locally as the Euclidean plane. Now how about a global statement? Things are quite easy in the real analytic case, but for milder smoothness hypotheses, the question was settled only in 1959 by Hartman and Nirenberg. They prove moreover that if M is complete then it has to be a complete cylinder (i.e. the lines perpendicular to a plane, passing through a plane curve). The starting idea, which is hard to work out completely and correctly, is that a family of

generatrices will develop singularities if extended far enough unless they are parallel. See do Carmo 1976 [451], section 5-8 for a very detailed and lucid exposition. Do not think mathematicians are now completely happy, they never are; see for example the very recent Fuchs & Tabachnikov 1999 [526].

1.6.8 The Isoperimetric Inequality in $\mathbb{E}^3$ à la Gromov

We now follow up our work on the planar case in §1.5. We consider domains $D \subset \mathbb{E}^3$ with compact boundary $M = \partial D$, and we ask

Question 10 *What is the best possible dimensionless ratio*

$$\frac{\text{Area}(M)^3}{\text{Vol}(D)^2}$$

for the area of $M = \partial D$ with respect to the volume of D , and which domains achieve this lower bound?

The expected answer is that round balls (with sphere as boundary) achieve the bound, and nothing else does.

Theorem 11 (Isoperimetric inequality in $\mathbb{E}^3$) *Among all domains $D \subset \mathbb{E}^3$ with twice continuously differentiable boundary, those which minimize the ratio*

$$\frac{\text{Area}(\partial D)^3}{\text{Vol}(D)^2}$$

are precisely the Euclidean balls.

Note 1.6.8.1 The plane is the only surface where uniqueness is easy to prove. There is a good reason for this: starting in dimension three, uniqueness in the isoperimetric problem cannot be true in general. For consider an object (which will turn out below to be a sphere) achieving the best possible ratio, and just add “hair” to it, that is to say, curves, etc. Then both the volume of the domain and the area of the boundary will not change. So uniqueness can be expected only under an additional hypothesis; the standard ones are either convexity or smoothness. From these remarks, and from what follows, one can see why the isoperimetric inequality—an amazing beautiful, useful and simply stated result—is usually absent from mathematics curricula. ♦

For a spectacular application of the isoperimetric inequality to convex bodies and the behavior of functions on spheres of high dimension (namely *Dvoretzky’s theorem* and its proof), see the note 7.1.1.2 on page 305 and the references therein.

In every dimension, the proof by Stokes’ theorem using the Knothe-Gromov trick, the one using Steiner’s symmetrization and the one using the Brunn-Minkowski inequality are valid to get the inequality. It is always more

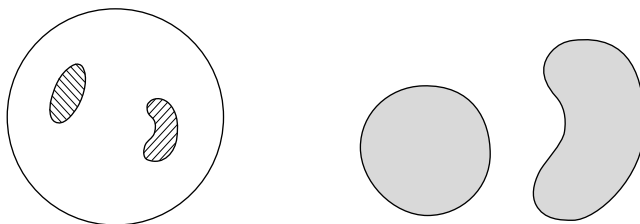


Fig. 1.75. (a) Isoperimetric inequality on the sphere. (b) Isoperimetric inequality in $\mathbb{E}^3$

difficult to prove that only spheres achieve equality, i.e. uniqueness. We now know how to prove the inequality and uniqueness using a technique found by Paul Levy in the 1930's. It remained buried up to 1980 when Gromov used it at full strength as follows. This proof is the more powerful one (see §1.5 and §§§7.1.2.2) and at the same time the most expensive. This is because it starts by asserting that there exist optimal domains D and that they have a boundary which, as a subset $M = \partial D$ of $\mathbb{E}^3$, is a smooth surface everywhere. This is known from geometric measure theory, although the required results emerged only in the 1960's (see §§14.7.2). Note that in higher dimensions everything works the same, with the proviso that geometric measure theory tells us that the solution in general will have singularities, but at most of codimension 6, which implies that they are of zero measure, and then the scheme below applies perfectly. A technical point: using the first variation equation 1.2 on page 4, we see that the point on M closest to a point of D will be a point for which the supporting tangent cone is contained in a half-space: this implies that such a point is regular (see §§§7.1.2.2 and the references there for more details).

The idea is fill up the inside of D starting from points of M and following the inner normal straight line. To be brief, we just speak of the normal $N_m M$ to M at m . Every point $d \in D$ is on at least one normal, namely take for $m \in M$ a point which is as close to d as possible. Then $d \in N_m M$.

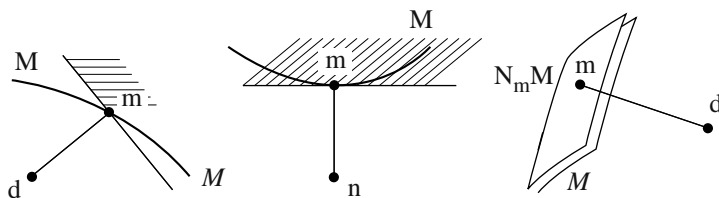


Fig. 1.76. Not a minimum; a local minimum

The infinitesimal volume element at distance t from m along a normal $N_m M$ is easily calculated to be

$$(1 - k_1(m)t)(1 - k_2(m)t)$$

with the effect that

$$\text{Vol}(D) = \int_M \int_0^{\text{Cut}(m)} (1 - k_1(m)t)(1 - k_2(m)t) dt dm \quad (1.18)$$

where $\text{Cut}(m)$ is the *cut value*, namely the length of the interval along the normal to M coming out of m at which we should stop in order to fill up D completely (see figure 1.77).

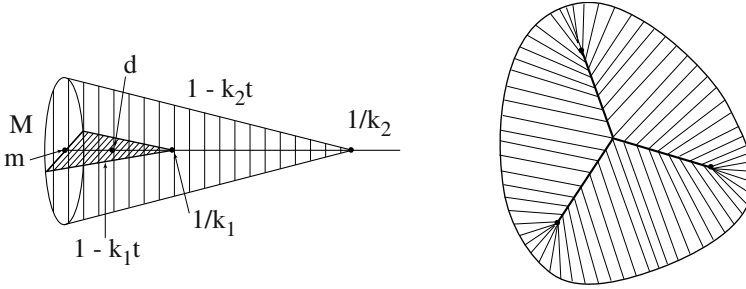


Fig. 1.77. Filling up a domain in the plane with normals from the boundary

The minimum property of M , in conjunction with the first variation formula 1.17 on page 59, shows that the mean curvature H is a constant, say h . From the elementary geometric-arithmetic mean inequality

$$xy \leq (x + y)^2 / 4$$

it follows that the integrand is bounded from above by $(1 - ht)^2$. A elementary geometric trick shows that the value $\text{Cut}(m)$ at which we stop integrating along $N_m M$ never exceeds $1/k_1(m)$ if $k_1(m) \leq k_2(m)$. This trick is essential and involves the notion of focal point (see Berger & Gostiaux 1988 [175], 2.7.11 or Gray 1990 [583], 8.1). It will be used at least two times below, in §§6.5.2 and in §§7.1.2.2. In conclusion one obtains

$$\text{Vol}(D) \leq \frac{h}{3} \text{Area}(M = \partial D) \quad (1.19)$$

The proof is now completed using the fact that $\text{Area}(\partial D)^3 / \text{Vol}(D)^2$ is also a minimum. One computes the first variation of that quotient for the normal variation $f = \text{constant}$. This yields

$$\text{Vol}(D) = \frac{h}{3} \text{Area}(\partial D).$$

Equality in equation 1.19 is now forced and tracing our steps backwards we find that $k_1 = k_2$ everywhere on M , i.e. all points are umbilics; we have seen already that this forces M to be a sphere.

1.6.8.1 Notes

A first remark is in order for the attentive reader. Putting together the mean curvature rigidity of the sphere mentioned in §§3.4.2 and the nonexistence of singularities in dimensions below 6 the above proof seems useless. But it is not that simple. First, the fact that surfaces of constant mean curvature are spheres is an extremely difficult one to prove; see §§3.4.2. Moreover, the above technique extends word for word in every dimension, and in general Riemannian manifolds with Ricci curvature bounded from below (see §§§7.1.2.2). So it is extremely powerful.

The reader might be interested to know that Schmidt 1939 [1105] proved that there is an isoperimetric inequality for domains of S^2 , the extremal figures being the spherical caps. He used symmetrization, but the above technique works as well. Schmidt proved his inequality in any dimension and not only for spheres but also for hyperbolic spaces: see §§4.3.2. Note also that our method “à la Gromov” works on spheres (and in any dimension) with easy changes in the integrand (the infinitesimal volume element) of 1.18 on the preceding page. You can guess which change to make if desired from theorem 111 on page 314. Steiner symmetrization also works because these spaces admit a lot of hyperplane symmetries (see theorem 40 on page 208). Also see §§§7.1.1.3.

One can be excused for the incorrect impression that the geometry of spheres is completely mastered today. Let us mention a famous open question. Girard’s formula 1.7 on page 32 implies that a spherical triangle all of whose angles are rational multiples of π has an area which is rational multiple of the total area of the sphere. Starting in dimension three, i.e. studying spheres $S^d \subset \mathbb{E}^{d+1}$, the question is to compute the volume of a simplex (the direct generalization of triangle in higher dimensions) as a function of the so-called dihedral angles, namely the angles along edges between the two adjacent faces. Schläfli found a formula in 1858, but one is unable to deduce from it the same rationality conclusion. The common view is that this time (in dimensions at least three) rational dihedral angles will not imply rational volume. But nobody can prove it today. For more on this in a general setting involving also volumes of simplices in hyperbolic spaces, see the survey Vinberg 1993 [1222] and the references there, as well as the references quoted in section 10.6 of Ziegler 1995 [1306].

Finally one can consider the volume of tubes around surfaces in $\mathbb{E}^3$ (comparing with §1.4 above). For this question, which has a nice answer and is of historical importance, see Gray 1990 [583], entirely devoted to tubes. With much less material, see chapter 6 of Berger & Gostiaux 1988 [175]. We will meet tubes again in §15.7.

1.7 Generic Surfaces

We want to say that a property of a surface (or more general mathematical object) is *stable* if the property continues to hold after small perturbation,

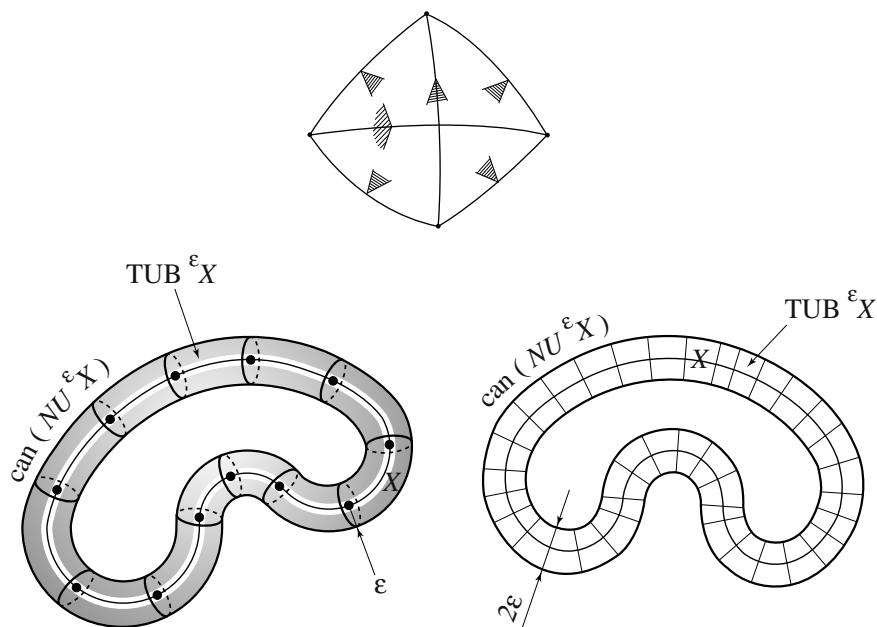


Fig. 1.78. (a) A tetrahedron in S^3 with its six dihedral angles. (b) Tubes around curves

and say it is *generic* if it always holds after (large or small) random perturbation. For example, positivity of Gauß curvature is stable, but not generic. Vanishing of Gauß curvature (inner flatness) is neither stable nor generic. The property of a parameterized smooth curve being immersed is generic. We will say that a generic parameterized smooth curve is immersed. Any property of a surface that can be observed by actual measurements must be stable, since the measurements will not be perfectly precise. If the property is generic, then we don't even have to make a measurement; we can expect it beforehand with near certainty.

Let us be more precise about these perturbations. Cut a sequence of points out of a line. Since the line is not countable, there will still be points left over, not belonging to the sequence. Similarly, if we cut a sequence of smooth curves out of a surface, there will be points left over. Call a subset of a topological space *skinny* if it is closed and nowhere dense. For example, a curve in a surface. A subset of a topological space is called *meager* if it is the union of a sequence of skinny sets, and *residual* if it is the complement of a meager set, i.e. the residue left over from cutting out a sequence of skinny sets. In our example of cutting curves out of a surface, each curve is skinny, the union of the curves is meager, and the complement residual. A topological space is said to be *fat* if it is not a meager subset of itself. A complete metric space is fat (the first skinny set in a sequence misses some closed ball, and the next

skinny set misses a smaller closed ball inside the previous ball, etc. converging down to a point or a closed ball of points). In particular, a finite set is fat. Similarly, a complete Frechet space is fat. A residual subset of a fat space is fat. A property of elements of a fat topological space is *generic* if it holds on a residual subset.

Whitney 1955 [1260] proved that the generic C^3 map from a surface to the plane has only cusp and fold singularities. The fold is the map

$$(x, y) \mapsto (x^2, y)$$

while the cusp is the map

$$(x, y) \mapsto (x^3 + xy, y)$$

and to say that these are the singularities of a map means that near any singular point, there are local coordinates (x, y) on the surface, and (u, v) on the plane, so that the map is represented in this way. These maps are drawn in figures 1.79 on the facing page and 1.80 on the next page. In each map, the singular points form a smooth curve. The image of this curve on the plane forms another curve, smooth except possibly with corners, the curve of singular values. One may strengthen this result (only slightly) to examine the preimages of singular values: generically, the preimage will contain at most two singular points, and if there are two, then the two corresponding tangent lines of the curve of singular points map to distinct tangent lines on the plane. Thus every compact smooth surface is obtained by stitching together finitely many pieces of cloth (i.e. regions of the plane) each of which has smooth edges, and finitely many corners. In effect, the curve of singular values is a sewing pattern for the surface (as in fashion magazines); compare figure 1.81 on page 70. But some of the corners of the cloth must be smoothed out (look at the cusp), while others (corresponding to double singular values) must remain corners when stitched together. Keep this in mind when we discuss triangulations of surfaces and the Gauß–Bonnet theorem.

Now consider the Euclidean geometry of a surface in $\mathbb{E}^3$. A point of positive Gauß curvature is called an *elliptic point*. Near an elliptic point, any tangent line to the surface will have only first order contact (since there are no zeros of the second fundamental form). At a point of negative Gauß curvature (called a *hyperbolic point*) such a line will have first order contact, except along the two asymptotic directions (null directions of the second fundamental form), where the tangent line achieves at least second order contact. These asymptotic directions form the tangents to the *asymptotic lines*. At points of vanishing Gauß curvature (*parabolic points*) we have collision of asymptotic directions, forming a single tangent line, achieving at least second order contact. The elliptic and hyperbolic points form open sets, and generically (in the natural topology on smooth surfaces) they are bounded by a smooth curve of parabolic points. Inside the region of hyperbolic points is a curve consisting of those points where the asymptotic lines have an inflection. For a little more detail,

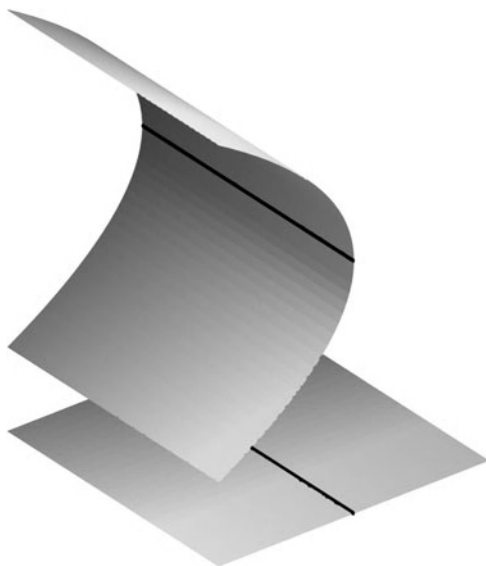


Fig. 1.79. A fold

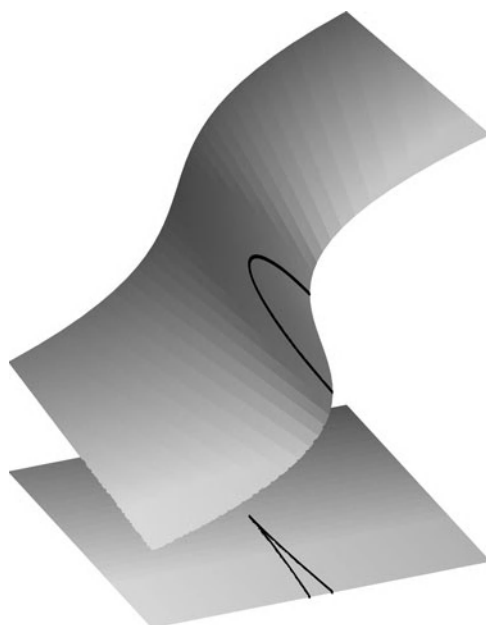


Fig. 1.80. A cusp

see Arnol'd 1992 [63]. For a lot more, especially on the singularities of wave fronts emanating from the surface, and on the umbilics of generic surfaces, see

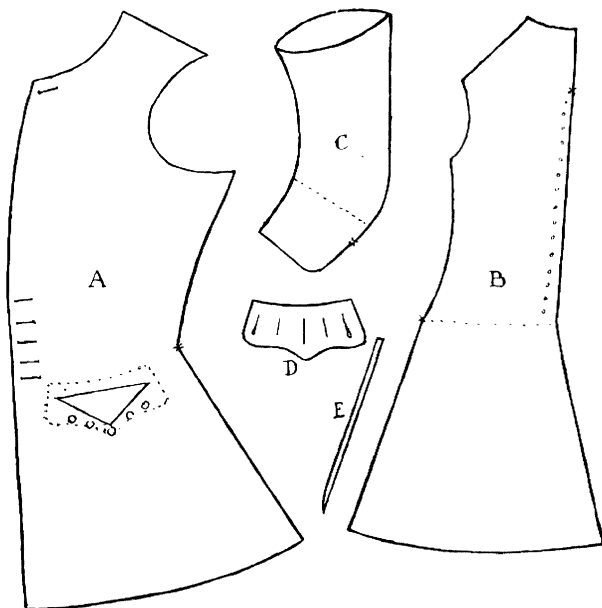


Fig. 1.81. A sewing pattern

Porteous 1994 [1039]. A brief discussion of the same topics will be found in Berger 2003 [173]. Generic surface geometry has been applied to the study of retinal imaging and the melting of ice and deforming of landscapes by wind and rain; see Donati & Stolfi 1997 [460].

1.8 Heat and Wave Analysis in $\mathbb{E}^2$

1.8.1 Classical equations of physics for a plane domain

We come back to the physics of a compact domain D in the Euclidean plane $\mathbb{E}^2$, thinking some more about billiard balls. In §§1.2.3 we studied the classical mechanics of a particle moving inside D . Now we want to study physics related to continuum and quantum mechanics. Namely, we consider the equations called respectively the *heat equation*, the *wave equation* and the *Schrödinger equation*.

These equations can be simply written using the notation

$$\Delta f = -\frac{\partial^2 f}{\partial x^2} - \frac{\partial^2 f}{\partial y^2}$$

for a function $f : D \rightarrow \mathbb{R}$. The operator $f \mapsto \Delta f$ is called the *Laplacian*. The heat equation describes what happens when you start with a given distribution of heat on D at a given time, and ask how the heat has spread at a later time.

The heat equation for the function f describing the temperature $f(x, y, t)$ at time t at the point (x, y) of D is

$$\Delta f = -\frac{1}{\nu} \frac{\partial f}{\partial t}$$

where ν is the *conductivity* of the material.

For the wave equation, you consider the cylinder built over D , and pour a thin layer of water in it. Then you try to describe the motion of the surface of the water.¹² The wave equation for the height $f(x, y, t)$ of the water after time t at the point $(x, y) \in D$ is

$$\Delta f = -\frac{1}{c^2} \frac{\partial^2 f}{\partial t^2}$$

where c is the *speed of sound* in the fluid. The wave equation coincides with the *vibrating membrane equation* which describes the normal motions of the domain D when it is considered as a vibrating membrane, a “drum”. It should be understood that the solutions of these equations in fact represent only a first order approximation of the motions under consideration. The wave equation also describes, in first approximation, the behavior of sounds in a flat object, see §§1.8.4.

The Schrödinger equation of a free particle uses complex valued functions. It is written

$$\frac{\hbar^2}{2m} \Delta f = i\hbar \frac{\partial f}{\partial t}$$

where $i = \sqrt{-1}$, $\hbar$ is *Planck’s constant*, and m is the *mass* of our free particle. Henceforth, we will take the constants appearing in any of these equations to be equal to 1. This can be achieved by rescaling the time variable (which may sound bizarre to an engineer, but is natural to a mathematician).

1.8.1.1 Bibliographical Note

The subject of classical continuum physics and quantum mechanics, even in the plane, is immense, and so is the bibliography. We gave many references during our journey, but the reader will find a commented bibliography at the end of Bérard 1986 [135], unfortunately dating from 1985.

1.8.2 Why the Eigenvalue Problem?

We want to solve the equations above, under various conditions on the boundary ∂D of D and on the initial condition at time $t = 0$. To solve such an equation depending both on the time t and the point $m = (x, y)$, our first clue is to

¹² The wave equation is just a first approximation. For better approximations, one needs to work much harder. For example, see Greenspan 1978 [598]. To our knowledge, no one has ever considered how to extend Greenspan’s work to curved surfaces, or to Riemannian geometry.

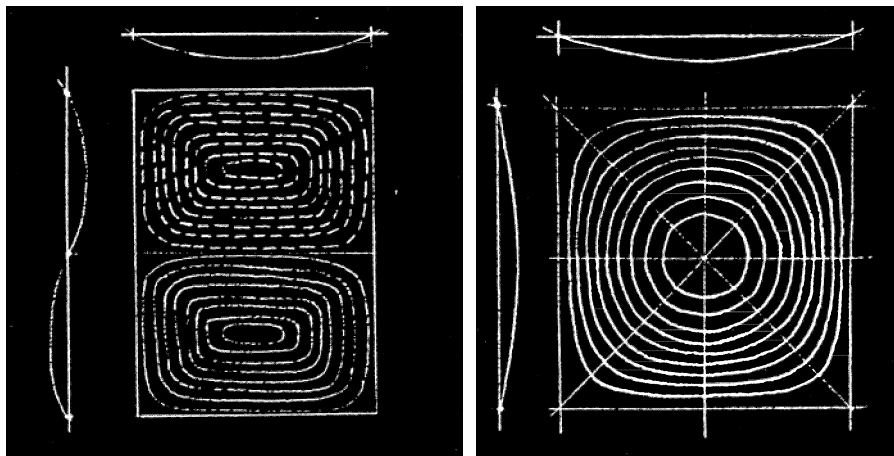


Fig. 1.82. The first two eigenfunctions of a square

use the fact that (roughly by the Stone–Weierstraß approximation theorem) we need only consider product functions $f(m, t) = g(m)h(t)$. Subsequently, one constructs series of them (as in Fourier series theory). For example, look at the heat equation. We find that g and h must satisfy:

$$\frac{\Delta g}{g} = -\frac{h'}{h}$$

where h' is the usual derivative in t .

Since the first fraction depends only on the point m and the second only on the time t their common value has to be a constant, call it λ . A function $g : D \rightarrow \mathbb{R}$ such that $\Delta g = \lambda g$ is called an *eigenfunction* of D and λ is called an *eigenvalue* of D . Of course, one should specify a boundary condition. Once these eigen-objects are calculated, we are almost done. Just note that the time dependence will give

$$h(t) = \begin{cases} e^{-\lambda t} & \text{for the heat equation} \\ e^{i\lambda t} & \text{for the Schrödinger equation} \\ e^{it\sqrt{\lambda}} & \text{for the wave equation.} \end{cases}$$

Physically, for a vibrating membrane, the product motions $g(m)h(t)$ are the *stationary* ones; they are the ones one can observe through *stroboscopy*.¹³

For all of these equations, the set of the eigenvalues λ of the Laplacian on D is an essential piece of data. For simplicity's sake, we will work only with the boundary condition $f(m, t) = 0$ for every $m \in D$ and every time t ; this is often called the *Dirichlet problem*. We will call the set of the corresponding

¹³ Recall that a stroboscope is an instrument which periodically flashes light, used for studying periodic motion.

eigenvalues the *spectrum* of D (for the Dirichlet problem) and denote it by $\text{Spec}(D)$. It is the analogue for quantum mechanics of the set of lengths of periodic trajectories in D which we met in §§1.2.3 for classical mechanics. This set of lengths is sometimes called the *length spectrum* of D .

The eigenvalues represent—in some sense—the energy levels of the domain D , so that the spectrum is of primary importance. We will see that the spectrum is much better understood than the length spectrum (see §§1.2.3) because there are systematic tools to address to it. Physicists are extremely interested in these energy levels: how do they look as a subset of $\mathbb{R}$? The first and simplest number to estimate is the analogue here of the counting length function introduced in §§1.2.3 for classical mechanics. Written $N(\lambda)$, it is defined to be the number of eigenvalues of Δ (for the Dirichlet problem on D) which are not larger than λ :

$$N(\lambda) = \# \{ \lambda_i \leq \lambda \}$$

This function, and its vague family resemblance to its cousin $CF(L)$, is far from understood (see §§1.8.6).

Affinities between these two spectra are still emerging from the shadows. They provoke curiosity in the mathematician, drawing comparison between a geometric notion and a notion coming from analysis. Some physicists explore this topic these days because it is one of the simplest models where one can test “explicitly” various conjectures on relations between classical and quantum mechanics.

In the sequel, we will amply revisit this question on compact Riemannian manifolds: see chapters 9 and 10. Treat our discussion here as an introduction and a motivation for later. Physicists are also interested in abstract Riemannian manifolds, even though naively they may seem unphysically abstract, because some of them are much simpler to handle than all but a few bounded regions of Euclidean space. The simplicity comes from having no boundary, enabling us to ignore boundary conditions, which are often the source of subtleties. The relations between periodic motions and vibrations (or waves) of a domain D transcend the similarities in the spectra. In kinematics, we have trajectories (and not only their length), while in vibrations we have the eigenfunctions, of which we can retain the nodal curves (the points of the domain where an eigenfunction vanishes). When you vibrate a membrane and cover it with dust, the dust collects in the nodal curves, because the membrane stays still at those points—at the first order. For known relations between spectra, nodal curves and periodic motions, we refer the reader to the recent book of Sarnak 1995 [1095] and the references therein.

Let us look for the eigenfunctions. As in Fourier series theory, (which corresponds essentially to D being an interval in the line $\mathbb{E}^1 = \mathbb{R}$), any reasonable function f vanishing on ∂D can be expressed as a converging series whose entries are eigenfunctions of the Dirichlet problem for D . Notation: write the totality of the spectrum as

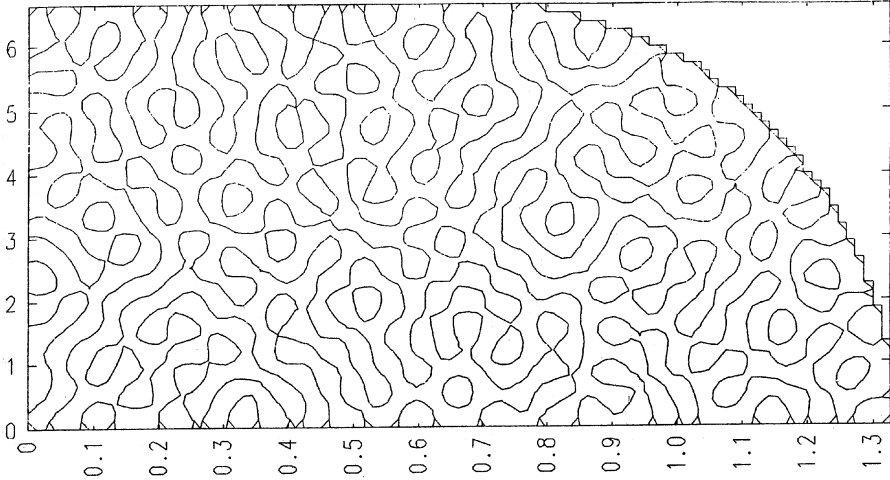


Fig. 1.83. A nodal set of Bunimovich's stadium

$$\text{Spec}(D) = \{\lambda_i\}_{i=1,2,\dots}$$

and the set of the corresponding eigenfunctions as $\{\phi_i\}_{i=1,2,\dots}$ normalized so that

$$\int_D \phi_i^2 = 1$$

for every $i = 1, 2, \dots$ and

$$\int_D \phi_i \phi_j = 0$$

for $i \neq j$. Beware that eigenvalues can have multiplicity, the multiplicity of λ being the dimension of the space of eigenfunctions with eigenvalue λ . This dimension is always finite. When writing the spectrum as above, it is generally understood that multiple eigenvalues are repeated a number of times equal to their multiplicity. Let us clarify a minor point: not only do the eigenfunctions ϕ_i have to be normalized, but they must also be orthogonal to one another. This is automatic when their eigenvalues are different (exercise for the reader), but for multiple eigenvalues one needs to choose an orthonormal basis of a finite dimensional Euclidean vector space.

The Fourier-like decomposition says that every function f vanishing along D is equal to the following series:

$$f = \sum_{i=1}^{\infty} c_i \phi_i$$

where

$$c_i = \int_D f(m) \phi_i(m) dm$$

for each $i = 1, 2, \dots$. When thinking of a vibrating membrane, the λ_i will be the frequencies of vibration, also called the *harmonics*; the lowest one is called the *fundamental tone*.

1.8.3 The First Way: the Minimax Technique

One way to hunt down the eigenfunctions is as follows. We get the first one by the so called *Dirichlet principle*: among functions vanishing along ∂D one looks for one minimizing the ratio

$$\frac{\int_D \|\text{grad } f(x, y)\|^2 dx dy}{\int_D f^2(x, y) dx dy}$$

where $\text{grad } f$ is the gradient of f :

$$\text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

This ratio is called the *Dirichlet quotient*.

Suppose that we have a function f which minimizes this ratio. Computing the derivative of this ratio with respect to ε for a variation $f + \varepsilon g$ of the function f and using Stokes' theorem together with the Lagrange multiplier technique, reveals

$$\Delta f = \lambda_1 f$$

for some constant λ_1 , and reveals that this λ_1 is the infimum of the Dirichlet quotient over all nonvanishing functions:

$$\lambda_1 = \inf \left\{ \frac{\int_D \|\text{grad } f\|^2 dx dy}{\int_D f^2 dx dy} \mid f \neq 0 \right\}. \quad (1.20)$$

This yields the first eigenfunction: $\phi_1 = f$, together with the first eigenvalue (which, by the way, is always of multiplicity one: you can prove it yourself). To dig up the next eigenvalue and its eigenfunctions, one applies the same trick, but restricting the set of functions f in consideration, allowing only functions which are orthogonal to the first eigenfunction:

$$\int_D f(x, y) \phi_1(x, y) dx dy = 0.$$

Proceed in this way, ad infinitum. See more on this in §9.4.

For rigorous proofs of the facts above, and also for facts which we did not quote explicitly, namely that the spectrum is a discrete, countable subset of the real line (of course made up entirely of positive numbers) and the fact that eigenvalues are of finite multiplicity, we refer the reader to Bérard 1986 [135], Chavel 1984 [325], Courant & Hilbert 1953 [406, 407], or any book on classical continuum physics or quantum mechanics. Note that those facts

were rigorously proven only in the 1920's. If you look at a proof using Sobolev inequalities, keep in mind that the best current derivation of those inequalities uses the isoperimetric inequality.

There are very few examples where the spectrum or the eigenfunctions can be determined explicitly. Two old standards are rectangles and disks. In both cases, separation of variables disentangles the eigenfunctions. Using again the Stone–Weierstraß theorem, and because the boundary condition agrees with the separation, on a rectangle one need only look for product functions $f(x, y) = g(x)h(y)$, and there will be no other eigenfunctions. If the rectangle has side lengths a and b respectively, then the eigenfunctions are

$$\sin \frac{\pi m x}{a} \sin \frac{\pi n y}{b} \quad (1.21)$$

where m and n are any integers, yielding the set of eigenvalues

$$\lambda = \pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right).$$

i.e. to obtain an eigenvalue at most λ we need m and n to be integer points inside a certain ellipse. We will see below that a simple expression yields an easy first order approximation of $N(\lambda)$ when $\lambda \rightarrow \infty$, but the second order term in λ is related to deep number theory and is still not completely understood today. It is believed that the number of integers m, n with

$$m^2 + n^2 \leq \lambda^2$$

is asymptotic to $\pi^2 \lambda^2 + O(\lambda^{\varepsilon+1/2})$ as $\lambda \rightarrow \infty$, for any $\varepsilon > 0$, but there is still no proof. This is called the Gauß circle problem; see §§1.8.5. However, it is known that $\pi^2 \lambda^2 + O(\lambda^{1/2})$ is too small.

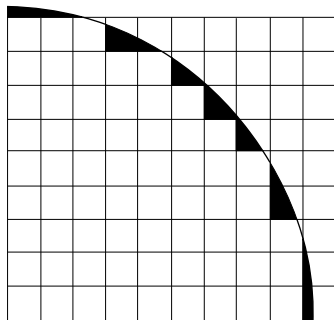


Fig. 1.84. The circle problem: find the integer points inside a circle

The eigentheory of the disk is treated by separating the variables using polar coordinates (ρ, θ) and looking for product functions $f(x, y) = g(\rho)h(\theta)$.

One calculates $h'' = -nh$ where n is any integer, and $g(\rho)$ satisfies the n -th Bessel equation. Everything is explicit, modulo knowing everything about Bessel functions; see any textbook in the realm of mathematical methods of physics, e.g. Morse & Feshbach 1953 [945], Courant & Hilbert 1953 [406, 407].

Be wary: vibrations of the disk are not easily grasped. Figure 1.85 is the intensity distribution of an eigenfunction of quite high frequency, taken from Sarnak 1995 [1095]. The very high concentration near the boundary is surprising. It explains the famous “whispering gallery effect”. The picture is less surprising if you know that higher Bessel functions vanish to higher order at the origin.

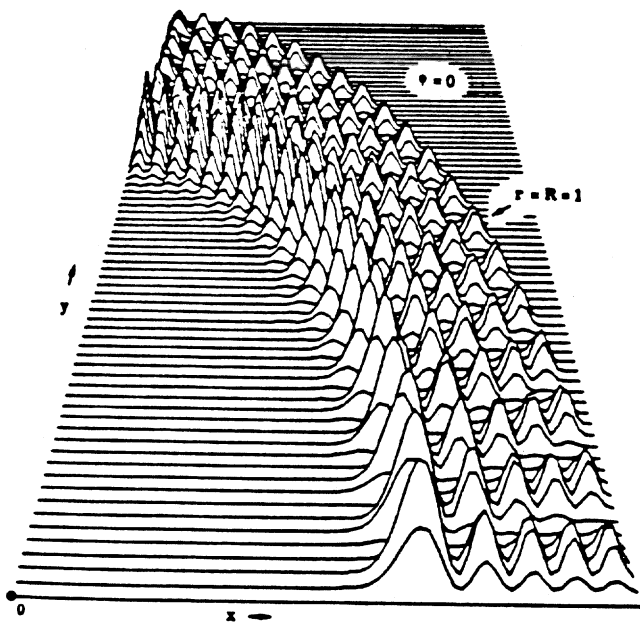


Fig. 1.85. Square of a high frequency eigenfunction on the disk

The eigenvalues (harmonics) of a vibrating string are just the integral multiples of the lowest one. This is never the case for a plane domain. Check it for yourself on the two above cases. This is why drums are never very musical, but the player tries when he beats a drum to generate a suitable mixture of various eigenfunctions, ideally only one. Figure 1.86 on the following page (picture and legend) is taken from Balian & Bloch 1970/1971/1972 [97, 98, 99, 100, 101], the pioneering papers which started the search (mentioned above) for relations between the two spectra. If you see some resemblance between the figure and certain musical instruments, it is not by chance.

To our knowledge the only other regions of the plane whose eigenfunctions, or even just eigenvalues, are explicitly computable are ellipses and isocles,

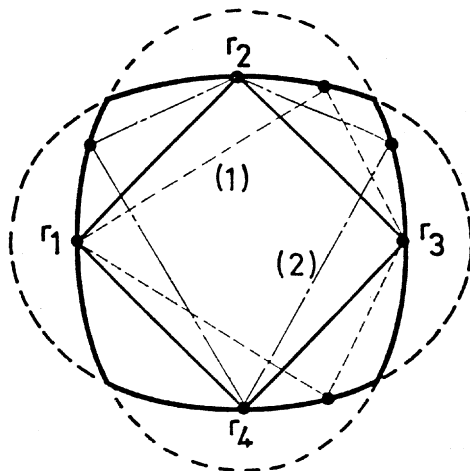


Fig. 1.86. From papers of Balian & Bloch

rectangular, equilateral and half equilateral triangles. The eigenvalues make themselves manifest as follows: for ellipses, separate variables using homofocal coordinates (see the figures in §§1.8.6 and the books of Courant & Hilbert 1953 [406, 407] and Morse & Feshbach 1953 [945]). For the triangles, use reflections on the boundary to extend the functions (vanishing at the boundary) to a square (for isosceles triangles) or to the whole plane with periodicity (for equilateral or half equilateral triangles). Periodic functions in the plane can be decomposed in Fourier series, but we will come back at length to this from the point of view of abstract manifolds. This will be achieved by replacing lattice-periodic objects in the plane by objects on an abstract (flat) torus: see §§4.3.3. Fourier analysis of periodic functions on the line or of functions on a circle are the same subject.

As a shocking example of our ignorance, one knows nothing about regular hexagons, not even the first eigenvalue. Try to understand why the reflection method does not work with regular hexagons. For more see Bérard 1980 [134], Pólya & Szegő 1951 [1037]. We did not really justify our use of the word “minimax;” it will be amply justified in §9.4.

1.8.4 Direct and Inverse Problems:

Can One Hear the Shape of a Drum?

Mathematicians are never trumped when they cannot determine mathematical objects explicitly. Under their own steam, with aid from physicists, they look for various replacements like approximations, estimates of the counting function $N(\lambda)$, etc. There are two ways to think of this. Direct problems are of the type:

Question 12 *I know more or less the shape of the domain D ; what can I deduce from that about its spectrum?*

Inverse problems work the opposite direction:

Question 13 *I know (more or less) the spectrum of a domain; what can I say about its shape?*

These two problems were addressed (more or less explicitly) in the 19th century and people have never stopped working on them. This book is an introduction to Riemannian geometry, and we will devote the whole of chapter 9 to these two problems in the Riemannian realm. Therefore, we will treat spectral geometry of Euclidean domains as mere motivation, and we will content ourselves with giving a few results, ideas and problems.

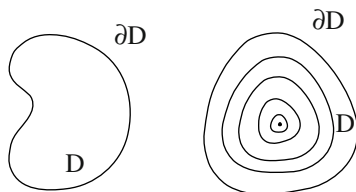
1.8.4.1 A Few Direct Problems

The main direct problem is the asymptotic expansion of $N(\lambda)$ and will be addressed in the next section in some detail because the technique extends automatically to Riemannian manifolds. A more naive problem is to consider the first eigenvalue λ_1 . For a vibrating membrane, λ_1 is called the *fundamental tone*, for obvious musical reasons. In the heat equation, λ_1 gives the dominant information (together with the first eigenfunction), i.e. the mode that decays with slowest exponential rate. Everything else quickly disappears. Controls on λ_1 are of practical importance to avoid “resonances”.

A natural question concerning the first eigenfunction ϕ_1 was open for a long time. Using Dirichlet’s principle, it is easy to see that the first eigenfunction of any domain never vanishes, except at the boundary. (After flipping sign if needed, take it to be positive.) But, as a sort of control, one expects that the shape of the boundary has some influence on the first eigenfunction, more precisely on its level curves $\{f = t\}$ when t runs from 0 to the maximum of f . Assume, for example that the domain is convex: is every level curve convex also? Brascamp & Lieb 1976 [256] proved that this is so. A similar problem was solved by them in the same paper. We want to know a little about how λ_1 depends on D . They proved that λ_1 is a concave function when one performs Minkowski addition of two domains.¹⁴

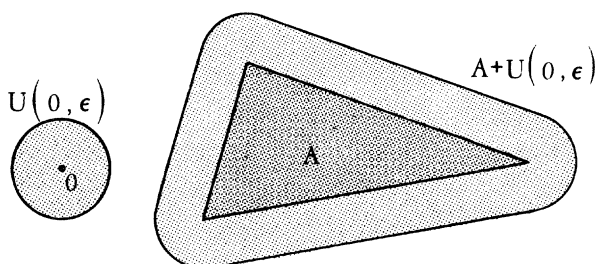
Other direct problems: to give systematic lower or upper bounds on the λ_i depending on the domain. See Bérard 1986 [135], Chavel 1984 [325], Chavel 1993 [326] and §§1.8.5 for examples. We note here that the minimax technique (explained in equation 1.20 on page 75, see also chapter 9) shows that upper bounds are in general quite easy to obtain, because one just exhibits some suitable function, calculating its Dirichlet quotient. For a lower bound, one has

¹⁴ A definition of Minkowski addition is presented in Berger 1994 [167] (11.1.3) or any book on convexity.

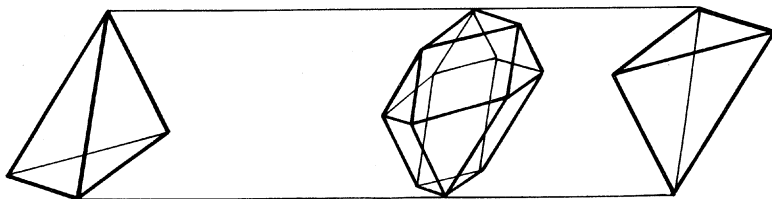


the level lines are all convex curves

Fig. 1.87. What are the level lines?



a A Minkowski sum in the plane



b A Minkowski sum in space

Fig. 1.88. Minkowski sums

to show an inequality valid for every function (in particular the eigenfunction under consideration). We will give the basic strategy later in this section.

Another natural idea when looking at the spectrum is to look at the ratio of the first two eigenvalues λ_1 and λ_2 . It was conjectured in 1955 by Payne, Pólya and Weinberger that the ratio λ_2/λ_1 is strictly smallest on disks. This was proven only in 1992: see Ashbaugh & Benguria 1994 [72] for this question, and the recent Ashbaugh & Benguria 1995 [73].

1.8.4.2 The Faber–Krahn Inequality

The last direct problem we shall ponder, but not the least, is the Faber–Krahn inequality: among all domains with equal area the one with lowest λ_1 is the disk:

$$\lambda_1(D) \geq \lambda_1(D_*) \quad (1.22)$$

where D_* denotes a disk with the same area as D . Musically it is quite intuitive. For the heat equation, a body with longer boundary should lose heat to its environment more quickly.

The assertion was stated without proof by Rayleigh in 1877; for history and earlier results see Rayleigh 1894 [1053] or Pólya & Szegő 1951 [1037]. Faber and Krahn proved it independently in 1920. The proof below extends with no difficulty to any dimension.

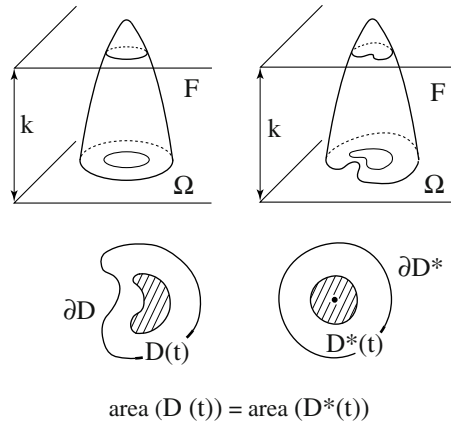


Fig. 1.89. Schwarz symmetrization

The underlying idea of the proof is too beautiful to be concealed. Moreover it is the archetype for the more general result which will see for Riemannian manifolds in chapter 9. Therefore, we are going to give a detailed proof. Here, we will symmetrize functions (as opposed to the geometric symmetrization of Steiner in §1.5) called *Schwarz symmetrization*. It is a kind of transplanting of functions defined on D to functions defined on a disk D_* such that $\text{Area}(D_*) = \text{Area}(D)$. See figure 1.89. Pick up the graph of some function $f : D \rightarrow \mathbb{R}$ and slice it horizontally. Watch the level curves $f^{-1}(t)$ when t goes from zero to the maximum of f . We now build up a function $f_* : D_* \rightarrow \mathbb{R}$ by forcing its level curves to be circles centered at the center of D_* , so that each level curve surrounds a disk whose area is the same as the area surrounded by the corresponding level curve of f . Perhaps we can make this clearer: let $D(t) = f^{-1}(t, \sup f]$ and define f_* by the condition that the disk $D_*(t) =$

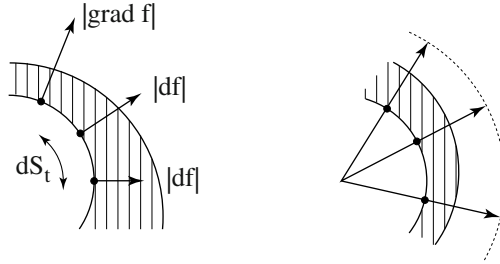
$(f_*)^{-1}(t, \sup f]$ is centered at the origin and that $\text{Area } D_*(t) = \text{Area } D(t)$. Note that $\sup f = \sup f_*$. By equation 1.20 on page 75 we only need to prove

$$\frac{\int_D \|\text{grad } f\|^2 dx dy}{\int_D f^2 dx dy} \geq \frac{\int_{D_*} \|\text{grad } f_*\|^2 dx dy}{\int_{D_*} f_*^2 dx dy}.$$

The basic idea is to replace the double integral in $dx dy$ by an integral first in dt and then along the level curves $f^{-1}(t)$. Figure 1.90 shows that

$$dx dy = \frac{ds dt}{\|\text{grad } f\|} \quad (1.23)$$

where ds denotes the arc length differential along the level curves. Two basic tools of the theory of integration are used here: Fubini's theorem and the change of variables, giving



the variation of area goes as $\frac{1}{\text{grad}(f)}$

Fig. 1.90. The variation of area across level curves

$$\text{Area } D(t) = \int_t^{\sup f} dt \int_{f^{-1}(t)} \frac{ds}{\|\text{grad } f\|} \quad (1.24)$$

The same equation applies to f_* and D_* , and because of the condition on the areas inside the level curves

$$\int_{f^{-1}(t)} \frac{ds}{\|\text{grad } f\|} = \int_{f_*^{-1}(t)} \frac{ds}{\|\text{grad } f_*\|}.$$

Applying equation 1.24 to D_* and f_* one gets

$$\int_D f^2 dx dy = \int_{D_*} f_*^2 dx dy.$$

To prove

$$\int_D \|\text{grad } f\|^2 dx dy \geq \int_{D_*} \|\text{grad } f_*\|^2 dx dy$$

we compute the derivative with respect to t of the integral

$$G(t) := \int_D \|\text{grad } f\|^2 \, dx \, dy$$

From equation 1.23 on the facing page one calculates

$$G'(t) = - \int_{f^{-1}(t)} \|\text{grad } f\| \, ds$$

and the Schwarz inequality reads

$$\begin{aligned} \left(\int_{f^{-1}(t)} \|\text{grad } f\| \, ds \right) \left(\int_{f^{-1}(t)} \frac{ds}{\|\text{grad } f\|} \right) &\geq \left(\int_{f^{-1}(t)} ds \right)^2 \\ &= (\text{length } f^{-1}(t))^2 \end{aligned} \quad (1.25)$$

The classical isoperimetric inequality applied to both $D(t)$ and $D_*(t)$ and their boundaries $f^{-1}(t)$ and $f_*^{-1}(t)$ give:

$$\text{length } f^{-1}(t) \geq \text{length } f_*^{-1}(t)$$

Write the analogue of 1.25 but for $D_*(t)$. We remark that $\|\text{grad } f_*\|$ is constant on the circles by the circular symmetry, so now we have equality in the Schwarz inequality. This shows that $G'(t) \geq G'_*(t)$ for every t so that

$$\int_D \|\text{grad } f\|^2 \, dx \, dy = G(0) \geq G_*(0) = \int_{D_*} \|\text{grad } f_*\|^2 \, dx \, dy$$

since $G(\sup f) = G_*(\sup f_*) = 0$, giving the finishing touch.

We hid a technical difficulty: the level curves need not be smooth. This is not too hard to overcome (see the remarks in Bérard 1986 [135], IV.6).

The Faber–Krahn inequality can also be obtained by Steiner line symmetrization (see §1.5); this symmetrization respects the area and diminishes λ_1 .

We could ask for more than the asymptotic expansion, namely how does the subset $\{\lambda_i\}_{i=1,2,\dots} \subset \mathbb{R}^+$ look? Physicists are interested in that question. They would like a sort of random repartition around Weyl's asymptotic formula. We will say few words on this in §9.11, see also Sarnak 1995 [1095]. The spectrum is a mystery even for triangles in the hyperbolic plane. Hyperbolic triangles are presently the physicists' favourite specimens, because hyperbolic geometry (negative curvature) is more realistic than Euclidean geometry for systems of particles.

1.8.4.3 Inverse Problems

The importance of inverse problems cannot be underestimated. For plane domains they provide the simplest model of determining the structural soundness

of an object by listening to its vibrations. Consider a vibrating object, anything from a space craft to a church bell. We need to test the strength of this object. Already in the middle ages, bell makers knew how to detect invisible cracks by sounding a bell (on the ground, before lifting it up to the belfry). How can one test the resistance to vibrations of large modern structures by non-destructive assays? This is a perplexing problem; to our knowledge there is no solid mathematical contribution on the subject. A small crack will not only change the boundary shape of our domain—one side of the crack will strike the other during vibrations, invalidating our use of the simple linear wave equation. On the other hand, presumably heat will not leak out of a thin crack very quickly, so perhaps the heat equation will still provide a reasonable approximation for a short time, unless cracks permeate the object, or take up significant volume. The effect on quantum mechanics is unclear. For references on nondestructive assays, see Bourguignon 1986 [238]. In the absence of mathematical results, engineers use both scale models and assays.

The spectrum determines (more or less) the length spectrum (for more precise statements, see the references). But we saw that periodic billiard motions are very difficult to study, and in particular recovery of something of the shape of a domain from its length spectrum is beyond reach today (see Guillemin & Melrose 1979 [669], from which we extract figure 1.91 on the facing page to help the reader feel the difficulties). There is basically only one exception: disks are characterized by their spectrum as will be proven twice in the next section. See also Guillemin & Melrose 1979 [668] for the spectral geometry of ellipses.

The Faber–Krahn inequality 1.22 on page 81 can also be seen as an inverse problem; another one is the following. Let us ask for more than only Faber–Krahn: their result tells us that in order for a domain to have a low fundamental tone, it should have large area. But this does not expose the shape of the domain. For example, consider rectangles where we have the explicit value $\lambda_1 = \pi^2 (1/a^2 + 1/b^2)$ for the fundamental tone. Then one sees that a large area ab with a small a will never yield a low fundamental tone. What one can feel is that both a and b should be large or, otherwise stated, the domain should contain in its interior a large disk (see figure 1.92 on the facing page). This was conjectured long ago, but proven only in 1977 by Hayman. We refer to Osserman 1977 [981] and Croke 1981 [412] for an improvement, and note that the optimal result is still an open question, namely the optimal constant and the corresponding shapes. In exchange, the minimum principle shows immediately that the fundamental tone of a domain is never larger than that of the maximal disk included in the domain.

Another long standing conjecture has been solved only recently. Consider the second eigenfunction ϕ_2 of a bounded plane domain D . What are the possibilities for its nodal lines? Using the minimax principle, Courant proved a long time ago that the nodal lines of the k -th eigenfunction ϕ_k cannot divide D into more than k different domains. Thus we have only two topological possibilities for ϕ_2 , as in figure 1.93 on the next page.

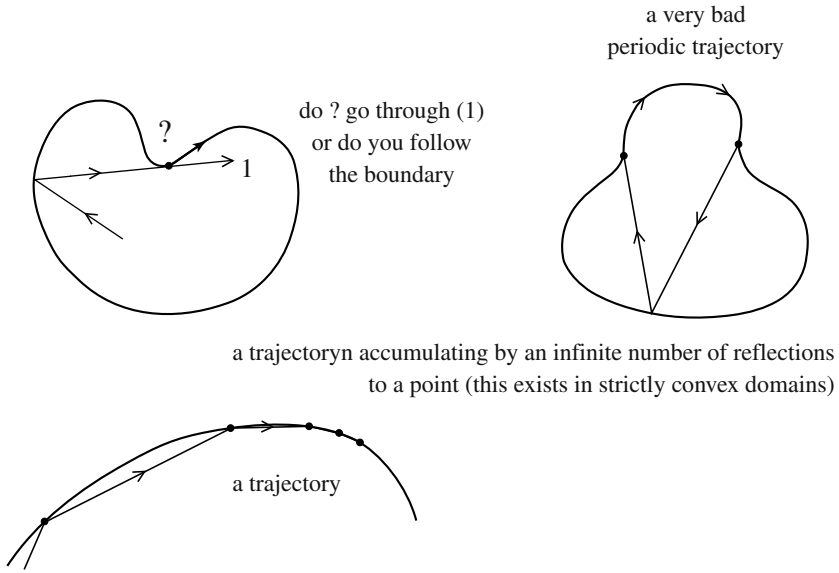
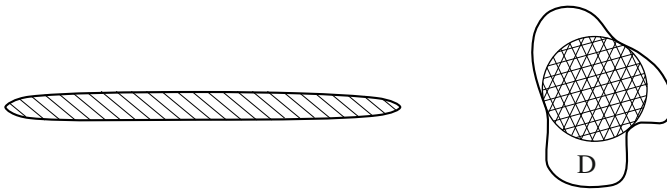
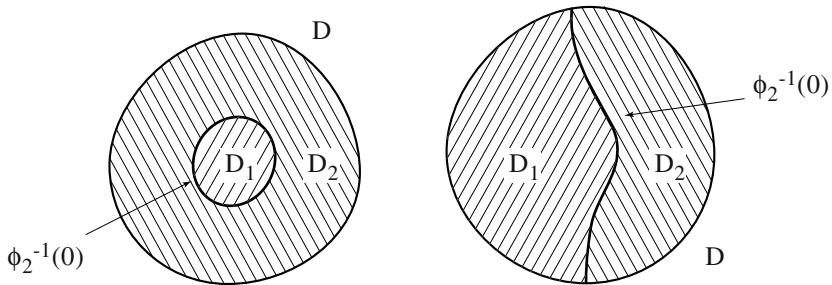


Fig. 1.91. From Guillemin & Melrose, 1979 [669]



a large area need not to produce a low tune nor a large diameter,
one needs to include a large disk

Fig. 1.92. Neither a large area nor a large diameter suffices to produce a low tone; one needs to include a large disk



excluded now

Fig. 1.93. Topology of the nodal lines of ϕ_2

In 1974, Payne conjectured that the only possibility is the one with a transverse nodal line (check it for yourself on the examples you know). This has just been proven in Melas 1992 [912], but only for convex domains. It is interesting that the proof makes an explicit use of Hayman's result mentioned above.

We come back to the main question: is the knowledge of the totality of $\text{Spec}(D)$ enough to fix the shape of D ? Formally, are two domains D and D' for which $\text{Spec}(D) = \text{Spec}(D')$ congruent, that is to say, is there an isometry of the Euclidean plane identifying D with D' ? This is exactly the musical question of the title of this section. It is not absurd to expect that only a countable sequence of numbers might yield the complete knowledge of a domain; think for example of a periodic function determined by its Fourier coefficients. Formulated in 1882 by Schuster, the negative answer was not uncovered until 1992, in Gordon, Webb, & Wolpert 1992 [579]. The two plane domains shown in figure 1.94 have the same spectrum. Simpler and more general examples have been found in Buser, Conway, Doyle & Semmler 1994 [294].

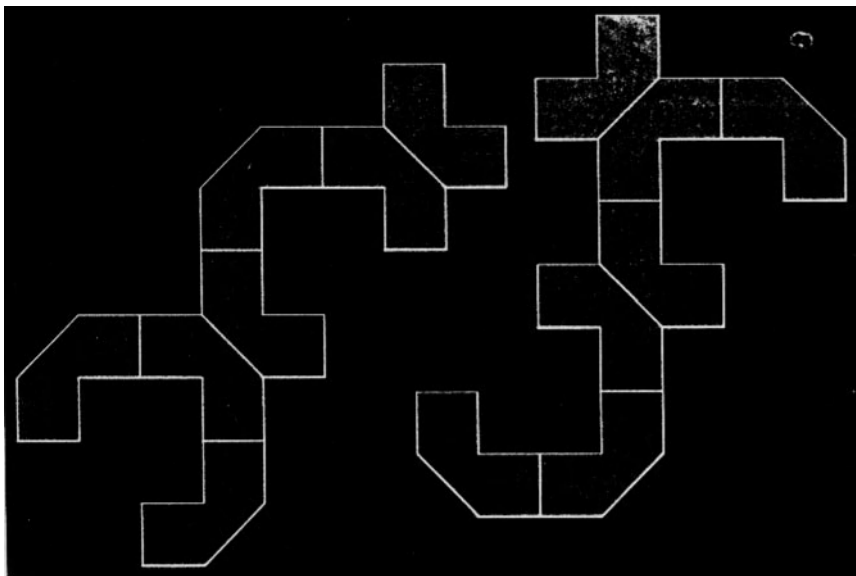


Fig. 1.94. Isospectral planar domains

You can never clip the wings of a mathematician; deciding the existence of isospectral domains does not end our inquiry. There is no uniqueness, but what is then the defect of uniqueness? Are there (up to congruence, of course) only a finite number of domains with identical spectra? Let us just mention that today there are compactness results (see Osgood, Phillips, & Sarnak 1988 [978]), but no example is known of a one parameter family of isospectral do-

mains (not all congruent). For the theory of isospectral Riemannian manifolds, see §9.12.

The counterexamples pictured in figure 1.94 on the preceding page are not convex. This question is addressed in Gordon & Webb 1994 [578] where convex counterexamples are built in four dimensions. But there are still no planar convex isospectral domains known.

1.8.5 Second Way: the Heat Equation

Today, one of the simplest tools to study the spectrum and the eigenfunctions is the heat equation. It is not the most powerful (the wave equation is more powerful), but the technique is much simpler. Both techniques, incidentally, extend to Riemannian manifolds with little alteration.

Henceforth, all functions vanish at the boundary ∂D .

We start with the *fundamental solution* of the heat equation,¹⁵ denoted by $K(m, n, t)$.

$$K : D \times D \times \mathbb{R}^+ \rightarrow \mathbb{R}.$$

It gives the solution $f(m, t)$ of the heat equation with given initial condition $f(m, 0) = g(m)$. The desired formula is the integral

$$f(m, t) = \int_D K(m, n, t) g(n) \, dn$$

and one can prove (see Bérard 1986 [135], Chavel 1984 [325] and also §9.7) that

$$K(m, n, t) = \sum_{i=1}^{\infty} \phi_i(m) \phi_i(n) e^{-\lambda_i t}$$

The reader can check that this ensures that f will satisfy the heat equation, as long as the above series converges. The series is known to converge, but that is not so easy to prove.

Another way to write the solution $f(m, t)$ with initial condition

$$g(m) = \sum_{i=1}^{\infty} a_i \phi_i(m)$$

is

$$f(m, t) = \sum_{i=1}^{\infty} a_i \phi_i(m) e^{-\lambda_i t}.$$

Our scheme to study the spectrum $\{\lambda_i\}$ is as follows. First, integrate the trace $K(m, m, t)$ over D , which yields

$$\int_D K(m, m, t) \, dm = \sum_{i=1}^{\infty} e^{-\lambda_i t}.$$

¹⁵ Also called the *heat kernel*.

It remains to get information on the diagonal values of K . To get some intuition, the fundamental solution can also be interpreted as follows: consider a singular initial data of heat at time zero consisting of a Dirac distribution located at the point m . Then $K(m, n, t)$ is the temperature after time t at the point n (see figure 1.95). In the case of the whole plane (no boundary D this time) this value is well known:

$$K_0(m, n, t) = \frac{1}{4\pi t} \exp\left(-\frac{\|m - n\|^2}{4t}\right). \quad (1.26)$$

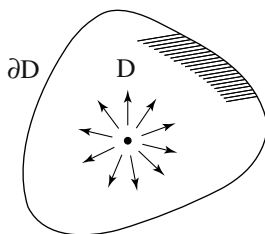


Fig. 1.95. It is infinitely cold outside; the particles diffusing heat away from the point m are not aware (at the first order at least) of the disaster that awaits them when they reach the boundary.

Note first that this implies that heat is diffused instantaneously, say with infinite speed, from any point (not very intuitive). Now, as Mark Kac says in Kac 1966 [775], in case there is a boundary ∂D , please do not worry. Your (correct) intuition is that the particles are not aware of the disaster that awaits them when they reach the boundary. It can be rigorously proven that their awareness is really not subtle, at least to the first order. The mathematical claim is that

$$K(m, m, t) \sim \frac{1}{4\pi t} \quad \text{as } t \rightarrow 0$$

whatever the point m is. We won't give a proof of this asymptotic. Integrating this over D :

$$\sum_{i=1}^{\infty} e^{-\lambda_i t} \sim \frac{\text{Area}(D)}{4\pi t} \quad \text{as } t \rightarrow 0$$

A very easy exercise will tell you that if you know the function

$$t \mapsto \sum_{i=1}^{\infty} e^{-\lambda_i t}$$

then you know $\text{Spec}(D)$. But do not deduce from this that you can easily estimate the counting function $N(\lambda)$ when $\lambda \rightarrow \infty$. Nevertheless, this is possible, but it is quite a hard theorem called the Hardy–Littlewood–Karamata Tauberian theorem:

$$N(\lambda) \sim \frac{\text{Area}(D)}{4\pi} \lambda$$

(see Ivrii 1998 [761]).

This asymptotic formula was proven as early as 1911 by Hermann Weyl; but note that his proof was completely different, not employing the heat equation. His more intuitive technique can be found in Bérard 1986 [135]; it uses minimax properties. Weyl's innovation is to approximate the domain D by an aggregate of squares (of smaller and smaller size, as in figure 1.96) and to paste eigenfunctions on these squares. To these functions we apply the minimax procedure, and can bound the eigenvalues of D on both sides by a limit argument. This time one does not need any Tauberian theorem. The first outcome of the minimax procedure is the strict monotonicity principle:

Proposition 14 *Consider two domains $D \subset D'$.*

$$\lambda_1(D) \geq \lambda_1(D')$$

Proof. Take the first eigenfunction ϕ_1 of D and extend it to be zero on $D' - D$. You are done by applying Dirichlet's principle to D' .

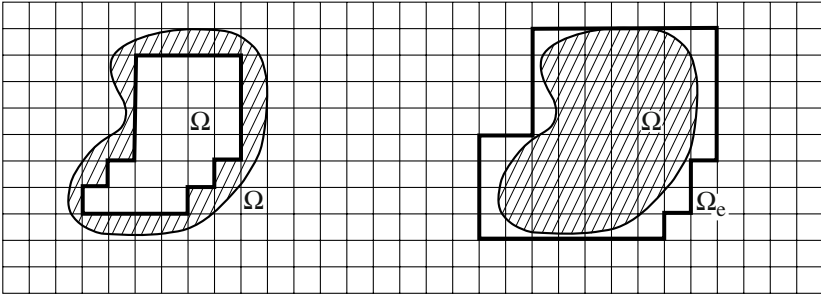


Fig. 1.96. Approximation by small squares

An immediate corollary is an asymptotic estimate for the eigenvalue λ_i :

$$\lambda_i \sim \frac{4\pi}{\text{Area}(D)} i \quad \text{as } i \rightarrow \infty$$

A second immediate corollary of Weyl's result is that you can hear if a domain is a disk because the spectrum will tell you the area of D , and then you just apply the Faber–Krahn inequality 1.22 on page 81.

How about a second order approximation? In 1954, Pleijel got the next order approximation. In his paper Kac 1966 [775], Kac works quite hard to get the third term, guessing that the right formula should be:

$$\sum_{i=1}^{\infty} e^{-\lambda_i t} \sim \frac{\text{Area}(D)}{2\pi t} - \frac{\text{length}(\partial D)}{\sqrt{2\pi t}} + \frac{1}{6}(1-r)$$

where r is the number of holes inside D . The second term is Pleijel's. Note that can one hear the area and the perimeter of D , hence the isoperimetric inequality yields again the fact that disks are characterized by their spectrum.

Kac could only prove the third term for polygons. It was proven the next year, 1967, in the very general context of Riemannian manifolds with boundary by McKean and Singer in their fundamental paper of 1967 [910]. You will read much more about it in chapter 9.

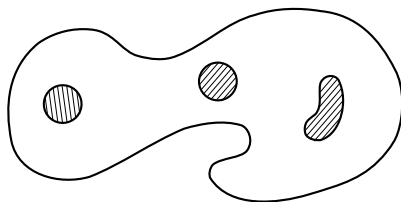


Fig. 1.97. One can hear the number of holes

The Tauberian theorem above shows that the first terms of $N(\lambda)$ and $\sum_{i=1}^{\infty} \exp(-\lambda_i t)$, can each be acquired from the other. But this does not apply to the following terms, so the preceding results say nothing concerning the second term of $N(\lambda)$. The second term of $N(\lambda)$ is still not completely understood today, even for squares; this is the famous *Gauß circle problem* in number theory (see Erdős, Gruber, & Hammer 1989 [491] and the references there on page 104, and also Bérard 1980 [133] for an introduction to the subject, and Huxley 1996 [752]). However, some results can be obtained, but one should use the wave equation instead. The fundamental solution is trickier than for the heat equation, but once it is understood, one purchases deeper results. In the case of Riemannian manifolds, optimal results can be obtained for the second order term. See chapter XXIX of Hörmander 1983 [738] and the very informative Trèves 1980 [1198, 1199], chapter 18. For more recent results and conjectures, see chapter 9.

1.8.5.1 Eigenfunctions

Another important question, having looked at eigenvalues, is to look at eigenfunctions, in particular where they vanish. The sets $\{\phi_i = 0\}$ are called the *nodal lines* of the domain. Nodal lines have been the subject of intensive studies on huge computers. They have many physical interpretations. Cover a vibrating membrane (a drum) with dust. Nodal lines are the places where the dust concentrates, precisely because (at least at the first order) in a given vibrating mode associated to an eigenfunction, nodal lines are the points which

do not move. Put your finger at a point m on the membrane; then the only possible modes occurring in the vibration will be those whose nodal line contains the point m . Nodal lines very often have a surprisingly “regular” behavior. This is only partially explained today; see references in chapter 9. Nodal lines have practical consequence. An example: when autoworkers attach the motor or the body to the underframe, bolts (or, more often these days, point solderings) are carefully placed on nodal lines of the underframe.

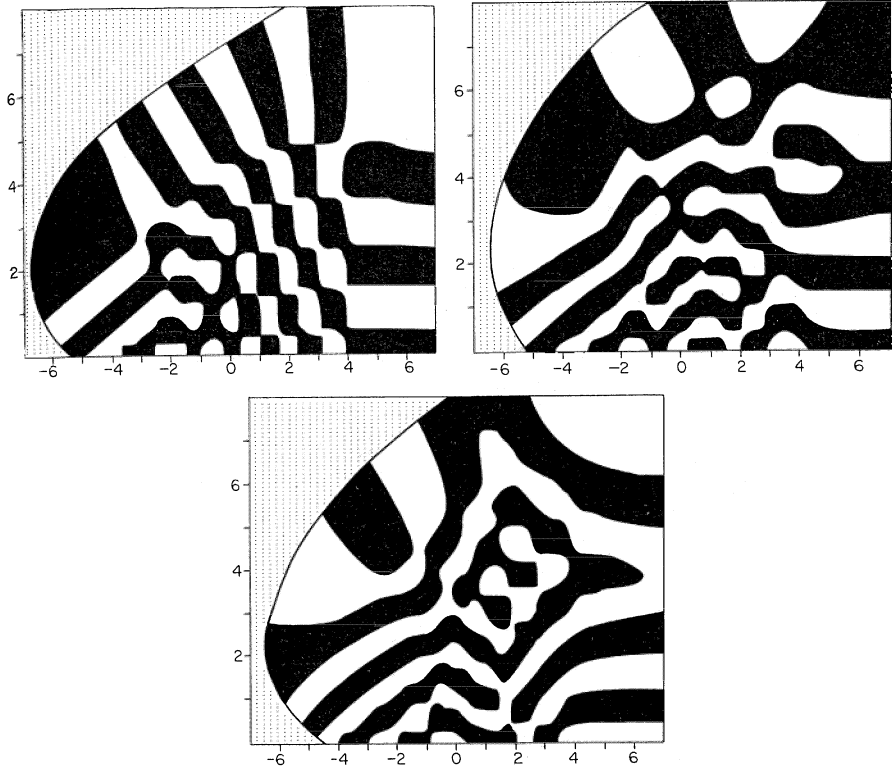


Fig. 1.98. Nodal lines of ergodic domains have surprisingly regular behaviour

1.8.6 Relations Between the Two Spectra

A plane domain D has both a length spectrum (for classical billiards) and a quantum mechanical spectrum (for quantum billiards). Formulas relating the two spectra, when they exist, are called *Poisson type formulas*. We will see the archetype of them, the classical Poisson formula 9.14 on page 389 for flat tori. The task of unearthing their mutual relations is that of connecting classical and quantum mechanics. It is not surprising that physicists published

the first papers on the subject: Balian & Bloch 1970/1971/1972/1974 [97, 98, 99, 100, 101]. They gave heuristic support to the conjecture that the spectrum determines the length spectrum. The other way around is more challenging. This is not surprising, since there are many more ways to address the spectrum than to address the length spectrum and the periodic trajectories. To our knowledge, this implication today is known only in the “generic” case; see Guillemin & Melrose 1979 [669]. We will not pursue this further, first because the case with boundary is much harder, and second, because we will study these relations on general Riemannian manifolds (but without boundary) in §9.9.

We mention an interesting fact, not well known, and basically the only result (with its extension in Colin de Verdière 1977 [390] to compact Riemannian manifolds) which goes from the length spectrum to the spectrum. It is due to V. F. Lazutkin and concerns smooth convex domains D of the plane. We recall first that the billiard trajectories inside an ellipse behave almost like those of a circle. A given trajectory remains constantly tangent to a homofocal ellipse (or hyperbola), as depicted in figure 1.99 on the facing page. By the way, it is unknown if only ellipses have this property. Back to an arbitrary smooth strictly convex domain: Lazutkin proved the existence of an infinite number of nested *caustics*. These are curves which are the envelopes of trajectories. Then he showed that to every caustic, there is an infinite sequence of numbers which approximate an infinite number of elements of the spectrum. The idea is beautiful: every tangent to the caustic defines an exterior half-plane. One then can perform, in a sense which has to be made precise, the sum of the waves sent outside from this half-plane. This sum is called a *quasimode* and will turn out to yield an approximation for a lot of eigenvalues. See chapters 9 and 10 for Riemannian manifolds, and Lazutkin 1993 [852] for a recent reference to Lazutkin’s many results.

Very recently, “visual” relations may have been discovered between periodic motions and nodal lines. The pictures in figure 1.100 on the next page are quite fascinating; they are taken from Gutzwiller 1990 [673] (a very informative book). On one hand, Gutzwiller conjectured that there are mathematical relations underlying the apparent visual relations between the two pictures, a phenomenon he called *scarring*. On the other hand, in the so-called *arithmetic case*, Sarnak proved in Sarnak 1995 [1095] that there is *no* scarring; also see §§§9.13.2.1.

There are some abstract Riemannian manifolds whose periodic geodesics are easier to track, and from them one can then get information on the spectrum; see the book Buser 1997 [293] for an exemplary case, and chapter 10.

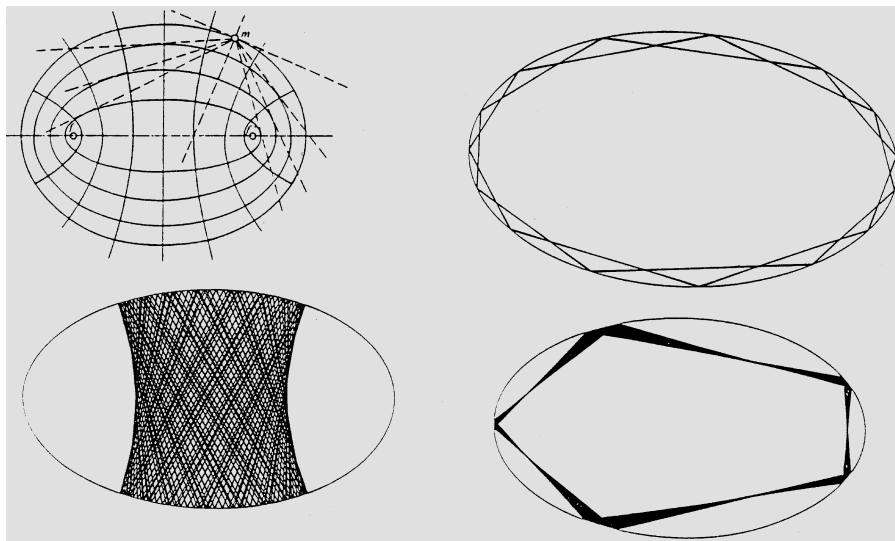


Fig. 1.99. Billiards in an ellipse

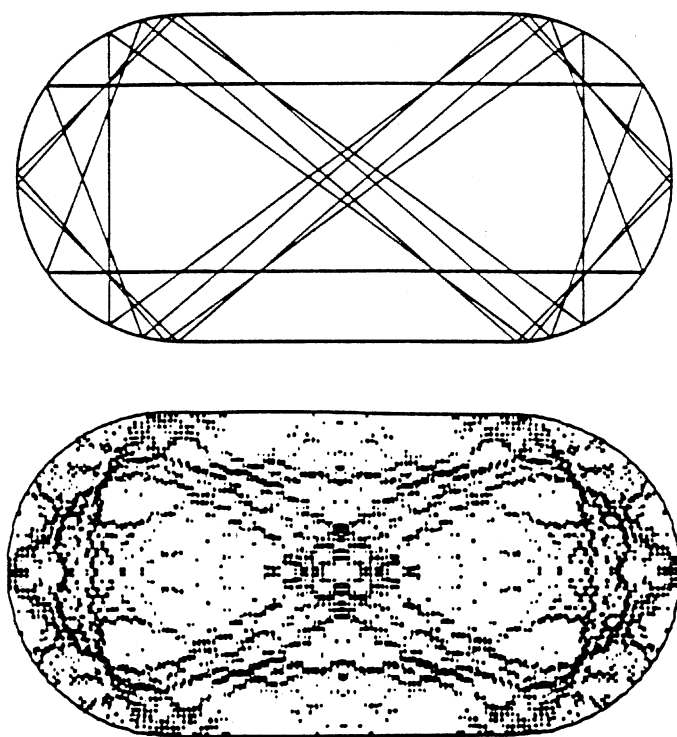


Fig. 1.100. Pictures from Gutzwiller 1990 [673]

1.9 Heat and Waves in $\mathbb{E}^3, \mathbb{E}^d$ and on the Sphere

1.9.1 Euclidean Spaces

In $\mathbb{E}^3$ or $\mathbb{E}^d$, there is little change from the plane, at the outset of our investigation. We replace Δ by:

$$\Delta = - \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} \quad (1.27)$$

where $\{x_i\}_{i=1,\dots,d}$ are Cartesian coordinates on $\mathbb{E}^d$. Stokes' theorem, used in the derivation of Dirichlet's principle, remains valid here:

$$\int_D \langle \text{grad } f, \text{grad } g \rangle = \int_D f \Delta g.$$

Note here something that we skipped above, namely the intrinsic character of the Laplacian Δ with respect to the Euclidean geometry of $\mathbb{E}^d$. In the language of modern analysis, one can say that the symbol of Δ is $-\|\cdot\|^2$. And if you add that the subsymbol vanishes, this completely identifies the Laplacian to an analyst.

Phenomena including the nature of the spectrum, expansion of a function as a generalized Fourier series in a basis of eigenfunctions, the expression via eigenfunctions of the fundamental solution of the heat equation, and Weyl's asymptotic formula, are identical but for the obvious changes. For example, the asymptotic expansion for $N(\lambda)$ becomes

$$N(\lambda) \sim \omega_d \frac{\lambda^{d/2}}{(2\pi)^d} \text{Vol}(D)$$

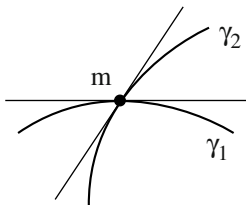
where ω_d is the volume of the unit ball in $\mathbb{E}^d$. From this one gets immediately

$$\lambda_i \sim \left(\frac{(2\pi)^d}{\omega_d \text{Vol}(D)} i \right)^{d/2}$$

We will look for explicit spectra of some special domains. Parallelipipeds are left to the reader. Let us now look at balls in $\mathbb{E}^3$. We aim to progress as on page 76, this time using spherical coordinates (ρ, θ, ϕ) where ρ is the distance to the origin, θ the latitude and ϕ the longitude, as in figure 1.101 on the next page. Recall that the change of coordinates is given by:

$$\begin{aligned} x &= \rho \sin \theta \cos \phi \\ y &= \rho \sin \theta \sin \phi \\ z &= \rho \cos \theta. \end{aligned}$$

We will now try to understand Δ on the spheres $\rho = \rho_0$.



how to compute the Laplacian of f on S^2

Fig. 1.101. (a) Spherical coordinates. (b) How to compute the Laplacian on the sphere

1.9.2 Spheres

It is very instructive to try to write Δ in spherical coordinates. Do not despair if you are stuck, it is a classical trap. One needs a trick to get out of it. You will find it, for example, in Berger, Gauduchon, & Mazet 1971 [174] (chapter III, G.V), Chavel 1984 [325], Courant & Hilbert 1953 [406], or in any text treating mathematical methods of physics. Even if you succeed, this equation looks a mess. You can get out of that mess, but you will not see the *spherical harmonics* very clearly. Moreover, you will suffer, because the representation of the sphere via latitude and longitude is very poor at the poles; see the pictures in your favourite atlas, or chapter 18 of Berger 1987 [164].

Why are things so complicated conceptually compared to a disk in the plane, where we separated variables, and looked for vibrations $h'' = -nh$ (n an integer)? The disk vibrations were gotten first through standard vibrations on the circle (and then for the radial directions, we had the Bessel equation). But the philosophy is that function theory on a circle is the same as for periodic functions on the real line, solved by Fourier analysis. What can replace the circle's Fourier analysis when we are on the sphere?

Approach the intrinsic geometry of the sphere, and consider spherical coordinates rather in the form (ρ, m) where $\rho > 0$ and $m \in S^2$, and work now with the inner geometry of S^2 , ignoring $\mathbb{E}^3$. We define the Laplace–Beltrami operator Δ (more precisely, Δ_{S^2} of S^2 if necessary to avoid confusion) for a function $f : S^2 \rightarrow \mathbb{R}$ by:

$$\Delta_{S^2} f(m) = - \left. \frac{d^2 (f \circ \gamma_1)}{dt^2} \right|_{t=0} - \left. \frac{d^2 (f \circ \gamma_2)}{dt^2} \right|_{t=0} \quad (1.28)$$

where γ_1, γ_2 are any two geodesics issuing from m at time $t = 0$, parameterized by arc length, and orthonormal at m . This is easily checked to be well defined (if you know some analysis, think of the symbol). Analogously, the reader can define Δ_{S^d} . A difficult computation yields, for a function $f : \mathbb{E}^d \rightarrow \mathbb{R}$

$$\Delta_{\mathbb{E}^d} f|_{S^{d-1}} = \Delta_{S^{d-1}} f - \frac{\partial^2 f}{\partial \rho^2} - (d-1) \frac{\partial f}{\partial \rho}$$

The eigenfunctions $\Delta f = \lambda f$ which are separated as $f(x, y, z) = g(\rho)h(m)$ must satisfy $\Delta_{S^2} h = \lambda h$. These functions on the sphere have been well known for a long time, and are called *spherical harmonics*. They are the subject of an immense literature, since they are next to the Fourier series for their importance in geometry and physics. They are the first examples of Fourier analysis on a Riemannian manifold considered abstractly, although this was not the way people looked at them originally. But we emphasized this aspect immediately, since we want to introduce you to Riemannian geometry and analysis in the general setting. For a spectacular geometric application of spherical harmonics to the extremely hard problem of balancing a large number of points evenly around the sphere, see Colin de Verdière 1989 [394] or the very good book Sarnak 1990 [1094]. For the use of spherical harmonics to estimate the kissing number (the maximum number of solid nonintersecting unit balls which can touch one given unit ball), see chapter 13 of Conway & Sloane 1999 [403]. We will meet the kissing number again on page 583.

The eigenvalues of S^d are the integers $k(d+k-1)$, with the multiplicity

$$\frac{(d+2k-1)(d+k-2)!}{k!(d+k-3)!}$$

where k runs through the integers. References can be Berger, Gauduchon & Mazet 1971 [174], Chavel 1984 [325]. The miracle is that the harmonic polynomials in $\mathbb{E}^d$, when restricted to S^{d-1} , are eigenfunctions of S^{d-1} and, by Stone–Weierstraß, they are numerous enough to prove that there are no other ones.

In particular, the first eigenfunctions are especially appealing; they are the restrictions of the linear forms of $\mathbb{E}^d$, and their eigenvalue is $\lambda_1 = d$. Interpreted in spherical geometry, they are the cosine of the distance function to some point. We will use this heavily in §§12.2.5.

Let us finish things quickly by saying that, spherical eigenvalues and eigenfunctions being known, this in turn solves Dirichlet’s problem for the vibrations of balls in $\mathbb{E}^3$ through Bessel type equations. Our analysis extends to any dimension d for balls in $\mathbb{E}^d$ via spherical harmonics for the spheres S^{d-1} . Again, for extremely few other domains D of $\mathbb{E}^d$ can the eigenvalues or the eigenfunctions be computed explicitly. Ellipsoids can be managed by the use of confocal quadrics; see Courant & Hilbert 1953 [406, 407], Morse & Feshbach 1953 [945]. And quite a recent result of Pierre Bérard works for a few very special polyhedra¹⁶ known as Weyl type domains, corresponding to Lie groups; see Bérard 1980 [134]. See also Bérard & Besson 1980 [138].

We will not say much now about spherical harmonics since you are going to meet them often later on and because they are not our principal concern. For those who like recent results and open problems (yes, there are some, even for such old and much belabored subjects) we mention first that Yves Meyer says among other things the following. We consider the wave equation

¹⁶ Polyhedra are rather called *polytopes* starting with dimension $d = 4$.

on spheres

$$\Delta_{S^2} f = -\frac{\partial^2 f}{\partial t^2}$$

as, for example, describing the evolution with time of a thin layer of water on the sphere. This would describe the motion of the sea, if there were no continents on the planet. It might not be a bad approximation, since oceans are the bulk of the Earth's surface. His first result backs intuition. Disturb the water with a big shock (a Dirac distribution, if you prefer) somewhere, say at the north pole. Then you will always find another big shock at the south pole after a time T which is no larger than 2π . But Meyer's second series of results oppose intuition: the big shock can move from the north to the south pole while remaining extremely small everywhere for all time between 0 and T , as in figure 1.102 on the next page. Worse: some apparently moderate shocks can, with larger and larger time, be very small almost everywhere and almost all the time, but at some times and some places can be as big as desired (everything there and then will break down). Meyer's results explained the following strange observation: quite recently a tidal wave was observed at Martinique, and came back there later, this being completely unnoticed everywhere else in between times.

1.9.3 Billiards in Higher Dimensions

The Faber–Krahn inequality extends to any dimension. Again, the proof involves the isoperimetric inequality (in any dimension). Contrarily, billiards in dimensions larger than two are still to be fully explored. There is one exception: billiards in concave regions, for which we refer to Katok, Strelcyn, Ledrappier & Przytycki 1986 [788]. Such regions have the best possible ergodic behavior.

But for nonconcave regions, even for polyhedra in three dimensional space, almost nothing is known concerning periodic motions or ergodicity, except for rectangular parallelepipeds, balls (of course) and the Bérard examples mentioned above. There is an unpublished result of Katok asserting that the length counting function is subexponential. But, even in the simplest possible cases, things are either not finished, or completely open. Let us mention only two cases. The first is that of the cube. Dynamics in a cube are comparable to dynamics in a square: a trajectory is periodic or everywhere dense. But how quickly a non-periodic trajectory darkens your computer screen is complicated. It depends on the simultaneous approximation of a pair of real numbers by rational numbers. This subject is partly open; in particular one does not know which triples of real numbers give the cubic analogue of the golden ratio (the trajectory in the square for which your computer screen gets dark most rapidly). But the coding of the faces successively hit by the ball is beginning to be unmasked: see Arnoux, Mauduit, Shiokawa, & Tamura 1994 [71]. For the regular tetrahedron, it seems that nobody today has the foggiest idea whether trajectories are ergodic or more like polygons with angles

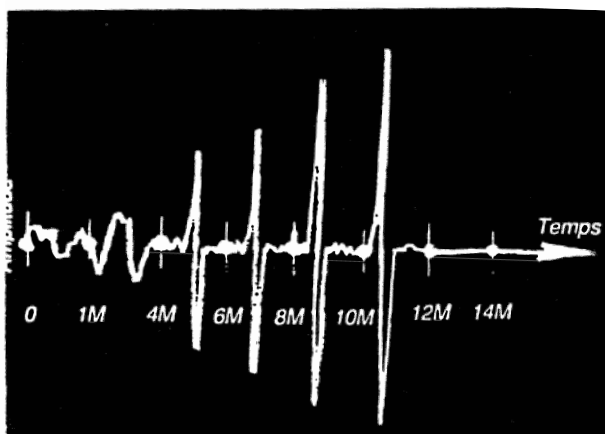
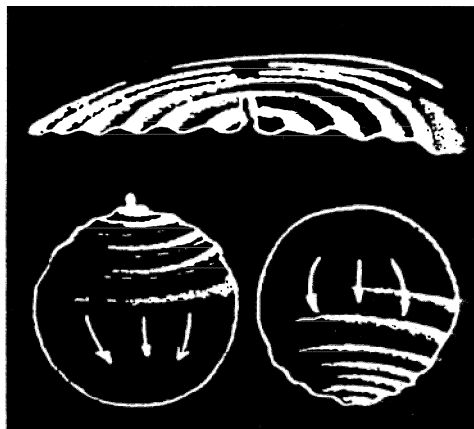


Fig. 1.102. Tidal waves returning to Martinique

rational multiples of π . The fact that there is no guess is surprising: today no one can program a computer to accurately simulate tetrahedral billiards. However, there are a few results yielding periodic trajectories in tetrahedra, based on the geometry of isometries of $\mathbb{E}^3$, but they are difficult to locate; Conway has some unpublished results, and there are published results of R. Hayward, M. Gardner, and D. Wells.

1.9.4 The Wave Equation Versus the Heat Equation

In any dimension, the wave equation is much harder to study than the heat equation. The drastic difference of behavior of waves according to the parity of the dimension of the space where waves propagate was observed long

ago. We will come back to these two equations in chapter 9. Briefly put, the heat equation corresponds to the Laplace transform, while the wave equation corresponds to the more powerful Fourier transform.